

Soccult

2026-05-18

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2     4.0.1      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
#load datasets
```

```
summary_data <- read_csv("experiment_summary.csv")
```

```
## Rows: 1798 Columns: 9
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (5): participant_id, consent, mode, phase, dyad_id
```

```
## dbl (4): iteration, target_angle, reaction_time_s, distance_px
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
frame_data <- read_csv("experiment_data.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
```

```
## e.g.:
```

```
##   dat <- vroom(...)
```

```
##   problems(dat)
```

```
## Rows: 316753 Columns: 17
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (5): participant_id, mode, phase, playback_axis, dyad_id
```

```
## dbl (12): iteration, target_angle, timestamp, cursor_x, cursor_y, h_vx, h_vy...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Create contribution-angle factor
```

```
summary_data <- summary_data %>%
  mutate(
    contribution_angle = as.factor(target_angle)
  )
```

```
# Check data
```

```
names(summary_data)
```

```
## [1] "participant_id"      "consent"           "mode"
## [4] "iteration"           "target_angle"      "phase"
## [7] "reaction_time_s"     "distance_px"        "dyad_id"
## [10] "contribution_angle"
```

```
names(frame_data)
```

```
## [1] "participant_id" "mode"           "iteration"      "target_angle"
## [5] "phase"          "timestamp"      "cursor_x"      "cursor_y"
## [9] "h_vx"           "h_vy"           "p_vx"          "p_vy"
## [13] "target_x"       "target_y"       "ai_axis"       "playback_axis"
## [17] "dyad_id"
```

```
# 1. Remove practice trials
```

```
summary_clean <- summary_data %>%
  filter(mode != "practice")
```

```
table(summary_clean$mode)
```

```
##
##      ai cooperative individual playback
##      480         240         480     480
```

```
#remove noemi
```

```
summary_clean <- summary_data %>%
  filter(!(participant_id == "noemi" & mode == "individual")) %>%
  filter(!(dyad_id == "jens_noemi" & mode == "cooperative"))
```

```
# Check trial counts
```

```
summary_clean %>%
  count(mode, phase)
```

```
## # A tibble: 10 x 3
##   mode      phase      n
##   <chr>    <chr>   <int>
## 1 ai      approach   240
## 2 ai      homing_in  240
## 3 cooperative approach   120
## 4 cooperative homing_in  120
## 5 individual approach   240
```

```
## 6 individual homing_in 240
## 7 playback approach 240
## 8 playback homing_in 240
## 9 practice approach 59
## 10 practice homing_in 59
```

Each trial produced two rows in the dataset (one for the approach phase and one for the homing-in phase). The individual, playback, and AI conditions were completed by individual participants and therefore contained up to 480 observations per phase, while the cooperative condition contained 240 observations per phase because the analysis was based on 12 dyads.

```
# Set contribution-angle order
summary_clean <- summary_clean %>%
  mutate(
    contribution_angle = factor(
      contribution_angle,
      levels = c("0", "22.5", "45", "67.5", "90")
    )
  )
```

```
# 2. Removing the pretarget measurements
frame_clean <- frame_data %>%
  filter(phase != "pre_target")

# Check
table(frame_clean$phase)
```

```
##
## approach homing_in
## 76409 190307
```

```
# 3. Define and label error trials in the dataset
```

```
# In Wahn et al. (2016), error trials were defined as trials in which the cursor entered the homing-in
```

```
summary_clean <- summary_clean %>%
  mutate(error_trial = FALSE)

# Check
table(summary_clean$error_trial)
```

```
##
## FALSE
## 1798
```

```
# After applying the operational definition to the dataset, no trials were classified as error trials i
```

```
# 4. Label conditions (making it factor variable)
summary_clean <- summary_clean %>%
  mutate(
    condition = factor(
      mode,
```

```

      levels = c("individual", "cooperative", "playback", "ai")
    )
  )

table(summary_clean$condition)

```

```

##
## individual cooperative playback ai
##      480      240      480      480

```

5. Label higher-level partner/adaptiveness type

```

summary_clean <- summary_clean %>%
  mutate(
    partner_type = case_when(
      condition == "individual" ~ "no_partner",
      condition == "cooperative" ~ "adaptive_human",
      condition == "ai" ~ "adaptive_ai",
      condition == "playback" ~ "nonadaptive_playback"
    )
  )

```

Convert it to a factor

```

summary_clean <- summary_clean %>%
  mutate(
    partner_type = factor(
      partner_type,
      levels = c(
        "no_partner",
        "adaptive_human",
        "adaptive_ai",
        "nonadaptive_playback"
      )
    )
  )

```

Check

```

table(summary_clean$partner_type)

```

```

##
##      no_partner      adaptive_human      adaptive_ai
##      480      240      480
## nonadaptive_playback
##      480

```

6. Converting phase to factor variable

```

summary_clean <- summary_clean %>%
  mutate(
    phase = factor(
      phase,
      levels = c("approach", "homing_in"),
      labels = c("Approach", "Homing-in")
    )
  )

```

```

)

# Check
table(summary_clean$phase)

##
## Approach Homing-in
##      899      899

# 7. Check contribution angles
table(summary_clean$contribution_angle)

##
##      0 22.5   45 67.5   90
## 358 366 360 352 362

# 8. Making sure iteration is numeric for further analysis
summary_clean <- summary_clean %>%
  mutate(
    iteration = as.numeric(iteration)
  )

# Check
glimpse(summary_clean)

## Rows: 1,798
## Columns: 13
## $ participant_id      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "mads", "ma~
## $ consent             <chr> "yes", "yes", "yes", "yes", "yes", "yes", "yes", "y~
## $ mode                <chr> "practice", "practice", "practice", "practice", "pr~
## $ iteration           <dbl> 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 1, 1, 2, 2, 3, 3, 4, ~
## $ target_angle        <dbl> 0.0, 0.0, 22.5, 22.5, 67.5, 67.5, 90.0, 90.0, 22.5, ~
## $ phase               <fct> Approach, Homing-in, Approach, Homing-in, Approach, ~
## $ reaction_time_s     <dbl> 1.2898, 2.4980, 2.3211, 1.4239, 2.4430, 1.5958, 2.1~
## $ distance_px         <dbl> 285.00, 155.00, 377.43, 65.00, 413.64, 70.00, 369.5~
## $ dyad_id             <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ contribution_angle  <fct> 0, 0, 22.5, 22.5, 67.5, 67.5, 90, 90, 22.5, 22.5, 6~
## $ error_trial         <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FA~
## $ condition           <fct> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, individual, ~
## $ partner_type        <fct> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, no_partner, ~

```

```

# 9. Add axis-control variable
frame_clean <- frame_clean %>%
  mutate(
    axis_control = case_when(

      # AI condition
      mode == "ai" & ai_axis == 1 ~ "AI_vertical",
      mode == "ai" & ai_axis == 0 ~ "AI_horizontal",

      # Playback condition

```

```

mode == "playback" & playback_axis == "vertical" ~ "Playback_vertical",
mode == "playback" & playback_axis == "horizontal" ~ "Playback_horizontal",

TRUE ~ NA_character_
)
)

# Convert to factor
frame_clean <- frame_clean %>%
  mutate(
    axis_control = factor(axis_control)
  )

# Check
table(frame_clean$axis_control, useNA = "ifany")

```

```

##
##      AI_horizontal      AI_vertical Playback_horizontal Playback_vertical
##           23099           24633           43397           33788
##           <NA>
##           141799

```

```

# 10. Use logged dyad ID
summary_clean <- summary_clean %>%
  mutate(
    dyad_id = as.character(dyad_id),
    dyad_id = na_if(dyad_id, ""),
    dyad_id = factor(dyad_id)
  )

# Check
table(summary_clean$dyad_id, useNA = "ifany")

```

```

##
##      666_    august_jens  jakob_niels  mads_lasse  odin_line  valeria_emil
##           40           40           40           40           40           40
##           <NA>
##           1558

```

```

# 11. Create trial-level dataset
trial_data <- summary_clean %>%
  select(
    participant_id,
    dyad_id,
    mode,
    condition,
    partner_type,
    iteration,
    phase,
    target_angle,
    contribution_angle,
    reaction_time_s,

```

```

    distance_px,
    error_trial
  )

# Check
trial_data %>%
  count(condition, phase)

```

```

## # A tibble: 10 x 3
##   condition phase      n
##   <fct>      <fct> <int>
## 1 individual Approach  240
## 2 individual Homing-in 240
## 3 cooperative Approach  120
## 4 cooperative Homing-in 120
## 5 playback   Approach  240
## 6 playback   Homing-in 240
## 7 ai         Approach  240
## 8 ai         Homing-in 240
## 9 <NA>       Approach   59
##10 <NA>       Homing-in   59

```

#This trial_data will be the main dataset for: RT analyses; DIST analyses; ANOVAs; t-tests; condition c

12. Rename dependent variables for analysis

```

trial_data <- trial_data %>%
  rename(
    RT   = reaction_time_s,
    DIST = distance_px
  )

# Check
glimpse(trial_data)

```

```

## Rows: 1,798
## Columns: 12
## $ participant_id    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "mads", "ma~
## $ dyad_id           <fct> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ mode              <chr> "practice", "practice", "practice", "practice", "pr~
## $ condition         <fct> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, individual,~
## $ partner_type      <fct> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, no_partner,~
## $ iteration         <dbl> 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 1, 1, 2, 2, 3, 3, 4, ~
## $ phase             <fct> Approach, Homing-in, Approach, Homing-in, Approach,~
## $ target_angle      <dbl> 0.0, 0.0, 22.5, 22.5, 67.5, 67.5, 90.0, 90.0, 22.5,~
## $ contribution_angle <fct> 0, 0, 22.5, 22.5, 67.5, 67.5, 90, 90, 22.5, 22.5, 6~
## $ RT                <dbl> 1.2898, 2.4980, 2.3211, 1.4239, 2.4430, 1.5958, 2.1~
## $ DIST              <dbl> 285.00, 155.00, 377.43, 65.00, 413.64, 70.00, 369.5~
## $ error_trial       <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FA~

```

13. Remove error trials from RT/DIST analyses

```

analysis_data <- trial_data %>%
  filter(error_trial == FALSE)

```

```
# Check
table(analysis_data$error_trial)
```

```
##
## FALSE
## 1798
```

Error-trial criteria: “Two types of behavior were classified as errors: If the cursor moved into the homing-in zone but then exited it again, and if the cursor moved outside of the screen borders.” (Wahn et al., 2016)

```
# 14. Create averaged participant-level dataset
participant_means <- analysis_data %>%
  group_by(
    participant_id,
    condition,
    partner_type,
    phase,
    contribution_angle
  ) %>%
  summarise(
    mean_RT = mean(RT, na.rm = TRUE),
    mean_DIST = mean(DIST, na.rm = TRUE),
    error_rate = mean(error_trial, na.rm = TRUE),
    n_trials = n(),
    .groups = "drop"
  )

# Check
glimpse(participant_means)
```

```
## Rows: 430
## Columns: 9
## $ participant_id      <chr> "666", "666", "666", "666", "666", "666", "666", "6~
## $ condition           <fct> individual, individual, individual, individual, ind~
## $ partner_type        <fct> no_partner, no_partner, no_partner, no_partner, no_~
## $ phase               <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle  <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ mean_RT             <dbl> 1.392000, 2.836275, 2.886925, 3.564550, 1.509950, 3~
## $ mean_DIST           <dbl> 285.0000, 423.7500, 448.7500, 446.2500, 285.0000, 3~
## $ error_rate          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ n_trials            <int> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, ~
```

```
# 15a. Aggregate to participant-level means
participant_means <- analysis_data %>%
  group_by(
    participant_id,
    condition,
    partner_type,
    phase,
    contribution_angle
  ) %>%
```



```

summarise(
  mean_RT = mean(RT, na.rm = TRUE),
  mean_DIST = mean(DIST, na.rm = TRUE),
  error_rate = mean(error_trial, na.rm = TRUE),
  n_trials = n(),
  .groups = "drop"
)

# Check
glimpse(participant_means)

```

```

## Rows: 430
## Columns: 9
## $ participant_id      <chr> "666", "666", "666", "666", "666", "666", "666", "6~
## $ condition           <fct> individual, individual, individual, individual, ind~
## $ partner_type        <fct> no_partner, no_partner, no_partner, no_partner, no~
## $ phase               <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle  <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ mean_RT             <dbl> 1.392000, 2.836275, 2.886925, 3.564550, 1.509950, 3~
## $ mean_DIST           <dbl> 285.0000, 423.7500, 448.7500, 446.2500, 285.0000, 3~
## $ error_rate          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ n_trials            <int> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, ~

```

15b. Aggregate cooperative condition to dyad-level means

```

dyad_means <- analysis_data %>%
  filter(condition == "cooperative") %>%
  group_by(
    dyad_id,
    phase,
    contribution_angle
  ) %>%
  summarise(
    mean_RT = mean(RT, na.rm = TRUE),
    mean_DIST = mean(DIST, na.rm = TRUE),
    error_rate = mean(error_trial, na.rm = TRUE),
    n_trials = n(),
    .groups = "drop"
  )

# Check
glimpse(dyad_means)

```

```

## Rows: 60
## Columns: 7
## $ dyad_id            <fct> 666_ , 666_ , 666_ , 666_ , 666_ , 6~
## $ phase              <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ mean_RT            <dbl> 1.377325, 2.763150, 1.863500, 3.114750, 1.528750, 1~
## $ mean_DIST          <dbl> 285.0000, 404.8900, 364.4400, 431.2500, 285.0000, 1~
## $ error_rate         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ n_trials           <int> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, ~

```

```

# 16. Calculate mean RT
mean_RT_data <- analysis_data %>%
  group_by(
    participant_id,
    condition,
    phase,
    contribution_angle
  ) %>%
  summarise(
    mean_RT = mean(RT, na.rm = TRUE),
    n_trials = n(),
    .groups = "drop"
  )

# Check
glimpse(mean_RT_data)

```

```

## Rows: 430
## Columns: 6
## $ participant_id      <chr> "666", "666", "666", "666", "666", "666", "666", "6~
## $ condition           <fct> individual, individual, individual, individual, ind~
## $ phase               <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle  <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ mean_RT             <dbl> 1.392000, 2.836275, 2.886925, 3.564550, 1.509950, 3~
## $ n_trials            <int> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, ~

```

```

mean_RT_data %>%
  count(condition, phase, contribution_angle)

```

```

## # A tibble: 50 x 4
##   condition phase      contribution_angle      n
##   <fct>      <fct>      <fct>          <int>
## 1 individual Approach    0              12
## 2 individual Approach  22.5            12
## 3 individual Approach  45              12
## 4 individual Approach  67.5            12
## 5 individual Approach  90              12
## 6 individual Homing-in  0              12
## 7 individual Homing-in 22.5            12
## 8 individual Homing-in 45              12
## 9 individual Homing-in 67.5            12
## 10 individual Homing-in 90              12
## # i 40 more rows

```

```

# 17. Calculate mean DIST
mean_DIST_data <- analysis_data %>%
  group_by(
    participant_id,
    condition,
    phase,
    contribution_angle
  ) %>%

```

```

summarise(
  mean_DIST = mean(DIST, na.rm = TRUE),
  n_trials = n(),
  .groups = "drop"
)

# Check
glimpse(mean_DIST_data)

## Rows: 430
## Columns: 6
## $ participant_id      <chr> "666", "666", "666", "666", "666", "666", "666", "6~
## $ condition           <fct> individual, individual, individual, individual, ind~
## $ phase               <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ mean_DIST           <dbl> 285.0000, 423.7500, 448.7500, 446.2500, 285.0000, 3~
## $ n_trials            <int> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, ~

```

```

mean_DIST_data %>%
  count(condition, phase, contribution_angle)

```

```

## # A tibble: 50 x 4
##   condition phase      contribution_angle      n
##   <fct>      <fct>      <fct>                <int>
## 1 individual Approach    0                  12
## 2 individual Approach  22.5                12
## 3 individual Approach  45                  12
## 4 individual Approach  67.5                12
## 5 individual Approach  90                  12
## 6 individual Homing-in  0                  12
## 7 individual Homing-in 22.5                12
## 8 individual Homing-in 45                  12
## 9 individual Homing-in 67.5                12
## 10 individual Homing-in 90                  12
## # i 40 more rows

```

18. Calculate collective benefit (RT)

```

# Mean individual RT per participant
individual_RT <- analysis_data %>%
  filter(condition == "individual") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(
    individual_RT = mean(RT, na.rm = TRUE),
    .groups = "drop"
  )

```

```

# Mean cooperative RT per dyad
cooperative_RT <- analysis_data %>%
  filter(condition == "cooperative") %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(

```

```

    cooperative_RT = mean(RT, na.rm = TRUE),
    .groups = "drop"
  )

# Best individual within each dyad
#dyad mapping
participant_dyad_map <- tibble::tribble(
  ~participant_id, ~dyad_id,
  "666",          "666_  ",
  "  ",          "666_  ",
  "august",       "august_jens",
  "jens",         "august_jens",
  "jakob",        "jakob_niels",
  "niels",        "jakob_niels",
  "odin",         "odin_line",
  "line",         "odin_line",
  "valeria",      "valeria_emil",
  "emil",         "valeria_emil",
  "lasse",        "mads_lasse",
  "mads",         "mads_lasse"
)

best_individual_RT <- individual_RT %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(
    best_individual_RT = min(individual_RT, na.rm = TRUE),
    .groups = "drop"
  )

# Calculate collective benefit
collective_benefit_RT <- cooperative_RT %>%
  left_join(
    best_individual_RT,
    by = c("dyad_id", "phase", "contribution_angle")
  ) %>%
  mutate(
    collective_benefit = best_individual_RT / cooperative_RT
  )

# Check
glimpse(collective_benefit_RT)

```

```

## Rows: 60
## Columns: 6
## $ dyad_id      <chr> "666_  ", "666_  ", "666_  ", "666_  ", "66~
## $ phase        <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ cooperative_RT <dbl> 1.377325, 2.763150, 1.863500, 3.114750, 1.528750, 1~
## $ best_individual_RT <dbl> 1.365425, 1.565950, 1.763900, 2.134950, 1.352375, 2~
## $ collective_benefit <dbl> 0.99136006, 0.56672638, 0.94655219, 0.68543222, 0.8~

```

```

#Collective benefit DIST

individual_DIST <- analysis_data %>%
  filter(condition == "individual") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(
    individual_DIST = mean(DIST, na.rm = TRUE),
    .groups = "drop"
  )

cooperative_DIST <- analysis_data %>%
  filter(condition == "cooperative") %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(
    cooperative_DIST = mean(DIST, na.rm = TRUE),
    .groups = "drop"
  )

#Best (lowest) individual DIST within each dyad
best_individual_DIST <- individual_DIST %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(
    best_individual_DIST = min(individual_DIST, na.rm = TRUE),
    .groups = "drop"
  )

#collective benefit (DIST)
collective_benefit_DIST <- cooperative_DIST %>%
  left_join(
    best_individual_DIST,
    by = c("dyad_id", "phase", "contribution_angle")
  ) %>%
  mutate(
    collective_benefit = best_individual_DIST / cooperative_DIST
  )

#check
glimpse(collective_benefit_DIST)

```

```

## Rows: 60
## Columns: 6
## $ dyad_id      <chr> "666_ ", "666_ ", "666_ ", "666_ ", "~
## $ phase        <fct> Approach, Approach, Approach, Approach, Approach,~
## $ contribution_angle <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, ~
## $ cooperative_DIST <dbl> 285.0000, 404.8900, 364.4400, 431.2500, 285.0000,~
## $ best_individual_DIST <dbl> 285.0000, 383.3350, 412.0350, 409.8725, 285.0000,~
## $ collective_benefit <dbl> 1.00000000, 0.94676332, 1.13059763, 0.95042899, 1~

```

Collective benefit interpretation: “Values above 1 indicate a collective benefit because dyad RT is lower than the faster member’s RT. This measure was calculated separately for Approach and Homing-in phase - averaged across all contribution angles as well as separately for each of the three contribution angles (30%, 50%, 70%)” - 0° and 90°: only one dyad member controls the cursor as only horizontal or vertical is

required - 45°: both members contributed equally - 50% - 22.5° and 67.5°: one of the members contributed more than the other: 30% vs 70%

```
# 19. Calculate skill ratio (RT)
skill_ratio_RT <- individual_RT %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(
    faster_partner = min(individual_RT, na.rm = TRUE),
    slower_partner = max(individual_RT, na.rm = TRUE),
    skill_ratio    = faster_partner / slower_partner,
    .groups = "drop"
  )
# Check
glimpse(skill_ratio_RT)
```

```
## Rows: 60
## Columns: 6
## $ dyad_id      <chr> "666_ ", "666_ ", "666_ ", "666_ ", "66~
## $ phase        <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ faster_partner <dbl> 1.365425, 1.565950, 1.763900, 2.134950, 1.352375, 2~
## $ slower_partner <dbl> 1.392000, 2.836275, 2.886925, 3.564550, 1.509950, 3~
## $ skill_ratio   <dbl> 0.9809088, 0.5521150, 0.6109961, 0.5989396, 0.89564~
```

```
#Skill ratio DIST
skill_ratio_DIST <- individual_DIST %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(
    faster_partner = min(individual_DIST, na.rm = TRUE),
    slower_partner = max(individual_DIST, na.rm = TRUE),
    skill_ratio    = faster_partner / slower_partner,
    .groups = "drop"
  )
# Check
glimpse(skill_ratio_DIST)
```

```
## Rows: 60
## Columns: 6
## $ dyad_id      <chr> "666_ ", "666_ ", "666_ ", "666_ ", "66~
## $ phase        <fct> Approach, Approach, Approach, Approach, Approach, H~
## $ contribution_angle <fct> 0, 22.5, 45, 67.5, 90, 0, 22.5, 45, 67.5, 90, 0, 22~
## $ faster_partner <dbl> 285.0000, 383.3350, 412.0350, 409.8725, 285.0000, 2~
## $ slower_partner <dbl> 285.0000, 423.7500, 448.7500, 446.2500, 285.0000, 3~
## $ skill_ratio   <dbl> 1.00000000, 0.90462537, 0.91818384, 0.91848179, 1.0~
```

Skill ratio interpretation: A value of 1 indicates that both members perform equally well individually, whereas an increasingly lower ratio indicates a larger interindividual performance difference.

#INDIVIDUAL and COLLECTIVE analysis

```

#correlating interindividual RT skill ratios with collective benefit for each of the contribution conditions
#join the two into one data frame
corr_data <- collective_benefit_RT %>%
  left_join(skill_ratio_RT, by = c("dyad_id", "phase", "contribution_angle"))

#Pearson r between skill_ratio and collective benefit
correlation_results <- corr_data %>%
  group_by(phase, contribution_angle) %>%
  summarise(
    n = sum(!is.na(skill_ratio) & !is.na(collective_benefit)),
    r = cor(skill_ratio, collective_benefit, use = "complete.obs"),
    p_value = tryCatch(
      cor.test(skill_ratio, collective_benefit)$p.value,
      error = function(e) NA_real_
    ),
    .groups = "drop"
  ) %>%
  mutate(
    sig = case_when(
      p_value < .001 ~ "***",
      p_value < .01 ~ "**",
      p_value < .05 ~ "*",
      p_value < .10 ~ "†",
      TRUE ~ ""
    )
  )
print(correlation_results)

```

```

## # A tibble: 10 x 6
##   phase      contribution_angle      n      r p_value sig
##   <fct>      <fct>          <int> <dbl>  <dbl> <chr>
## 1 Approach    0                6 0.486  0.328 ""
## 2 Approach   22.5                6 0.590  0.218 ""
## 3 Approach   45                 6 0.839  0.0366 "*"
## 4 Approach   67.5                6 0.847  0.0333 "*"
## 5 Approach   90                 6 0.507  0.304 ""
## 6 Homing-in  0                 6 0.583  0.224 ""
## 7 Homing-in  22.5                6 0.563  0.245 ""
## 8 Homing-in  45                 6 0.344  0.505 ""
## 9 Homing-in  67.5                6 0.562  0.246 ""
## 10 Homing-in 90                 6 0.257  0.623 ""

```

```

#plot
library(ggplot2)

ggplot(corr_data, aes(x = skill_ratio, y = collective_benefit)) +
  geom_point(size = 2.5, alpha = 0.75, aes(colour = phase)) +
  geom_smooth(method = "lm", se = TRUE, colour = "grey30", linewidth = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey60") +
  facet_grid(phase ~ contribution_angle,
    labeller = labeller(
      contribution_angle = function(x) paste0(x, "°")
    )
  )

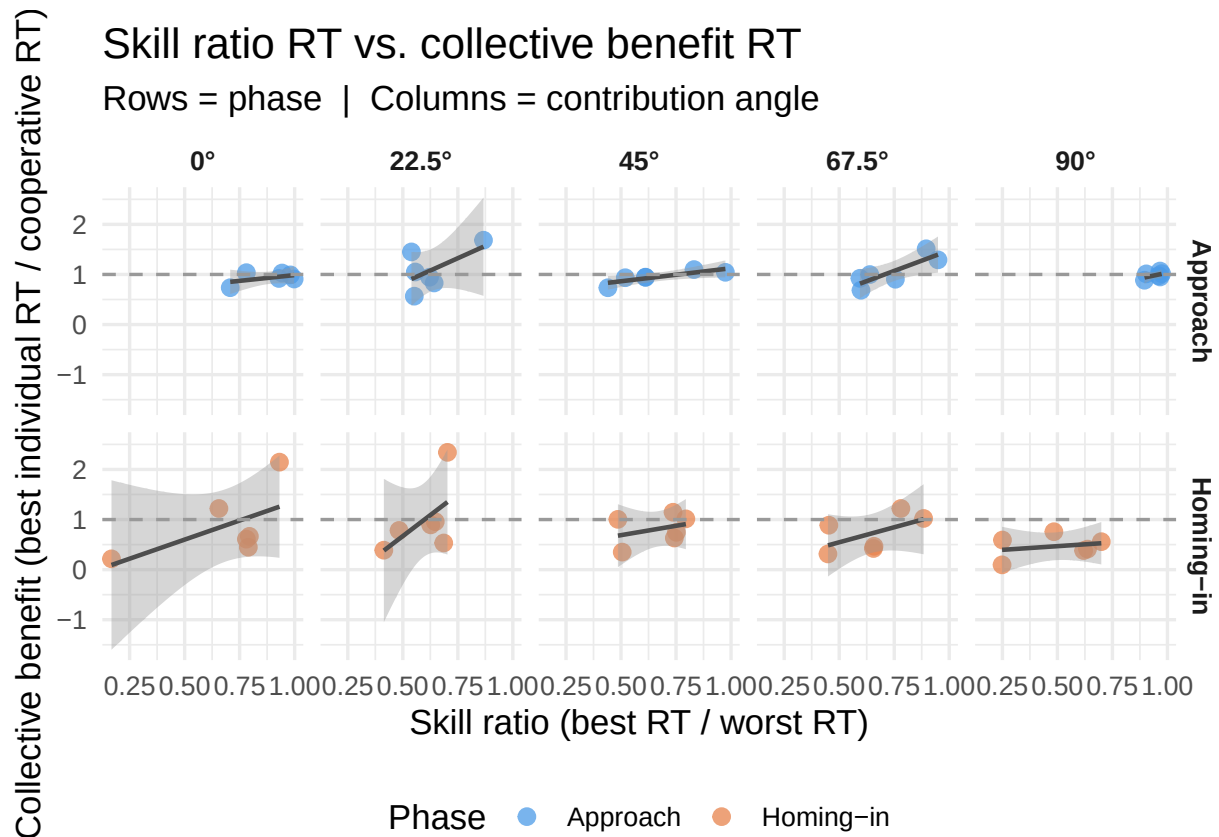
```

```

    )) +
  scale_colour_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  labs(
    title = "Skill ratio RT vs. collective benefit RT",
    subtitle = "Rows = phase | Columns = contribution angle",
    x = "Skill ratio (best RT / worst RT)",
    y = "Collective benefit (best individual RT / cooperative RT)",
    colour = "Phase"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom",
        strip.text = element_text(face = "bold"))

```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```

#correlating interindividual DIST skill ratios with collective benefit for each of the contribution condi

#join the two into one data frame
corr_data2 <- collective_benefit_DIST %>%
  left_join(skill_ratio_DIST, by = c("dyad_id", "phase", "contribution_angle"))

#Pearson r between skill_ratio and collective benefit
correlation_results2 <- corr_data2 %>%
  group_by(phase, contribution_angle) %>%

```



```

summarise(
  n = sum(!is.na(skill_ratio) & !is.na(collective_benefit)),
  r = cor(skill_ratio, collective_benefit, use = "complete.obs"),
  p_value = tryCatch(
    cor.test(skill_ratio, collective_benefit)$p.value,
    error = function(e) NA_real_
  ),
  .groups = "drop"
) %>%
mutate(
  sig = case_when(
    p_value < .001 ~ "***",
    p_value < .01 ~ "**",
    p_value < .05 ~ "*",
    p_value < .10 ~ "†",
    TRUE ~ ""
  )
)
print(correlation_results2)

```

```

## # A tibble: 10 x 6
##   phase      contribution_angle    n      r p_value sig
##   <fct>      <fct>          <int>  <dbl>  <dbl> <chr>
## 1 Approach    0                6  0.459   0.360 ""
## 2 Approach   22.5                6 -0.0589  0.912 ""
## 3 Approach   45                6  0.812   0.0496 "*"
## 4 Approach   67.5                6  0.354   0.491 ""
## 5 Approach   90                6 -0.490   0.324 ""
## 6 Homing-in  0                6  0.667   0.147 ""
## 7 Homing-in  22.5                6  0.0592  0.911 ""
## 8 Homing-in  45                6  0.369   0.472 ""
## 9 Homing-in  67.5                6  0.815   0.0482 "*"
## 10 Homing-in 90                6  0.330   0.522 ""

```

```

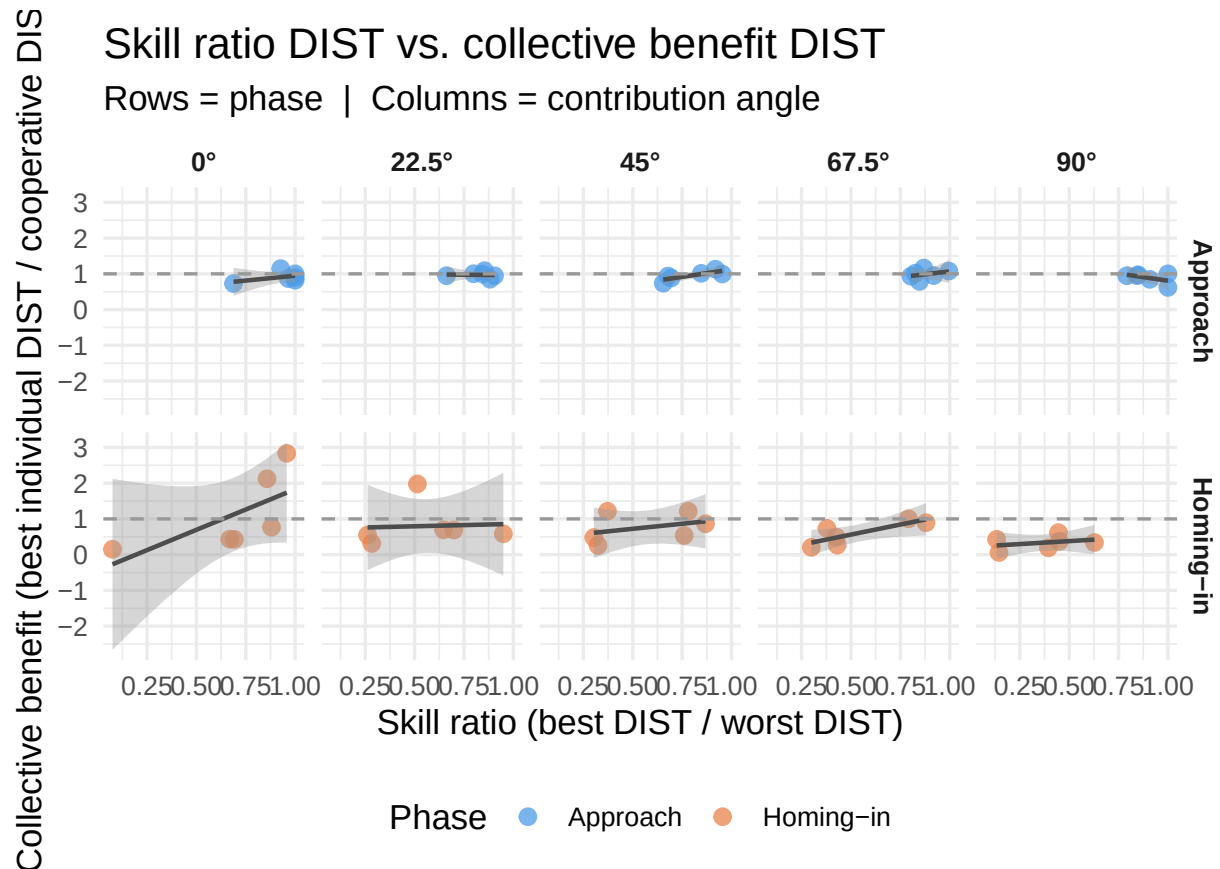
#plot
library(ggplot2)

ggplot(corr_data2, aes(x = skill_ratio, y = collective_benefit)) +
  geom_point(size = 2.5, alpha = 0.75, aes(colour = phase)) +
  geom_smooth(method = "lm", se = TRUE, colour = "grey30", linewidth = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey60") +
  facet_grid(phase ~ contribution_angle,
    labeller = labeller(
      contribution_angle = function(x) paste0(x, "°")
    )) +
  scale_colour_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  labs(
    title = "Skill ratio DIST vs. collective benefit DIST",
    subtitle = "Rows = phase | Columns = contribution angle",
    x = "Skill ratio (best DIST / worst DIST)",
    y = "Collective benefit (best individual DIST / cooperative DIST)",
    colour = "Phase"
  ) +

```

```
theme_minimal(base_size = 13) +
theme(legend.position = "bottom",
      strip.text = element_text(face = "bold"))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Interpretation: Wahn et al. predicted that the collective benefit systematically increases with larger skill ratios, suggesting that dyad members who have increasingly similar individual performance levels benefit more from performing jointly - particularly when both members contribute equally to the joint performance. If however the better individual's dominance compensates for a larger interindividual difference (as in the 70% contribution condition (22.5° and 67.5°)), they predicted a reduced correlation relative to the other contribution conditions. - With only 6 dyads, we have very low power per angle x phase cell, so we have to treat the correlations as descriptive/exploratory and flag that in the write-up (we could collapse across angles and just group by phase for more statistical power, but I don't think we should)

```
#Checking for social facilitation
#Comparing RT in the individual and joint condition for the baseline angles 0° and 90°
#done separately for Approach and Homing-in

library(effects)
```

```
## Warning: package 'effectsize' was built under R version 4.4.3
```

```

#filter to baseline angles
temp_sf <- analysis_data %>%
  filter(contribution_angle %in% c(0, 90),
         condition %in% c("individual", "cooperative")) %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  mutate(dyad_id = coalesce(dyad_id.x, dyad_id.y)) %>%
  select(-dyad_id.x, -dyad_id.y)

social_fac_data <- temp_sf %>%
  group_by(participant_id, dyad_id, phase, contribution_angle, condition) %>%
  summarise(
    mean_RT = mean(RT, na.rm = TRUE),
    .groups = "drop"
  )

#summary table
social_fac_summary <- social_fac_data %>%
  group_by(phase, contribution_angle, condition) %>%
  summarise(
    n = n(),
    M_RT = mean(mean_RT, na.rm = TRUE),
    SE_RT = sd(mean_RT, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )
print(social_fac_summary)

```

```

## # A tibble: 8 x 6
##   phase      contribution_angle condition      n  M_RT  SE_RT
##   <fct>      <fct>              <fct>    <int> <dbl> <dbl>
## 1 Approach  0                individual    12  1.35 0.0533
## 2 Approach  0                cooperative     6  1.36 0.0917
## 3 Approach  90               individual    12  1.40 0.0249
## 4 Approach  90               cooperative     6  1.39 0.0412
## 5 Homing-in 0                individual    12  4.02 0.755
## 6 Homing-in 0                cooperative     6  4.14 0.677
## 7 Homing-in 90               individual    12  2.78 0.510
## 8 Homing-in 90               cooperative     6  5.24 2.03

```

```

#paired t-tests + Wilcoxon + Cohen's d per phase x angle

#individual: average both members per dyad
temp_ind <- social_fac_data %>%
  filter(condition == "individual") %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  mutate(dyad_id = coalesce(dyad_id.x, dyad_id.y)) %>%
  select(-dyad_id.x, -dyad_id.y)

individual_sf <- temp_ind %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(ind_RT = mean(mean_RT, na.rm = TRUE), .groups = "drop")

#cooperative
cooperative_sf <- social_fac_data %>%

```

```

filter(condition == "cooperative") %>%
group_by(dyad_id, phase, contribution_angle) %>%
summarise(coop_RT = mean(mean_RT, na.rm = TRUE), .groups = "drop")

#join and run both t-test and Wilcoxon with Cohen's d
sf_tests <- individual_sf %>%
left_join(cooperative_sf, by = c("dyad_id", "phase", "contribution_angle")) %>%
group_by(phase, contribution_angle) %>%
summarise(
  n = n(),
  M_ind = mean(ind_RT, na.rm = TRUE),
  M_coop = mean(coop_RT, na.rm = TRUE),
  mean_diff = mean(coop_RT - ind_RT, na.rm = TRUE),
  t_stat = tryCatch(t.test(coop_RT, ind_RT, paired = TRUE)$statistic, error = function(e) NA),
  t_p_value = tryCatch(t.test(coop_RT, ind_RT, paired = TRUE)$p.value, error = function(e) NA),
  W_stat = tryCatch(wilcox.test(coop_RT, ind_RT, paired = TRUE)$statistic, error = function(e) NA),
  W_p_value = tryCatch(wilcox.test(coop_RT, ind_RT, paired = TRUE)$p.value, error = function(e) NA),
  cohens_d = tryCatch(
    repeated_measures_d(coop_RT, ind_RT, method = "z")$d,
    error = function(e) NA
  ),
  .groups = "drop"
) %>%
mutate(
  t_sig = case_when(
    t_p_value < .001 ~ "****",
    t_p_value < .01 ~ "***",
    t_p_value < .05 ~ "**",
    t_p_value < .10 ~ "+",
    TRUE ~ ""
  ),
  W_sig = case_when(
    W_p_value < .001 ~ "****",
    W_p_value < .01 ~ "***",
    W_p_value < .05 ~ "**",
    W_p_value < .10 ~ "+",
    TRUE ~ ""
  )
)
print(sf_tests)

```

```

## # A tibble: 4 x 13
##   phase contribution_angle      n M_ind M_coop mean_diff t_stat t_p_value W_stat
##   <fct> <fct>          <int> <dbl> <dbl>      <dbl> <dbl>      <dbl> <dbl>
## 1 Appro~ 0              6  1.35  1.36    0.0167  0.282    0.789    11
## 2 Appro~ 90             6  1.40  1.39   -0.0150 -0.457    0.667     9
## 3 Homin~ 0              6  4.02  4.14    0.113   0.114    0.914    13
## 4 Homin~ 90             6  2.78  5.24    2.46    1.29    0.253    17
## # i 4 more variables: W_p_value <dbl>, cohens_d <dbl>, t_sig <chr>, W_sig <chr>

```

```

#plot
temp_plot <- social_fac_data %>%
left_join(participant_dyad_map, by = "participant_id") %>%

```

```

mutate(dyad_id = coalesce(dyad_id.x, dyad_id.y)) %>%
select(-dyad_id.x, -dyad_id.y)

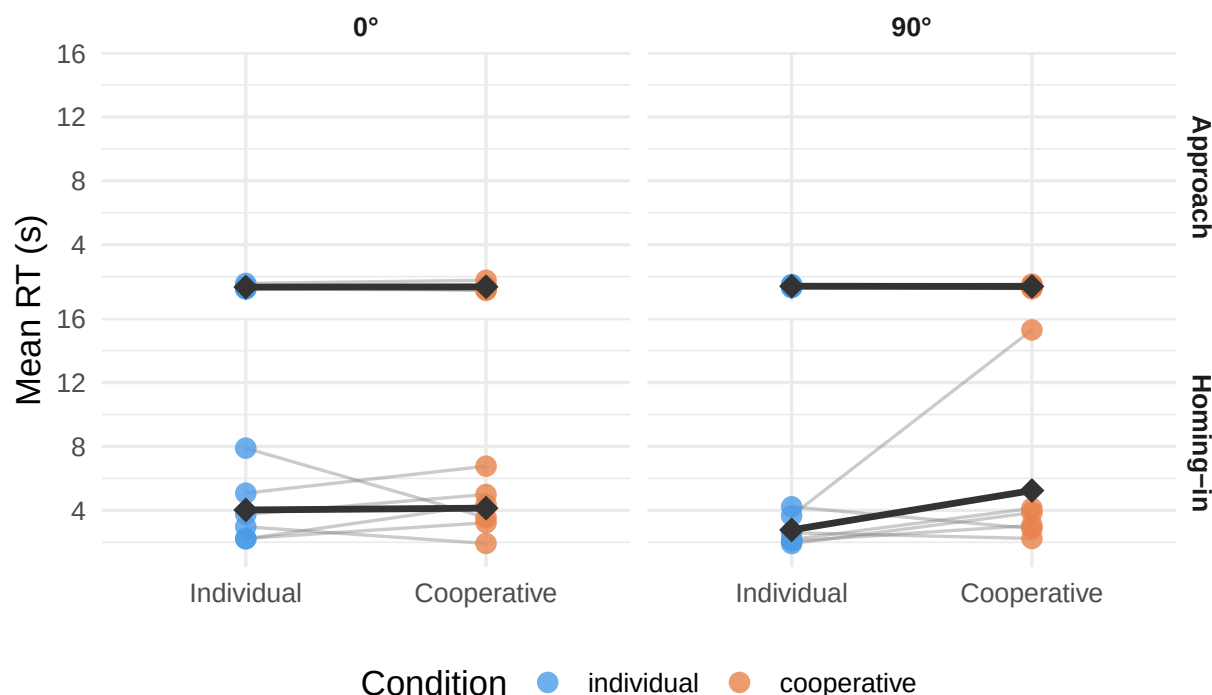
temp_plot %>%
  group_by(dyad_id, phase, contribution_angle, condition) %>%
  summarise(mean_RT = mean(mean_RT, na.rm = TRUE), .groups = "drop") %>%
  mutate(contribution_angle = factor(paste0(contribution_angle, "°"),
    levels = c("0°", "90°"))) %>%

  ggplot(aes(x = condition, y = mean_RT, colour = condition, group = dyad_id)) +
  geom_line(alpha = 0.4, colour = "grey50") +
  geom_point(size = 3, alpha = 0.8) +
  stat_summary(aes(group = 1), fun = mean, geom = "line",
    linewidth = 1.2, colour = "grey20") +
  stat_summary(aes(group = 1), fun = mean, geom = "point",
    size = 4, shape = 18, colour = "grey20") +
  facet_grid(phase ~ contribution_angle) +
  scale_colour_manual(values = c("individual" = "#4C9BE8",
    "cooperative" = "#E8834C")) +
  scale_x_discrete(labels = c("individual" = "Individual",
    "cooperative" = "Cooperative")) +
  labs(
    title = "Social facilitation check",
    subtitle = "Baseline angles only | Rows = phase | Columns = angle",
    x = NULL,
    y = "Mean RT (s)",
    colour = "Condition"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom",
    strip.text = element_text(face = "bold"))

```

Social facilitation check

Baseline angles only | Rows = phase | Columns = angle



Interpretation:

1. Approach phase: At 0°, there was no significant difference between individual and cooperative RT ($W = 11$, $p = 1.000$, $d = 0.12$). Similarly, at 90°, cooperative and individual RTs did not differ significantly ($W = 9$, $p = .844$, $d = -0.19$), with the small negative d indicating cooperative RT was marginally faster than individual.
2. Homing-in phase: At 0°, no significant difference was found between conditions ($W = 13$, $p = .688$, $d = 0.05$). At 90°, cooperative RT ($M = 5.24$ s) was notably slower than individual RT ($M = 2.78$ s), though this difference did not reach significance ($W = 17$, $p = .219$, $d = 0.53$), representing a medium effect size.

```
#Checking for social facilitation
#Comparing DIST in the individual and joint condition for the baseline angles 0° and 90°
#done separately for Approach and Homing-in

library(effects)

#filter to baseline angles
temp_sf_dist <- analysis_data %>%
  filter(contribution_angle %in% c(0, 90),
         condition %in% c("individual", "cooperative")) %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  mutate(dyad_id = coalesce(dyad_id.x, dyad_id.y)) %>%
  select(-dyad_id.x, -dyad_id.y)

social_fac_data_dist <- temp_sf_dist %>%
```

```

group_by(participant_id, dyad_id, phase, contribution_angle, condition) %>%
  summarise(
    mean_DIST = mean(DIST, na.rm = TRUE),
    .groups = "drop"
  )

#summary table
social_fac_summary_dist <- social_fac_data_dist %>%
  group_by(phase, contribution_angle, condition) %>%
  summarise(
    n = n(),
    M_DIST = mean(mean_DIST, na.rm = TRUE),
    SE_DIST = sd(mean_DIST, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )
print(social_fac_summary_dist)

```

```

## # A tibble: 8 x 6
##   phase      contribution_angle condition      n M_DIST SE_DIST
##   <fct>      <fct>              <fct>    <int> <dbl>  <dbl>
## 1 Approach  0                individual    12  305.   12.0
## 2 Approach  0                cooperative     6  326.   18.5
## 3 Approach  90               individual    12  303.    7.99
## 4 Approach  90               cooperative     6  330.   26.9
## 5 Homing-in 0                individual    12  377.   95.7
## 6 Homing-in 0                cooperative     6  300.   49.7
## 7 Homing-in 90               individual    12  209.   52.3
## 8 Homing-in 90               cooperative     6  448.  189.

```

#paired t-tests + Wilcoxon + Cohen's d per phase x angle

```

#individual: average both members per dyad
temp_ind_dist <- social_fac_data_dist %>%
  filter(condition == "individual") %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  mutate(dyad_id = coalesce(dyad_id.x, dyad_id.y)) %>%
  select(-dyad_id.x, -dyad_id.y)

individual_sf_dist <- temp_ind_dist %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(ind_DIST = mean(mean_DIST, na.rm = TRUE), .groups = "drop")

#cooperative
cooperative_sf_dist <- social_fac_data_dist %>%
  filter(condition == "cooperative") %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(coop_DIST = mean(mean_DIST, na.rm = TRUE), .groups = "drop")

#join and run both t-test and Wilcoxon with Cohen's d
sf_tests_dist <- individual_sf_dist %>%
  left_join(cooperative_sf_dist, by = c("dyad_id", "phase", "contribution_angle")) %>%
  group_by(phase, contribution_angle) %>%
  summarise(

```

```

n          = n(),
M_ind      = mean(ind_DIST, na.rm = TRUE),
M_coop     = mean(coop_DIST, na.rm = TRUE),
mean_diff  = mean(coop_DIST - ind_DIST, na.rm = TRUE),
t_stat     = tryCatch(t.test(coop_DIST, ind_DIST, paired = TRUE)$statistic, error = function(e)
t_p_value  = tryCatch(t.test(coop_DIST, ind_DIST, paired = TRUE)$p.value,      error = function(e)
W_stat     = tryCatch(wilcox.test(coop_DIST, ind_DIST, paired = TRUE)$statistic, error = function(e)
W_p_value  = tryCatch(wilcox.test(coop_DIST, ind_DIST, paired = TRUE)$p.value, error = function(e)
cohens_d   = tryCatch(
  repeated_measures_d(coop_DIST, ind_DIST, method = "z")$d,
  error = function(e) NA
),
.groups    = "drop"
) %>%
mutate(
  t_sig = case_when(
    t_p_value < .001 ~ "***",
    t_p_value < .01  ~ "**",
    t_p_value < .05  ~ "*",
    t_p_value < .10  ~ "+",
    TRUE           ~ ""
  ),
  W_sig = case_when(
    W_p_value < .001 ~ "***",
    W_p_value < .01  ~ "**",
    W_p_value < .05  ~ "*",
    W_p_value < .10  ~ "+",
    TRUE           ~ ""
  )
)

```

```

## Warning: There were 4 warnings in `summarise()`.
## The first warning was:
## i In argument: `W_stat = tryCatch(...)` .
## i In group 1: `phase = Approach` `contribution_angle = 0` .
## Caused by warning in `wilcox.test.default()` :
## ! cannot compute exact p-value with zeroes
## i Run `dplyr::last_dplyr_warnings()` to see the 3 remaining warnings.

```

```
print(sf_tests_dist)
```

```

## # A tibble: 4 x 13
##   phase contribution_angle      n M_ind M_coop mean_diff t_stat t_p_value W_stat
##   <fct> <fct>             <int> <dbl> <dbl>      <dbl> <dbl>      <dbl> <dbl>
## 1 Appro~ 0                6  305.  326.      20.4  1.21      0.280    11
## 2 Appro~ 90              6  303.  330.      26.9  0.874     0.422     9
## 3 Homin~ 0                6  377.  300.     -76.6 -0.715     0.507     9
## 4 Homin~ 90              6  209.  448.     238.  1.39      0.224    18
## # i 4 more variables: W_p_value <dbl>, cohens_d <dbl>, t_sig <chr>, W_sig <chr>

```

```

#plot
temp_plot_dist <- social_fac_data_dist %>%

```



```

left_join(participant_dyad_map, by = "participant_id") %>%
mutate(dyad_id = coalesce(dyad_id.x, dyad_id.y)) %>%
select(-dyad_id.x, -dyad_id.y)

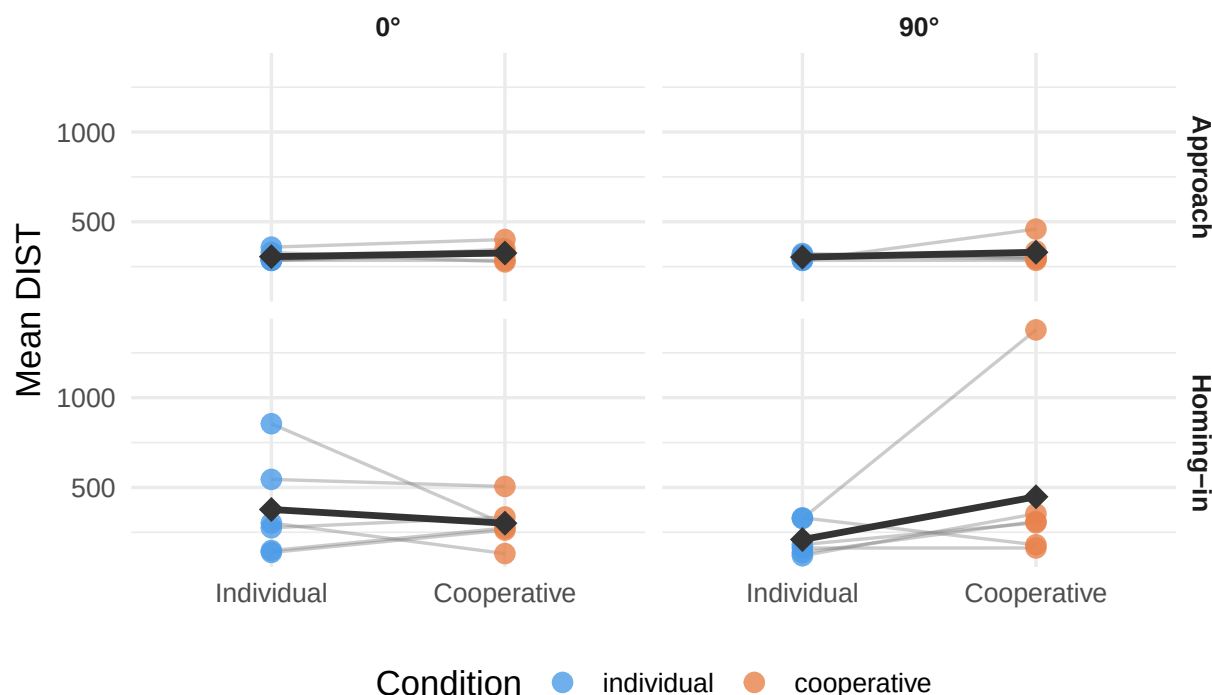
temp_plot_dist %>%
  group_by(dyad_id, phase, contribution_angle, condition) %>%
  summarise(mean_DIST = mean(mean_DIST, na.rm = TRUE), .groups = "drop") %>%
  mutate(contribution_angle = factor(paste0(contribution_angle, "°"),
    levels = c("0°", "90°"))) %>%

  ggplot(aes(x = condition, y = mean_DIST, colour = condition, group = dyad_id)) +
  geom_line(alpha = 0.4, colour = "grey50") +
  geom_point(size = 3, alpha = 0.8) +
  stat_summary(aes(group = 1), fun = mean, geom = "line",
    linewidth = 1.2, colour = "grey20") +
  stat_summary(aes(group = 1), fun = mean, geom = "point",
    size = 4, shape = 18, colour = "grey20") +
  facet_grid(phase ~ contribution_angle) +
  scale_colour_manual(values = c("individual" = "#4C9BE8",
    "cooperative" = "#E8834C")) +
  scale_x_discrete(labels = c("individual" = "Individual",
    "cooperative" = "Cooperative")) +
  labs(
    title = "Social facilitation check (DIST)",
    subtitle = "Baseline angles only | Rows = phase | Columns = angle",
    x = NULL,
    y = "Mean DIST",
    colour = "Condition"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom",
    strip.text = element_text(face = "bold"))

```

Social facilitation check (DIST)

Baseline angles only | Rows = phase | Columns = angle



Interpretation:

1. Approach: At 0°, dyads travelled slightly more distance cooperatively ($M = 326$) than individually ($M = 305$), a mean difference of 20.4 units, though this was non-significant ($W = 11$, $p = .418$, $d = 0.42$). At 90°, a similar pattern emerged — cooperative DIST ($M = 330$) exceeded individual DIST ($M = 303$) by 26.9 units, again non-significant ($W = 9$, $p = .787$, $d = 0.30$). Both effects are small-to-medium in size but in the direction of a coordination cost — dyads travel more than individuals.
2. Homing-in: At 0°, the pattern reverses: cooperative DIST ($M = 300$) was actually lower than individual DIST ($M = 377$), a mean difference of -76.6 units ($W = 9$, $p = .844$, $d = -0.25$). This hints at a slight collective benefit in path efficiency, though it is non-significant and small. At 90°, the largest effect emerges — cooperative DIST ($M = 448$) was 238 units greater than individual DIST ($M = 209$), a medium effect ($d = 0.48$) that nonetheless failed to reach significance ($W = 18$, $p = .156$), suggesting a meaningful coordination cost at maximal asymmetry during Homing-in.

```
#Determining whether joint control was beneficial
#Is collective RT benefit larger than 1?
#one-sample t-tests

one_sample_tests_RT <- collective_benefit_RT %>%
  group_by(phase, contribution_angle) %>%
  summarise(
    n = sum(!is.na(collective_benefit)),
    M_benefit = mean(collective_benefit, na.rm = TRUE),
    SE_benefit = sd(collective_benefit, na.rm = TRUE) / sqrt(n),
    t_stat = tryCatch(t.test(collective_benefit, mu = 1)$statistic, error = function(e) NA),
    p_value = tryCatch(t.test(collective_benefit, mu = 1)$p.value, error = function(e) NA),
```

```

cohens_d = tryCatch(
  repeated_measures_d(collective_benefit, mu = 1, method = "z")$d,
  error = function(e) (mean(collective_benefit, na.rm = TRUE) - 1) /
    sd(collective_benefit, na.rm = TRUE) # fallback: manual d
),
.groups = "drop"
) %>%
mutate(
  sig = case_when(
    p_value < .001 ~ "***",
    p_value < .01 ~ "**",
    p_value < .05 ~ "*",
    p_value < .10 ~ "†",
    TRUE ~ ""
  )
)

print(one_sample_tests_RT)

```

```

## # A tibble: 10 x 9
##   phase contribution_angle      n M_benefit SE_benefit t_stat p_value cohens_d
##   <fct> <fct>          <int>   <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
## 1 Appro~ 0              6    0.937    0.0457 -1.38    0.225   -0.564
## 2 Appro~ 22.5           6    1.09     0.168    0.522    0.624    0.213
## 3 Appro~ 45             6    0.950    0.0507 -0.988    0.369   -0.403
## 4 Appro~ 67.5           6    1.05     0.122    0.433    0.683    0.177
## 5 Appro~ 90             6    0.984    0.0258 -0.607    0.570   -0.248
## 6 Homin~ 0              6    0.884    0.287   -0.405    0.702   -0.165
## 7 Homin~ 22.5           6    0.983    0.286   -0.0612  0.954   -0.0250
## 8 Homin~ 45             6    0.815    0.121   -1.53     0.187   -0.624
## 9 Homin~ 67.5           6    0.722    0.152   -1.84     0.126   -0.750
## 10 Homin~ 90            6    0.465    0.0925 -5.78    0.00218 -2.36
## # i 1 more variable: sig <chr>

```

```

#plot
collective_benefit_RT %>%
  mutate(contribution_angle = factor(paste0(contribution_angle, "°"),
    levels = c("0°", "22.5°", "45°", "67.5°", "90°"))) %>%
  ggplot(aes(x = contribution_angle, y = collective_benefit, colour = phase)) +
  # individual dyad points
  geom_point(alpha = 0.6, size = 2.5,
    position = position_dodge(width = 0.4)) +
  # mean ± SE bars
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
    position = position_dodge(width = 0.4), linewidth = 0.8) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
    position = position_dodge(width = 0.4)) +
  stat_summary(fun = mean, geom = "line", aes(group = phase),
    position = position_dodge(width = 0.4), linewidth = 0.8) +
  # null hypothesis line
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
  # significance stars from one-sample tests
  geom_text(data = one_sample_tests_RT,

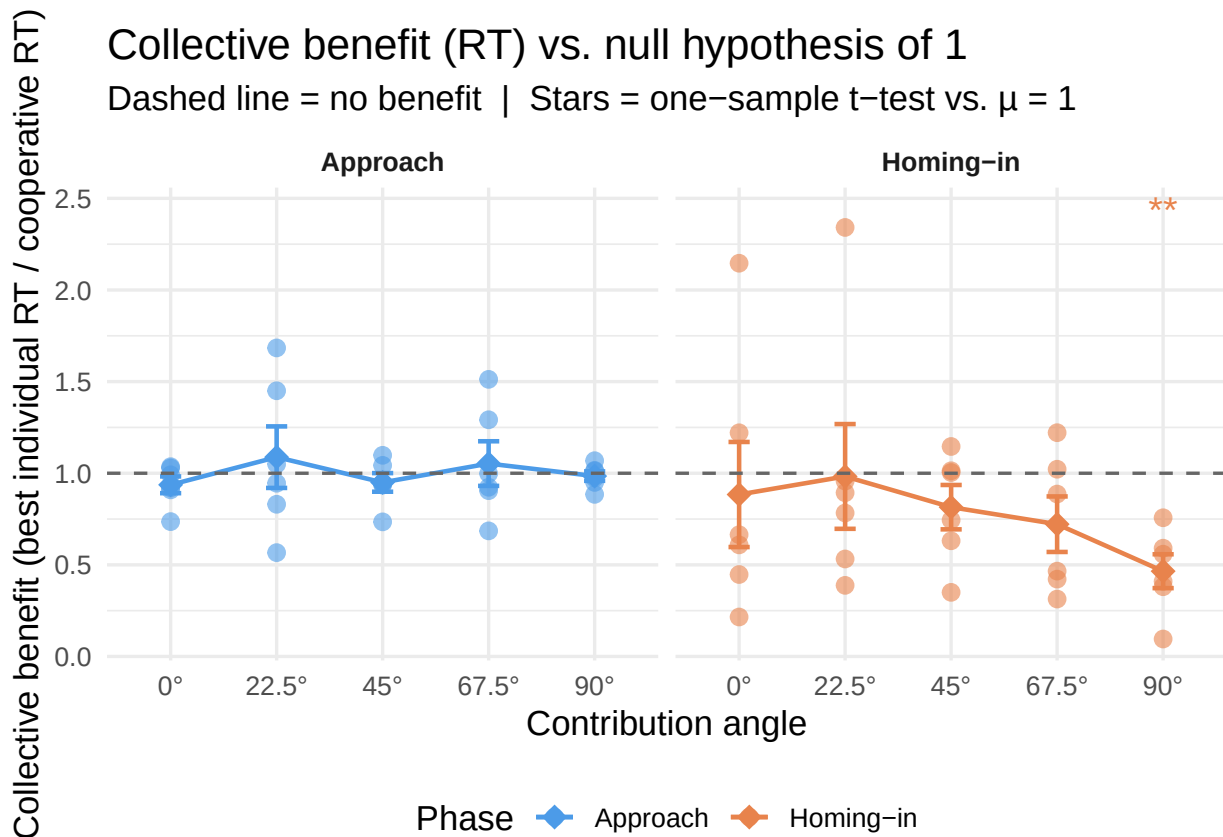
```

```

aes(x = factor(paste0(contribution_angle, "°"),
  levels = c("0°", "22.5°", "45°", "67.5°", "90°")),
  y = max(collective_benefit_RT$collective_benefit, na.rm = TRUE) + 0.1,
  label = sig, colour = phase),
  position = position_dodge(width = 0.4),
  size = 5, show.legend = FALSE) +
facet_wrap(~ phase) +
scale_colour_manual(values = c("Approach" = "#4C9BE8",
  "Homing-in" = "#E8834C")) +

labs(
  title = "Collective benefit (RT) vs. null hypothesis of 1",
  subtitle = "Dashed line = no benefit | Stars = one-sample t-test vs.  $\mu = 1$ ",
  x = "Contribution angle",
  y = "Collective benefit (best individual RT / cooperative RT)",
  colour = "Phase"
) +
theme_minimal(base_size = 13) +
theme(legend.position = "bottom",
  strip.text = element_text(face = "bold"))

```



```

#Determining whether joint control was beneficial
#Is collective DIST benefit larger than 1?
#one-sample t-tests

one_sample_tests_DIST <- collective_benefit_DIST %>%

```

```

group_by(phase, contribution_angle) %>%
  summarise(
    n      = sum(!is.na(collective_benefit)),
    M_benefit = mean(collective_benefit, na.rm = TRUE),
    SE_benefit = sd(collective_benefit, na.rm = TRUE) / sqrt(n),
    t_stat    = tryCatch(t.test(collective_benefit, mu = 1)$statistic, error = function(e) NA),
    p_value   = tryCatch(t.test(collective_benefit, mu = 1)$p.value,   error = function(e) NA),
    cohens_d  = tryCatch(
      (mean(collective_benefit, na.rm = TRUE) - 1) / sd(collective_benefit, na.rm = TRUE),
      error = function(e) NA
    ),
    .groups   = "drop"
  ) %>%
  mutate(
    sig = case_when(
      p_value < .001 ~ "***",
      p_value < .01  ~ "**",
      p_value < .05  ~ "*",
      p_value < .10  ~ "†",
      TRUE          ~ ""
    )
  )

print(one_sample_tests_DIST)

```

```

## # A tibble: 10 x 9
##   phase    contribution_angle      n M_benefit SE_benefit t_stat p_value cohens_d
##   <fct>    <fct>          <int>   <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 Approa~ 0                6    0.910    0.0596 -1.50  1.93e-1 -0.614
## 2 Approa~ 22.5            6    0.971    0.0323 -0.890 4.14e-1 -0.363
## 3 Approa~ 45             6    0.950    0.0543 -0.925 3.97e-1 -0.378
## 4 Approa~ 67.5           6    0.988    0.0526 -0.222 8.33e-1 -0.0906
## 5 Approa~ 90             6    0.889    0.0578 -1.91  1.14e-1 -0.780
## 6 Homing~ 0                6    1.12     0.446    0.277 7.93e-1  0.113
## 7 Homing~ 22.5           6    0.802    0.242   -0.819 4.50e-1 -0.335
## 8 Homing~ 45             6    0.762    0.164   -1.46  2.05e-1 -0.594
## 9 Homing~ 67.5           6    0.600    0.135   -2.96  3.14e-2 -1.21
## 10 Homing~ 90            6    0.335    0.0786 -8.47  3.77e-4 -3.46
## # i 1 more variable: sig <chr>

```

```

#plot
collective_benefit_DIST %>%
  mutate(contribution_angle = factor(paste0(contribution_angle, "°"),
    levels = c("0°", "22.5°", "45°", "67.5°", "90°"))) %>%
  ggplot(aes(x = contribution_angle, y = collective_benefit, colour = phase)) +
  # individual dyad points
  geom_point(alpha = 0.6, size = 2.5,
    position = position_dodge(width = 0.4)) +
  # mean ± SE bars
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
    position = position_dodge(width = 0.4), linewidth = 0.8) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
    position = position_dodge(width = 0.4)) +

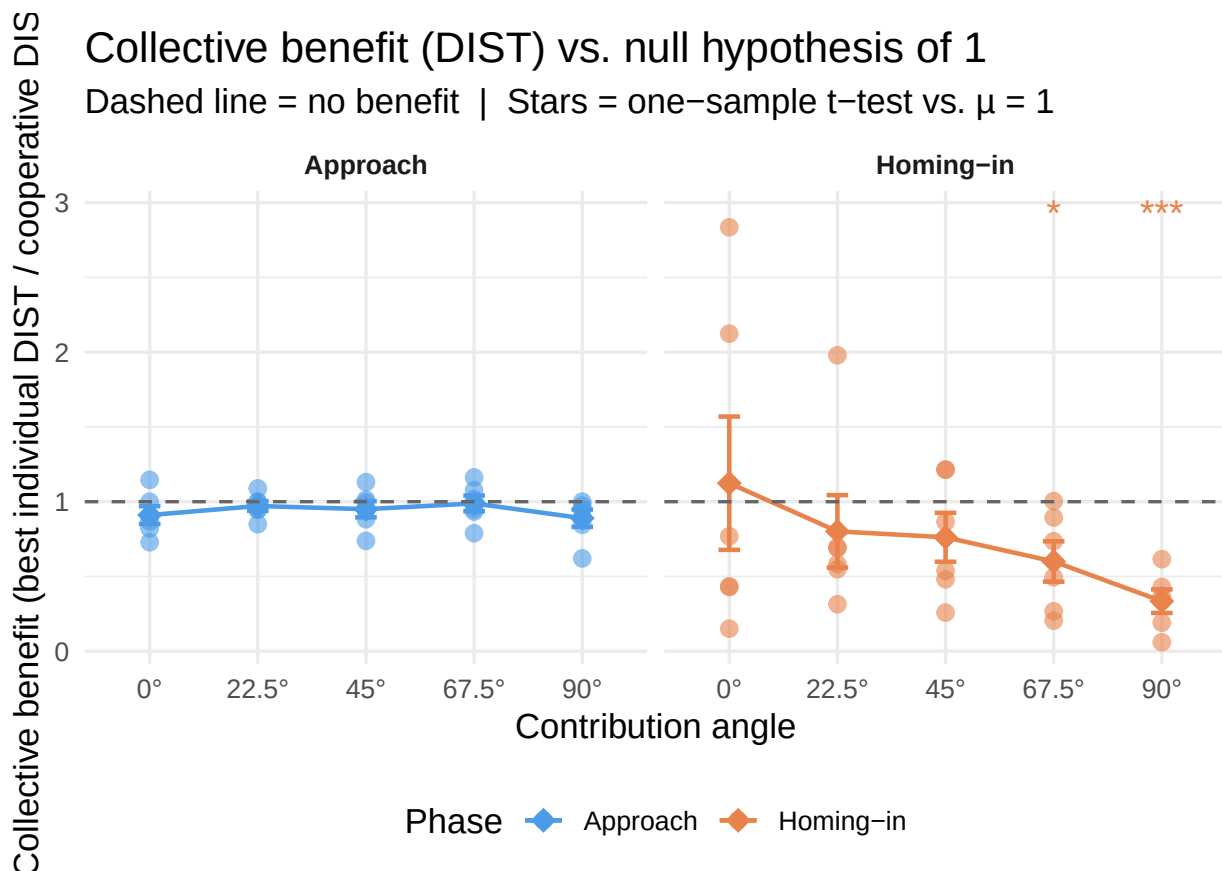
```

```

stat_summary(fun = mean, geom = "line", aes(group = phase),
             position = position_dodge(width = 0.4), linewidth = 0.8) +
# null hypothesis line
geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
# significance stars from one-sample tests
geom_text(data = one_sample_tests_DIST,
          aes(x = factor(paste0(Contribution_angle, "°"),
                        levels = c("0°", "22.5°", "45°", "67.5°", "90°")),
              y = max(collective_benefit_DIST$collective_benefit, na.rm = TRUE) + 0.1,
              label = sig, colour = phase),
          position = position_dodge(width = 0.4),
          size = 5, show.legend = FALSE) +
facet_wrap(~ phase) +
scale_colour_manual(values = c("Approach" = "#4C9BE8",
                              "Homing-in" = "#E8834C")) +

labs(
  title = "Collective benefit (DIST) vs. null hypothesis of 1",
  subtitle = "Dashed line = no benefit | Stars = one-sample t-test vs.  $\mu = 1$ ",
  x = "Contribution angle",
  y = "Collective benefit (best individual DIST / cooperative DIST)",
  colour = "Phase"
) +
theme_minimal(base_size = 13) +
theme(legend.position = "bottom",
      strip.text = element_text(face = "bold"))

```



```
#2x3 Repeated measures ANOVA: phase x contribution angle
#at contribution angles
#separately for RT and DIST
```

```
library(ez)
```

```
## Warning: package 'ez' was built under R version 4.4.3
```

```
library(effectsize)
```

```
#RT
```

```
anova_data_RT <- collective_benefit_RT %>%
  filter(contribution_angle %in% c(22.5, 45, 67.5)) %>%
  mutate(
    phase = factor(phase),
    contribution_angle = factor(contribution_angle)
  )
```

```
#check for missing cells
```

```
anova_data_RT %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(n = n(), .groups = "drop") %>%
  print()
```

```
## # A tibble: 36 x 4
##   dyad_id    phase    contribution_angle    n
##   <chr>      <fct>      <fct>                <int>
## 1 666_      Approach  22.5                  1
## 2 666_      Approach  45                   1
## 3 666_      Approach  67.5                 1
## 4 666_      Homing-in 22.5                  1
## 5 666_      Homing-in 45                   1
## 6 666_      Homing-in 67.5                 1
## 7 august_jens Approach  22.5                  1
## 8 august_jens Approach  45                   1
## 9 august_jens Approach  67.5                 1
## 10 august_jens Homing-in 22.5                  1
## # i 26 more rows
```

```
#ANOVA
```

```
anova_RT <- ezANOVA(
  data      = anova_data_RT,
  dv        = collective_benefit,
  wid       = dyad_id,
  within    = .(phase, contribution_angle),
  type      = 3,
  detailed  = TRUE,
  return_aov = TRUE
)
```

```
## Warning: Converting "dyad_id" to factor for ANOVA.
```

```
print(anova_RT)
```

```
## $ANOVA
##           Effect DFn DFd           SSn           SSd           F
## 1      (Intercept)    1    5 31.46523753 1.278520 123.0533777
## 2           phase    1    5  0.32613243 1.077189   1.5138131
## 3  contribution_angle    2   10 0.18084142 1.042036   0.8677313
## 4 phase:contribution_angle    2   10 0.09017701 1.549420   0.2910025
##           p p<.05           ges
## 1 0.0001037482 * 0.86413518
## 2 0.2732816588   0.06184602
## 3 0.4492612163   0.03526545
## 4 0.7536335212   0.01790171
##
## $`Mauchly's Test for Sphericity`
##           Effect           W           p p<.05
## 3      contribution_angle 0.7161419 0.5128593
## 4 phase:contribution_angle 0.3500099 0.1225070
##
## $`Sphericity Corrections`
##           Effect           GGe           p[GG] p[GG]<.05           HFe           p[HF]
## 3      contribution_angle 0.7789023 0.4300020           1.0671718 0.4492612
## 4 phase:contribution_angle 0.6060643 0.6518677           0.6960071 0.6801753
##           p[HF]<.05
## 3
## 4
##
## $aov
##
## Call:
## aov(formula = formula(aov_formula), data = data)
##
## Grand Mean: 0.9348981
##
## Stratum 1: dyad_id
##
## Terms:
##           Residuals
## Sum of Squares    1.27852
## Deg. of Freedom      5
##
## Residual standard error: 0.5056718
##
## Stratum 2: dyad_id:phase
##
## Terms:
##           phase Residuals
## Sum of Squares 0.3261324 1.0771886
## Deg. of Freedom      1      5
##
## Residual standard error: 0.4641527
## 2 out of 3 effects not estimable
## Estimated effects are balanced
```



```
##
## Stratum 3: dyad_id:contribution_angle
##
## Terms:
##               contribution_angle Residuals
## Sum of Squares      0.1808414 1.0420358
## Deg. of Freedom           2         10
##
## Residual standard error: 0.3228058
## 2 out of 4 effects not estimable
## Estimated effects may be unbalanced
##
## Stratum 4: dyad_id:phase:contribution_angle
##
## Terms:
##               phase:contribution_angle Residuals
## Sum of Squares      0.090177 1.549420
## Deg. of Freedom           2         10
##
## Residual standard error: 0.3936267
## Estimated effects may be unbalanced
```

```
#effect sizes (partial eta squared is given by ezANOVA,
# but also compute Cohen's f for completeness)
effectsize::eta_squared(anova_RT$aov, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type I)
##
## Group | Parameter | Eta2 (partial) | 95% CI
## -----|-----|-----|-----
## dyad_id:phase | phase | 0.23 | [0.00, 1.00]
## dyad_id:contribution_angle | contribution_angle | 0.15 | [0.00, 1.00]
## dyad_id:phase:contribution_angle | phase:contribution_angle | 0.05 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
#post-hoc pairwise comparisons if contribution_angle is significant
#collapsed across phase
pairwise_RT_data <- anova_data_RT %>%
  group_by(dyad_id, contribution_angle) %>%
  summarise(mean_benefit = mean(collective_benefit, na.rm = TRUE), .groups = "drop")

pairwise_RT <- pairwise.t.test(
  x = pairwise_RT_data$mean_benefit,
  g = pairwise_RT_data$contribution_angle,
  paired = TRUE,
  p.adjust.method = "bonferroni"
)
print(pairwise_RT)
```

```
##
## Pairwise comparisons using paired t tests
##
```

```
## data: pairwise_RT_data$mean_benefit and pairwise_RT_data$contribution_angle
##
##      22.5 45
## 45    0.58 -
## 67.5 1.00 1.00
##
## P value adjustment method: bonferroni
```

```
#DIST
```

```
anova_data_DIST <- collective_benefit_DIST %>%
  filter(contribution_angle %in% c(22.5, 45, 67.5)) %>%
  mutate(
    phase          = factor(phase),
    contribution_angle = factor(contribution_angle)
  )
```

```
#check for missing cells
```

```
anova_data_DIST %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(n = n(), .groups = "drop") %>%
  print()
```

```
## # A tibble: 36 x 4
##   dyad_id      phase      contribution_angle      n
##   <chr>        <fct>        <fct>                <int>
## 1 666_        Approach  22.5                  1
## 2 666_        Approach  45                   1
## 3 666_        Approach  67.5                 1
## 4 666_        Homing-in 22.5                 1
## 5 666_        Homing-in 45                 1
## 6 666_        Homing-in 67.5                 1
## 7 august_jens Approach  22.5                  1
## 8 august_jens Approach  45                   1
## 9 august_jens Approach  67.5                 1
## 10 august_jens Homing-in 22.5                 1
## # i 26 more rows
```

```
#run ANOVA
```

```
anova_DIST <- ezANOVA(
  data      = anova_data_DIST,
  dv        = collective_benefit,
  wid       = dyad_id,
  within    = .(phase, contribution_angle),
  type      = 3,
  detailed  = TRUE,
  return_aov = TRUE
)
```

```
## Warning: Converting "dyad_id" to factor for ANOVA.
```

```
print(anova_DIST)
```

```
## $ANOVA
##           Effect DFn DFd          SSn          SSd          F
## 1      (Intercept)    1    5 25.73205074 1.2015459 107.0789315
## 2           phase    1    5  0.55650606 0.7105896   3.9158050
## 3  contribution_angle    2   10 0.05299215 0.7148942   0.3706294
## 4 phase:contribution_angle    2   10 0.08829194 0.6850683   0.6444025
##           p p<.05          ges
## 1 0.000145058 * 0.88596333
## 2 0.104733404   0.14385191
## 3 0.699397567   0.01574762
## 4 0.545456516   0.02596524
##
## $`Mauchly's Test for Sphericity`
##           Effect          W          p p<.05
## 3      contribution_angle 0.3939489 0.1551957
## 4 phase:contribution_angle 0.4670618 0.2181467
##
## $`Sphericity Corrections`
##           Effect          GGe          p[GG] p[GG]<.05          HFe          p[HF]
## 3      contribution_angle 0.6226452 0.6098761          0.7286505 0.6394193
## 4 phase:contribution_angle 0.6523420 0.4914376          0.7885800 0.5152327
##           p[HF]<.05
## 3
## 4
##
## $aov
##
## Call:
## aov(formula = formula(aov_formula), data = data)
##
## Grand Mean: 0.8454461
##
## Stratum 1: dyad_id
##
## Terms:
##           Residuals
## Sum of Squares    1.201546
## Deg. of Freedom      5
##
## Residual standard error: 0.4902134
##
## Stratum 2: dyad_id:phase
##
## Terms:
##           phase Residuals
## Sum of Squares  0.5565061 0.7105896
## Deg. of Freedom      1      5
##
## Residual standard error: 0.3769853
## 2 out of 3 effects not estimable
## Estimated effects are balanced
```

```
##
## Stratum 3: dyad_id:contribution_angle
##
## Terms:
##               contribution_angle Residuals
## Sum of Squares      0.0529922 0.7148942
## Deg. of Freedom           2      10
##
## Residual standard error: 0.2673751
## 2 out of 4 effects not estimable
## Estimated effects may be unbalanced
##
## Stratum 4: dyad_id:phase:contribution_angle
##
## Terms:
##               phase:contribution_angle Residuals
## Sum of Squares      0.0882919 0.6850683
## Deg. of Freedom           2      10
##
## Residual standard error: 0.2617381
## Estimated effects may be unbalanced

#effect sizes
effectsize::eta_squared(anova_DIST$aov, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type I)
##
## Group | Parameter | Eta2 (partial) | 95% CI
## -----|-----|-----|-----
## dyad_id:phase | phase | 0.44 | [0.00, 1.00]
## dyad_id:contribution_angle | contribution_angle | 0.07 | [0.00, 1.00]
## dyad_id:phase:contribution_angle | phase:contribution_angle | 0.11 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
#post-hoc pairwise comparisons
pairwise_DIST_data <- anova_data_DIST %>%
  group_by(dyad_id, contribution_angle) %>%
  summarise(mean_benefit = mean(collective_benefit, na.rm = TRUE), .groups = "drop")

pairwise_DIST <- pairwise.t.test(
  x = pairwise_DIST_data$mean_benefit,
  g = pairwise_DIST_data$contribution_angle,
  paired = TRUE,
  p.adjust.method = "bonferroni"
)
print(pairwise_DIST)
```

```
##
## Pairwise comparisons using paired t tests
##
## data: pairwise_DIST_data$mean_benefit and pairwise_DIST_data$contribution_angle
##
```

```
##      22.5 45
## 45    1    -
## 67.5 1    1
##
## P value adjustment method: bonferroni
```

```
#plot
```

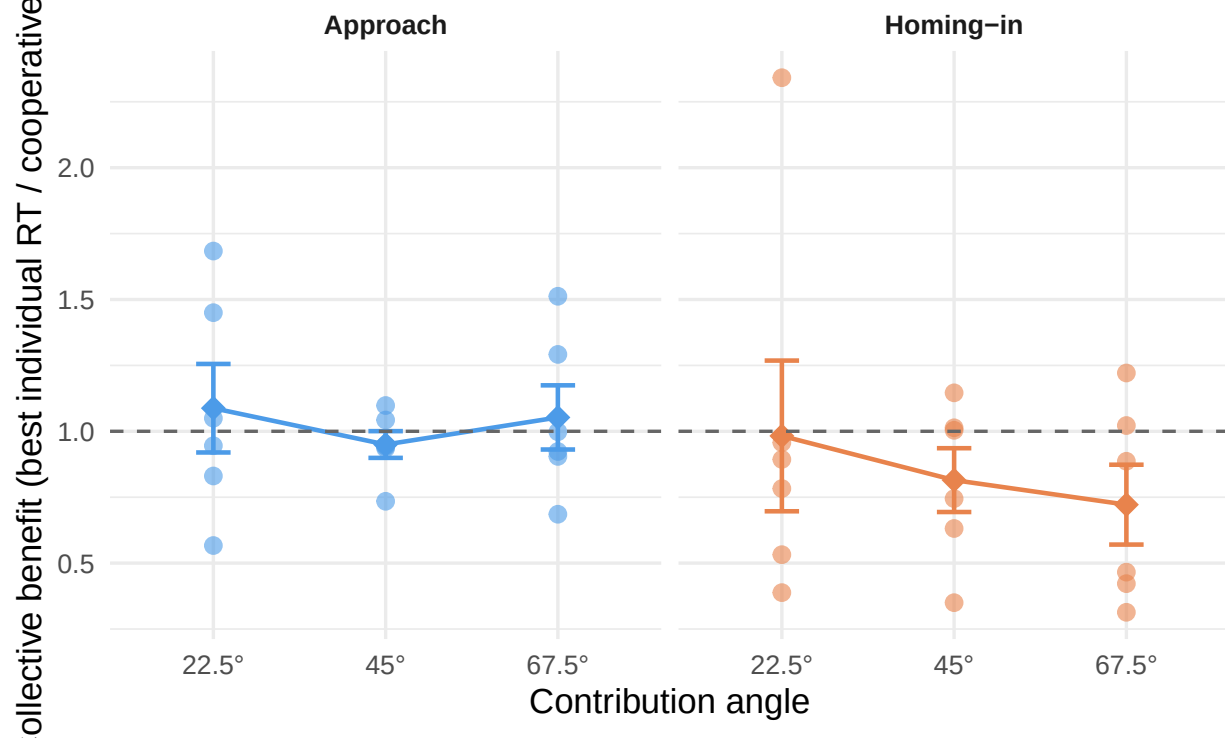
```
plot_anova <- function(data, title_label, y_label) {
  data %>%
    mutate(contribution_angle = factor(paste0(contribution_angle, "°"),
                                          levels = c("22.5°", "45°", "67.5°"))) %>%
    ggplot(aes(x = contribution_angle, y = collective_benefit, colour = phase)) +
    geom_point(alpha = 0.6, size = 2.5) +
    stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2, linewidth = 0.8) +
    stat_summary(fun = mean, geom = "point", size = 4, shape = 18) +
    stat_summary(fun = mean, geom = "line", aes(group = 1), linewidth = 0.8) +
    geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
    facet_wrap(~ phase) +
    scale_colour_manual(values = c("Approach" = "#4C9BE8",
                                   "Homing-in" = "#E8834C")) +

    labs(
      title = title_label,
      subtitle = "22.5°, 45°, 67.5° only | Dashed line = no benefit",
      x = "Contribution angle",
      y = y_label
    ) +
    theme_minimal(base_size = 13) +
    theme(legend.position = "none",
          strip.text = element_text(face = "bold"))
}

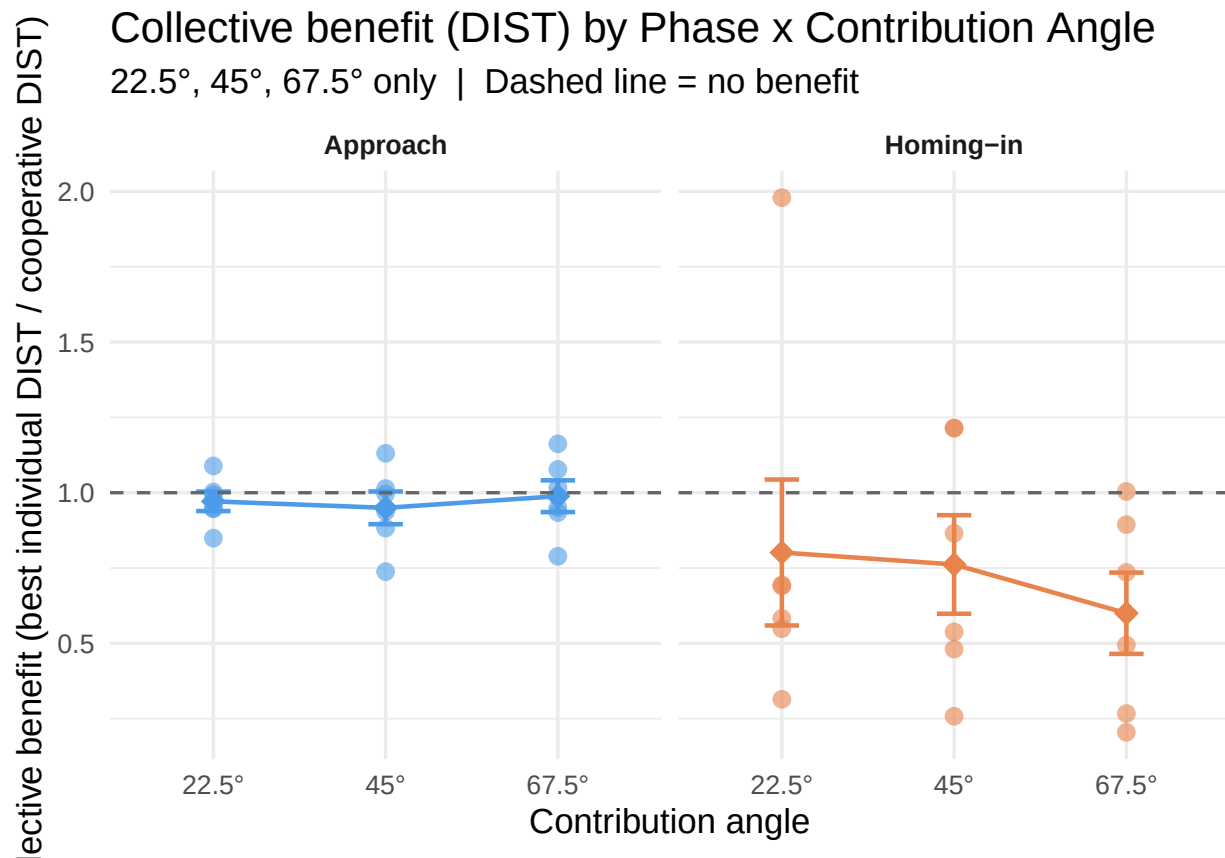
plot_anova(anova_data_RT,
            "Collective benefit (RT) by Phase x Contribution Angle",
            "Collective benefit (best individual RT / cooperative RT)")
```

Collective benefit (RT) by Phase x Contribution Angle

22.5°, 45°, 67.5° only | Dashed line = no benefit



```
plot_anova(anova_data_DIST,
  "Collective benefit (DIST) by Phase x Contribution Angle",
  "Collective benefit (best individual DIST / cooperative DIST)")
```



```
#Steiger test for comparing correlations between angles both DIST and RT
library(cocor)
```

```
## Warning: package 'cocor' was built under R version 4.4.3
```

```
run_steiger <- function(corr_data, correlation_results, measure_label) {

  angles <- c(22.5, 45, 67.5)
  phases <- c("Approach", "Homing-in")

  # all pairs where 67.5° is compared to the other two
  comparisons <- list(
    c(67.5, 22.5),
    c(67.5, 45)
  )

  for (phase_label in phases) {

    d <- corr_data %>% filter(phase == phase_label)

    r_vals <- correlation_results %>%
      filter(phase == phase_label) %>%
      select(contribution_angle, r) %>%
      deframe()

  }

}
```

```

n <- d %>%
  filter(contribution_angle == angles[1]) %>%
  summarise(n = sum(!is.na(skill_ratio) & !is.na(collective_benefit))) %>%
  pull(n)

get_r <- function(a1, a2) {
  x <- d %>% filter(contribution_angle == a1) %>% pull(collective_benefit)
  y <- d %>% filter(contribution_angle == a2) %>% pull(collective_benefit)
  cor(x, y, use = "complete.obs")
}

for (pair in comparisons) {
  a1 <- pair[1]; a2 <- pair[2]
  cat("\n=====", measure_label, "|", phase_label, ":", a1, "° vs", a2, "° =====\n")
  print(cocor.dep.groups.overlap(
    r.jk = r_vals[as.character(a1)],
    r.jh = r_vals[as.character(a2)],
    r.kh = get_r(a1, a2),
    n = n
  ))
}
}

```

```

#RT
run_steiger(corr_data, correlation_results, "RT")

```

```

##
## ===== RT | Approach : 67.5 ° vs 22.5 ° =====

## Warning in sqrt(2 * R(r.jk, r.jh, r.kh)): NaNs produced

## Warning in sqrt(((n - 1) * (1 + r.kh))/(2 * ((n - 1)/(n - 3)) * R(r.jk, , NaNs
## produced

## Warning in sqrt(2 - 2 * covariance): NaNs produced

## Warning in sqrt(2 * R(r.jk, r.jh, r.kh) + ((r.jk - r.jh)^2 * (1 - r.kh)^3)/(4
## * : NaNs produced

##
## Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.8471 and r.jh = 0.59
## Difference: r.jk - r.jh = 0.2572
## Related correlation: r.kh = 0.965
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)

```



```

## z = 1.7251, p-value = 0.0845
## Null hypothesis retained
##
## hotelling1940: Hotelling's t (1940)
## t = NaN, df = 3, p-value = NaN
## Null hypothesis retained
##
## williams1959: Williams' t (1959)
## t = NaN, df = 3, p-value = NaN
## Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
## z = 1.7251, p-value = 0.0845
## Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
## z = NaN, p-value = NaN
## Null hypothesis retained
##
## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)
## t = NaN, df = 3, p-value = NaN
## Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
## z = 2.6107, p-value = 0.0090
## Null hypothesis rejected
##
## meng1992: Meng, Rosenthal, and Rubin's z (1992)
## z = 2.5674, p-value = 0.0102
## Null hypothesis rejected
## 95% confidence interval for r.jk - r.jh: 0.1345 1.0022
## Null hypothesis rejected (Interval does not include 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
## z = 2.5078, p-value = 0.0121
## Null hypothesis rejected
##
## zou2007: Zou's (2007) confidence interval
## 95% confidence interval for r.jk - r.jh: -0.1121 1.1351
## Null hypothesis retained (Interval includes 0)
##
##
## ===== RT | Approach : 67.5 ° vs 45 ° =====

## Warning in sqrt((1 - r.jk^2)^2 + (1 - r.jh^2)^2 - 2 * k): NaNs produced

## Warning in sqrt(2 * R(r.jk, r.jh, r.kh)): NaNs produced

## Warning in sqrt(((n - 1) * (1 + r.kh))/(2 * ((n - 1)/(n - 3)) * R(r.jk, : NaNs
## produced

## Warning in sqrt((1 - r.jk^2)^2 + (1 - r.jh^2)^2 - 2 * r.kh^3 - (2 * r.kh - :
## NaNs produced

```

```

## Warning in sqrt(2 - 2 * covariance): NaNs produced

## Warning in sqrt(2 * R(r.jk, r.jh, r.kh) + (((r.jk - r.jh)^2 * (1 - r.kh)^3)/(4
## * : NaNs produced

## Warning in sqrt(2 - 2 * covariance): NaNs produced
## Warning in sqrt(2 - 2 * covariance): NaNs produced

## Warning in sqrt((r.jk - l1)^2 + (u2 - r.jh)^2 - 2 * c * (r.jk - l1) * (u2 - :
## NaNs produced

## Warning in sqrt((u1 - r.jk)^2 + (r.jh - l2)^2 - 2 * c * (u1 - r.jk) * (r.jh - :
## NaNs produced

##
##   Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.8471 and r.jh = 0.8393
## Difference: r.jk - r.jh = 0.0078
## Related correlation: r.kh = -0.4338
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)
##   z = NaN, p-value = NaN
##   Null hypothesis retained
##
## hotelling1940: Hotelling's t (1940)
##   t = NaN, df = 3, p-value = NaN
##   Null hypothesis retained
##
## williams1959: Williams' t (1959)
##   t = NaN, df = 3, p-value = NaN
##   Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
##   z = NaN, p-value = NaN
##   Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
##   z = NaN, p-value = NaN
##   Null hypothesis retained
##
## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)
##   t = NaN, df = 3, p-value = NaN
##   Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
##   z = NaN, p-value = NaN
##   Null hypothesis retained
##

```

```

## meng1992: Meng, Rosenthal, and Rubin's z (1992)
##   z = 0.0276, p-value = 0.9780
##   Null hypothesis retained
##   95% confidence interval for r.jk - r.jh: -1.8892 1.9433
##   Null hypothesis retained (Interval includes 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
##   z = NaN, p-value = NaN
##   Null hypothesis retained
##
## zou2007: Zou's (2007) confidence interval
##   95% confidence interval for r.jk - r.jh: NaN NaN
##   Null hypothesis retained
##
##
## ===== RT | Homing-in : 67.5 ° vs 22.5 ° =====
##
##   Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.5617 and r.jh = 0.5626
## Difference: r.jk - r.jh = -9e-04
## Related correlation: r.kh = -0.1651
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)
##   z = -0.0020, p-value = 0.9984
##   Null hypothesis retained
##
## hotelling1940: Hotelling's t (1940)
##   t = -0.0020, df = 3, p-value = 0.9985
##   Null hypothesis retained
##
## williams1959: Williams' t (1959)
##   t = -0.0016, df = 3, p-value = 0.9988
##   Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
##   z = -0.0020, p-value = 0.9984
##   Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
##   z = -0.0014, p-value = 0.9989
##   Null hypothesis retained
##
## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)
##   t = -0.0020, df = 3, p-value = 0.9985
##   Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
##   z = -0.0014, p-value = 0.9989
##   Null hypothesis retained

```

```

##
## meng1992: Meng, Rosenthal, and Rubin's z (1992)
##   z = -0.0014, p-value = 0.9989
##   Null hypothesis retained
##   95% confidence interval for r.jk - r.jh: -1.7868 1.7843
##   Null hypothesis retained (Interval includes 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
##   z = -0.0014, p-value = 0.9989
##   Null hypothesis retained
##
## zou2007: Zou's (2007) confidence interval
##   95% confidence interval for r.jk - r.jh: -1.1746 1.1731
##   Null hypothesis retained (Interval includes 0)
##
##
## ===== RT | Homing-in : 67.5 ° vs 45 ° =====
##
##   Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.5617 and r.jh = 0.3438
## Difference: r.jk - r.jh = 0.2179
## Related correlation: r.kh = 0.5234
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)
##   z = 0.6336, p-value = 0.5264
##   Null hypothesis retained
##
## hotelling1940: Hotelling's t (1940)
##   t = 0.4684, df = 3, p-value = 0.6714
##   Null hypothesis retained
##
## williams1959: Williams' t (1959)
##   t = 0.4653, df = 3, p-value = 0.6734
##   Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
##   z = 0.6336, p-value = 0.5264
##   Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
##   z = 0.4550, p-value = 0.6491
##   Null hypothesis retained
##
## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)
##   t = 0.4684, df = 3, p-value = 0.6714
##   Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
##   z = 0.4522, p-value = 0.6511

```

```
## Null hypothesis retained
##
## meng1992: Meng, Rosenthal, and Rubin's z (1992)
## z = 0.4499, p-value = 0.6528
## Null hypothesis retained
## 95% confidence interval for r.jk - r.jh: -0.9296 1.4835
## Null hypothesis retained (Interval includes 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
## z = 0.4510, p-value = 0.6520
## Null hypothesis retained
##
## zou2007: Zou's (2007) confidence interval
## 95% confidence interval for r.jk - r.jh: -0.7028 1.1092
## Null hypothesis retained (Interval includes 0)
```

```
#DIST
run_steiger(corr_data2, correlation_results2, "DIST")
```

```
##
## ===== DIST | Approach : 67.5 ° vs 22.5 ° =====
##
## Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.3539 and r.jh = -0.0589
## Difference: r.jk - r.jh = 0.4128
## Related correlation: r.kh = 0.1309
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)
## z = 0.8223, p-value = 0.4109
## Null hypothesis retained
##
## hottelling1940: Hotelling's t (1940)
## t = 0.5837, df = 3, p-value = 0.6004
## Null hypothesis retained
##
## williams1959: Williams' t (1959)
## t = 0.5822, df = 3, p-value = 0.6013
## Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
## z = 0.8223, p-value = 0.4109
## Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
## z = 0.5668, p-value = 0.5708
## Null hypothesis retained
##
## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)
## t = 0.5827, df = 3, p-value = 0.6010
```

```

## Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
## z = 0.5600, p-value = 0.5755
## Null hypothesis retained
##
## meng1992: Meng, Rosenthal, and Rubin's z (1992)
## z = 0.5534, p-value = 0.5800
## Null hypothesis retained
## 95% confidence interval for r.jk - r.jh: -1.0902 1.9480
## Null hypothesis retained (Interval includes 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
## z = 0.5597, p-value = 0.5757
## Null hypothesis retained
##
## zou2007: Zou's (2007) confidence interval
## 95% confidence interval for r.jk - r.jh: -0.8015 1.2959
## Null hypothesis retained (Interval includes 0)
##
##
## ===== DIST | Approach : 67.5 ° vs 45 ° =====
##
## Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.3539 and r.jh = 0.8121
## Difference: r.jk - r.jh = -0.4582
## Related correlation: r.kh = -0.0198
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)
## z = -1.1506, p-value = 0.2499
## Null hypothesis retained
##
## hotelling1940: Hotelling's t (1940)
## t = -1.2317, df = 3, p-value = 0.3058
## Null hypothesis retained
##
## williams1959: Williams' t (1959)
## t = -0.9952, df = 3, p-value = 0.3930
## Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
## z = -1.1506, p-value = 0.2499
## Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
## z = -0.8840, p-value = 0.3767
## Null hypothesis retained
##
## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)

```

```

## t = -1.2152, df = 3, p-value = 0.3112
## Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
## z = -0.8757, p-value = 0.3812
## Null hypothesis retained
##
## meng1992: Meng, Rosenthal, and Rubin's z (1992)
## z = -0.8810, p-value = 0.3783
## Null hypothesis retained
## 95% confidence interval for r.jk - r.jh: -2.4611 0.9347
## Null hypothesis retained (Interval includes 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
## z = -0.8835, p-value = 0.3770
## Null hypothesis retained
##
## zou2007: Zou's (2007) confidence interval
## 95% confidence interval for r.jk - r.jh: -1.4872 0.5746
## Null hypothesis retained (Interval includes 0)
##
##
## ===== DIST | Homing-in : 67.5 ° vs 22.5 ° =====
##
## Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.815 and r.jh = 0.0592
## Difference: r.jk - r.jh = 0.7558
## Related correlation: r.kh = 0.0329
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)
## z = 1.7654, p-value = 0.0775
## Null hypothesis retained
##
## hotelling1940: Hotelling's t (1940)
## t = 1.6271, df = 3, p-value = 0.2022
## Null hypothesis retained
##
## williams1959: Williams' t (1959)
## t = 1.5140, df = 3, p-value = 0.2273
## Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
## z = 1.7654, p-value = 0.0775
## Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
## z = 1.3317, p-value = 0.1829
## Null hypothesis retained
##

```

```

## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)
##   t = 1.5966, df = 3, p-value = 0.2086
##   Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
##   z = 1.2884, p-value = 0.1976
##   Null hypothesis retained
##
## meng1992: Meng, Rosenthal, and Rubin's z (1992)
##   z = 1.2642, p-value = 0.2062
##   Null hypothesis retained
##   95% confidence interval for r.jk - r.jh: -0.5958 2.7610
##   Null hypothesis retained (Interval includes 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
##   z = 1.2683, p-value = 0.2047
##   Null hypothesis retained
##
## zou2007: Zou's (2007) confidence interval
##   95% confidence interval for r.jk - r.jh: -0.3542 1.6196
##   Null hypothesis retained (Interval includes 0)
##
##
## ===== DIST | Homing-in : 67.5 ° vs 45 ° =====
##
##   Results of a comparison of two overlapping correlations based on dependent groups
##
## Comparison between r.jk = 0.815 and r.jh = 0.3686
## Difference: r.jk - r.jh = 0.4464
## Related correlation: r.kh = 0.5669
## Group size: n = 6
## Null hypothesis: r.jk is equal to r.jh
## Alternative hypothesis: r.jk is not equal to r.jh (two-sided)
## Alpha: 0.05
##
## pearson1898: Pearson and Filon's z (1898)
##   z = 1.4160, p-value = 0.1568
##   Null hypothesis retained
##
## hotelling1940: Hotelling's t (1940)
##   t = 1.4620, df = 3, p-value = 0.2399
##   Null hypothesis retained
##
## williams1959: Williams' t (1959)
##   t = 1.4344, df = 3, p-value = 0.2469
##   Null hypothesis retained
##
## olkin1967: Olkin's z (1967)
##   z = 1.4160, p-value = 0.1568
##   Null hypothesis retained
##
## dunn1969: Dunn and Clark's z (1969)
##   z = 1.2507, p-value = 0.2110
##   Null hypothesis retained

```



```
##
## hendrickson1970: Hendrickson, Stanley, and Hills' (1970) modification of Williams' t (1959)
##   t = 1.4607, df = 3, p-value = 0.2402
##   Null hypothesis retained
##
## steiger1980: Steiger's (1980) modification of Dunn and Clark's z (1969) using average correlations
##   z = 1.2051, p-value = 0.2282
##   Null hypothesis retained
##
## meng1992: Meng, Rosenthal, and Rubin's z (1992)
##   z = 1.1765, p-value = 0.2394
##   Null hypothesis retained
##   95% confidence interval for r.jk - r.jh: -0.5027 2.0127
##   Null hypothesis retained (Interval includes 0)
##
## hittner2003: Hittner, May, and Silver's (2003) modification of Dunn and Clark's z (1969) using a back
##   z = 1.1685, p-value = 0.2426
##   Null hypothesis retained
##
## zou2007: Zou's (2007) confidence interval
##   95% confidence interval for r.jk - r.jh: -0.2919 1.3841
##   Null hypothesis retained (Interval includes 0)
```

```
#Making plots for individual differences as predictor for RT
#recode contribution angles to match paper's % labels
library(patchwork)
```

```
## Warning: package 'patchwork' was built under R version 4.4.3
```

```
corr_data_plot <- corr_data %>%
  mutate(
    angle_label = case_when(
      contribution_angle == 22.5 ~ "30%",
      contribution_angle == 45   ~ "50%",
      contribution_angle == 67.5 ~ "70%"
    ),
    angle_label = factor(angle_label, levels = c("30%", "50%", "70%"))
  ) %>%
  filter(!is.na(angle_label)) # <-- drops the NA group

plot_fig4 <- function(data, phase_label, panel_label) {
  data %>%
    filter(phase == phase_label) %>%
    ggplot(aes(x = skill_ratio, y = collective_benefit,
               colour = angle_label, fill = angle_label)) +
    geom_point(size = 2.5, alpha = 0.8) +
    geom_smooth(method = "lm", se = TRUE, linewidth = 1.2, alpha = 0.15,
               aes(group = angle_label)) +
    geom_hline(yintercept = 1, linetype = "dashed", colour = "grey50") +
    scale_colour_manual(
      name = "Action contribution\nof better member",
      values = c("30%" = "#4C9BE8", "50%" = "#E8834C", "70%" = "#6DBE6D")
    ) +
```

```

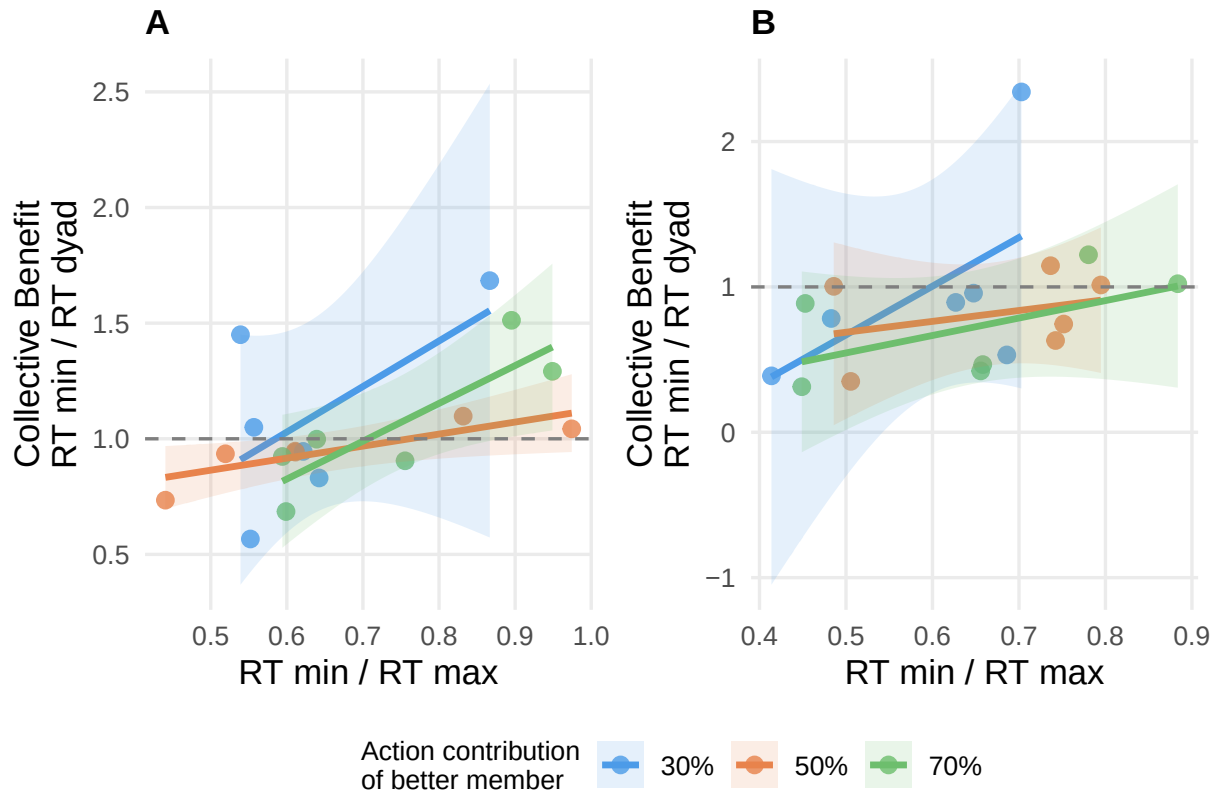
scale_fill_manual(
  name = "Action contribution\nof better member",
  values = c("30%" = "#4C9BE8", "50%" = "#E8834C", "70%" = "#6DBE6D")
) +
labs(
  title = panel_label,
  x = "RT min / RT max",
  y = "Collective Benefit\nRT min / RT dyad"
) +
theme_minimal(base_size = 13) +
theme(
  legend.position = "bottom",
  legend.title = element_text(size = 10),
  plot.title = element_text(face = "bold", size = 13),
  panel.grid.minor = element_blank()
)
}

p_approach <- plot_fig4(corr_data_plot, "Approach", "A")
p_homingin <- plot_fig4(corr_data_plot, "Homing-in", "B")

p_approach + p_homingin +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'

```



```
#Making plots for individual differences as predictor for DIST
corr_data2_plot <- corr_data2 %>%
  mutate(
    angle_label = case_when(
      contribution_angle == 22.5 ~ "30%",
      contribution_angle == 45 ~ "50%",
      contribution_angle == 67.5 ~ "70%"
    ),
    angle_label = factor(angle_label, levels = c("30%", "50%", "70%"))
  ) %>%
  filter(!is.na(angle_label))

plot_fig4_DIST <- function(data, phase_label, panel_label) {
  data %>%
    filter(phase == phase_label) %>%
    ggplot(aes(x = skill_ratio, y = collective_benefit,
               colour = angle_label, fill = angle_label)) +
    geom_point(size = 2.5, alpha = 0.8) +
    geom_smooth(method = "lm", se = TRUE, linewidth = 1.2, alpha = 0.15,
               aes(group = angle_label)) +
    geom_hline(yintercept = 1, linetype = "dashed", colour = "grey50") +
    scale_colour_manual(
      name = "Action contribution\nof better member",
      values = c("30%" = "#4C9BE8", "50%" = "#E8834C", "70%" = "#6DBE6D")
    ) +
    scale_fill_manual(
```

```

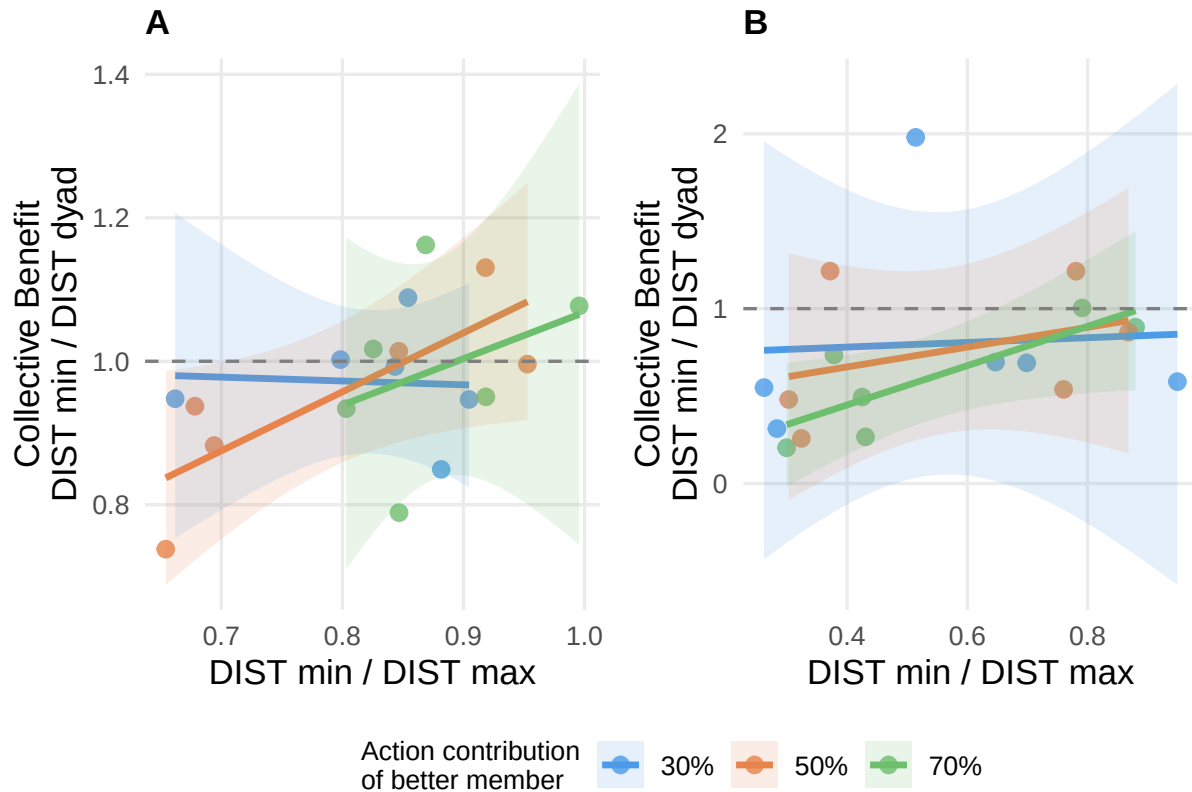
    name = "Action contribution\nof better member",
    values = c("30%" = "#4C9BE8", "50%" = "#E8834C", "70%" = "#6DBE6D")
  ) +
  labs(
    title = panel_label,
    x = "DIST min / DIST max",
    y = "Collective Benefit\nDIST min / DIST dyad"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "bottom",
    legend.title = element_text(size = 10),
    plot.title = element_text(face = "bold", size = 13),
    panel.grid.minor = element_blank()
  )
}

p_approach_DIST <- plot_fig4_DIST(corr_data2_plot, "Approach", "A")
p_homingin_DIST <- plot_fig4_DIST(corr_data2_plot, "Homing-in", "B")

p_approach_DIST + p_homingin_DIST +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'

```



#AI and PLAYBACK analysis

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
#Performance relative to individual baseline (RT and DIST)
ind_means_RT <- analysis_data %>%
  filter(condition == "individual") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(ind_RT = mean(RT, na.rm = TRUE), .groups = "drop")

ai_means_RT <- analysis_data %>%
  filter(condition == "ai") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(ai_RT = mean(RT, na.rm = TRUE), .groups = "drop")

pb_means_RT <- analysis_data %>%
  filter(condition == "playback") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(pb_RT = mean(RT, na.rm = TRUE), .groups = "drop")

# Ratio: individual / condition (>1 means condition is better)
ratio_ai_RT <- ind_means_RT %>%
  inner_join(ai_means_RT, by = c("participant_id", "phase", "contribution_angle")) %>%
  mutate(ratio = ind_RT / ai_RT)
```

```

ratio_pb_RT <- ind_means_RT %>%
  inner_join(pb_means_RT, by = c("participant_id", "phase", "contribution_angle")) %>%
  mutate(ratio = ind_RT / pb_RT)

#one-sample t-test: is ratio different from 1?
run_ratio_tests <- function(df, ratio_col = "ratio", label) {
  df %>%
    group_by(phase, contribution_angle) %>%
    summarise(
      condition = label,
      n = sum(!is.na(.data[[ratio_col]])),
      M_ratio = mean(.data[[ratio_col]], na.rm = TRUE),
      SE_ratio = sd(.data[[ratio_col]], na.rm = TRUE) / sqrt(n),
      t_stat = tryCatch(t.test(.data[[ratio_col]], mu = 1)$statistic, error = function(e) NA),
      p_value = tryCatch(t.test(.data[[ratio_col]], mu = 1)$p.value, error = function(e) NA),
      cohens_d = tryCatch(
        (mean(.data[[ratio_col]], na.rm = TRUE) - 1) / sd(.data[[ratio_col]], na.rm = TRUE),
        error = function(e) NA
      ),
      .groups = "drop"
    ) %>%
    mutate(
      sig = case_when(
        p_value < .001 ~ "***",
        p_value < .01 ~ "**",
        p_value < .05 ~ "*",
        p_value < .10 ~ "+",
        TRUE ~ ""
      )
    )
}

ratio_tests_ai_RT <- run_ratio_tests(ratio_ai_RT, label = "AI")
ratio_tests_pb_RT <- run_ratio_tests(ratio_pb_RT, label = "Playback")

cat(" AI vs Individual (RT) \n")

```

```
## AI vs Individual (RT)
```

```
print(ratio_tests_ai_RT)
```

```
## # A tibble: 10 x 10
##   phase      contribution_angle condition      n M_ratio SE_ratio t_stat  p_value
##   <fct>      <fct>              <chr>    <int>  <dbl>   <dbl>   <dbl>   <dbl>
## 1 Approach    0                AI         12   1.17   0.0577   2.92   0.0139
## 2 Approach  22.5              AI         12   2.05   0.207    5.08   0.000353
## 3 Approach  45                AI         12   1.56   0.125    4.45   0.000977
## 4 Approach  67.5              AI         12   2.19   0.287    4.13   0.00168
## 5 Approach  90                AI         12   1.24   0.146    1.65   0.128
## 6 Homing-in  0                AI         12   1.59   0.432    1.37   0.197
## 7 Homing-in 22.5              AI         12   2.74   0.334    5.20   0.000296
## 8 Homing-in 45                AI         12   2.06   0.216    4.91   0.000462
```

```
## 9 Homing-in 67.5 AI 12 1.50 0.194 2.56 0.0267
## 10 Homing-in 90 AI 12 1.16 0.181 0.867 0.404
## # i 2 more variables: cohens_d <dbl>, sig <chr>
```

```
cat("\n Playback vs Individual (RT) \n")
```

```
##
## Playback vs Individual (RT)
```

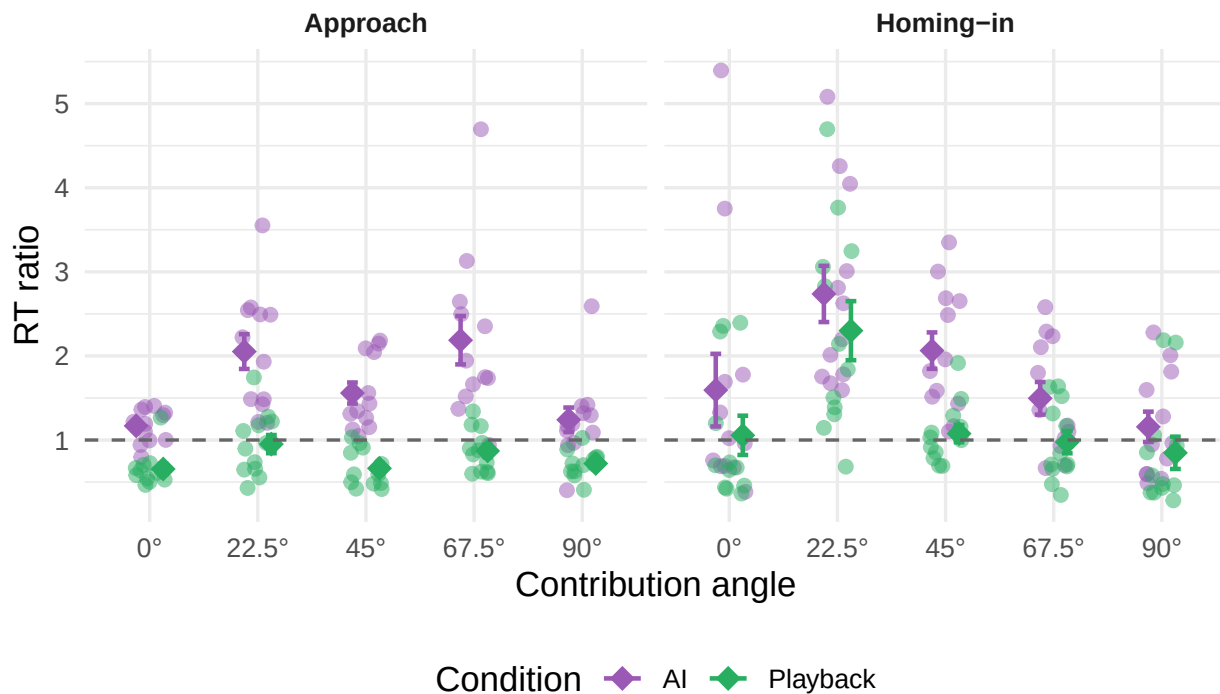
```
print(ratio_tests_pb_RT)
```

```
## # A tibble: 10 x 10
##   phase      contribution_angle condition      n M_ratio SE_ratio t_stat p_value
##   <fct>      <fct>              <chr>    <int> <dbl>    <dbl> <dbl>    <dbl>
## 1 Approach    0 Playback          12 0.655 0.0602 -5.74 1.31e-4
## 2 Approach  22.5 Playback          12 0.951 0.108 -0.452 6.60e-1
## 3 Approach  45 Playback          12 0.663 0.0642 -5.25 2.73e-4
## 4 Approach  67.5 Playback          12 0.870 0.0730 -1.78 1.03e-1
## 5 Approach  90 Playback          12 0.721 0.0453 -6.17 6.97e-5
## 6 Homing-in 0 Playback          12 1.06 0.234 0.238 8.16e-1
## 7 Homing-in 22.5 Playback          12 2.30 0.351 3.70 3.48e-3
## 8 Homing-in 45 Playback          12 1.08 0.103 0.738 4.76e-1
## 9 Homing-in 67.5 Playback          12 0.975 0.132 -0.192 8.51e-1
## 10 Homing-in 90 Playback          12 0.848 0.192 -0.794 4.44e-1
## # i 2 more variables: cohens_d <dbl>, sig <chr>
```

```
#plot
bind_rows(
  ratio_ai_RT %>% mutate(condition = "AI"),
  ratio_pb_RT %>% mutate(condition = "Playback")
) %>%
  mutate(
    contribution_angle = factor(paste0(contribution_angle, "°"),
                                levels = c("0°", "22.5°", "45°", "67.5°", "90°"))
  ) %>%
  ggplot(aes(x = contribution_angle, y = ratio, colour = condition)) +
  geom_jitter(width = 0.15, alpha = 0.5, size = 2) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
               position = position_dodge(0.5), linewidth = 0.8) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
               position = position_dodge(0.5)) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
  facet_wrap(~ phase) +
  scale_colour_manual(values = c("AI" = "#9B59B6", "Playback" = "#27AE60")) +
  labs(
    title = "RT ratio (Individual / Condition) by contribution angle",
    subtitle = "Dashed line = no difference from individual | >1 = condition faster",
    x = "Contribution angle", y = "RT ratio", colour = "Condition"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold"))
```

RT ratio (Individual / Condition) by contribution angle

Dashed line = no difference from individual | >1 = condition faster



```
#Performance relative to individual baseline (RT and DIST)
#same as above just for DIST
ind_means_DIST <- analysis_data %>%
  filter(condition == "individual") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(ind_DIST = mean(DIST, na.rm = TRUE), .groups = "drop")

ai_means_DIST <- analysis_data %>%
  filter(condition == "ai") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(ai_DIST = mean(DIST, na.rm = TRUE), .groups = "drop")

pb_means_DIST <- analysis_data %>%
  filter(condition == "playback") %>%
  group_by(participant_id, phase, contribution_angle) %>%
  summarise(pb_DIST = mean(DIST, na.rm = TRUE), .groups = "drop")

ratio_ai_DIST <- ind_means_DIST %>%
  inner_join(ai_means_DIST, by = c("participant_id", "phase", "contribution_angle")) %>%
  mutate(ratio = ind_DIST / ai_DIST)

ratio_pb_DIST <- ind_means_DIST %>%
  inner_join(pb_means_DIST, by = c("participant_id", "phase", "contribution_angle")) %>%
  mutate(ratio = ind_DIST / pb_DIST)

ratio_tests_ai_DIST <- run_ratio_tests(ratio_ai_DIST, label = "AI")
```



```
ratio_tests_pb_DIST <- run_ratio_tests(ratio_pb_DIST, label = "Playback")
```

```
cat("  AI vs Individual (DIST)  \n")
```

```
##  AI vs Individual (DIST)
```

```
print(ratio_tests_ai_DIST)
```

```
## # A tibble: 10 x 10
##   phase      contribution_angle condition      n M_ratio SE_ratio t_stat p_value
##   <fct>      <fct>              <chr>    <int> <dbl>    <dbl> <dbl>   <dbl>
## 1 Approach  0                AI        12  0.893  0.0569 -1.89  0.0858
## 2 Approach 22.5            AI        12  1.21   0.0517  4.01  0.00205
## 3 Approach 45             AI        12  1.05   0.0556  0.897 0.389
## 4 Approach 67.5           AI        12  1.22   0.0382  5.66  0.000148
## 5 Approach 90             AI        12  0.932  0.0423 -1.61  0.137
## 6 Homing-in 0             AI        12  2.38   1.02    1.35  0.203
## 7 Homing-in 22.5          AI        12  2.70   0.386   4.40  0.00107
## 8 Homing-in 45            AI        12  2.08   0.226   4.77  0.000578
## 9 Homing-in 67.5          AI        12  1.40   0.222   1.82  0.0959
## 10 Homing-in 90           AI        12  1.17   0.282   0.592 0.566
## # i 2 more variables: cohens_d <dbl>, sig <chr>
```

```
cat("\n Playback vs Individual (DIST)  \n")
```

```
##
```

```
##  Playback vs Individual (DIST)
```

```
print(ratio_tests_pb_DIST)
```

```
## # A tibble: 10 x 10
##   phase      contribution_angle condition      n M_ratio SE_ratio t_stat p_value
##   <fct>      <fct>              <chr>    <int> <dbl>    <dbl> <dbl>   <dbl>
## 1 Approach  0                Playback    12  0.846  0.0497 -3.10  1.00e-2
## 2 Approach 22.5            Playback    12  1.04   0.0595  0.622 5.47e-1
## 3 Approach 45             Playback    12  0.927  0.0626 -1.17  2.66e-1
## 4 Approach 67.5           Playback    12  1.03   0.0463  0.611 5.54e-1
## 5 Approach 90             Playback    12  0.838  0.0158 -10.3  5.76e-7
## 6 Homing~~ 0             Playback    12  1.48   0.453   1.05  3.15e-1
## 7 Homing~~ 22.5          Playback    12  3.35   0.674   3.49  5.07e-3
## 8 Homing~~ 45            Playback    12  1.52   0.221   2.36  3.77e-2
## 9 Homing~~ 67.5          Playback    12  1.26   0.217   1.21  2.50e-1
## 10 Homing~~ 90           Playback    12  0.991  0.263  -0.0328 9.74e-1
## # i 2 more variables: cohens_d <dbl>, sig <chr>
```

```
bind_rows(
  ratio_ai_DIST %>% mutate(condition = "AI"),
  ratio_pb_DIST %>% mutate(condition = "Playback")
) %>%
  mutate(
```

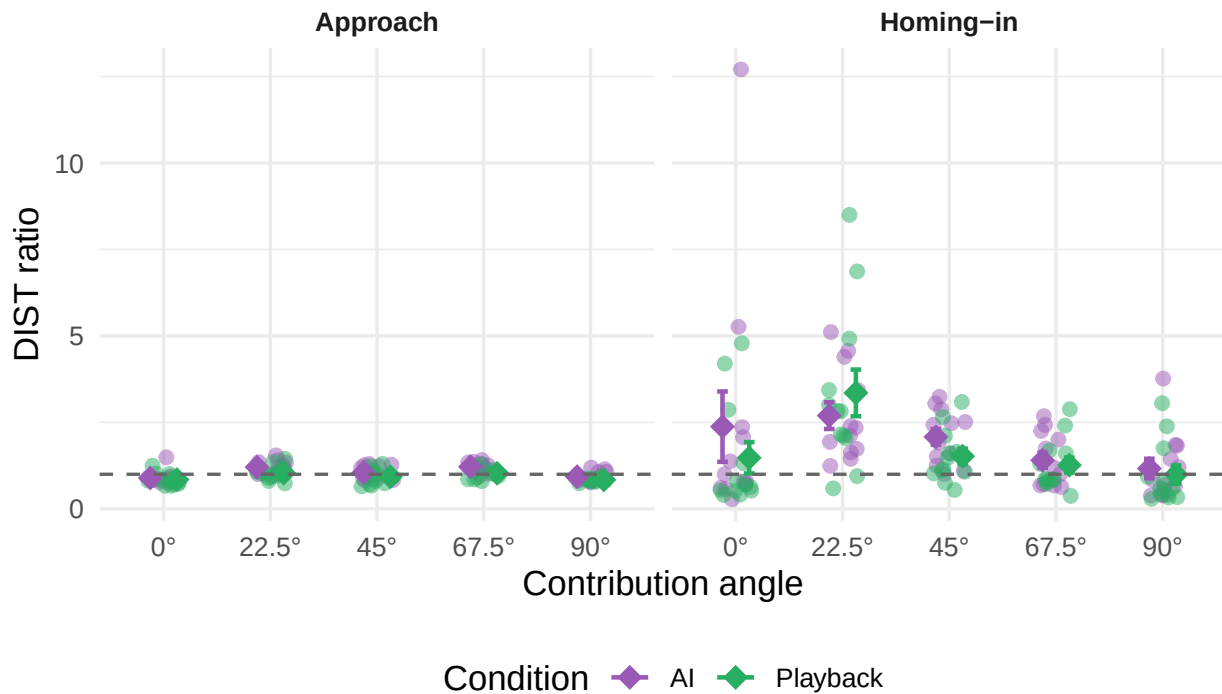
```

contribution_angle = factor(paste0(contribution_angle, "°"),
                             levels = c("0°", "22.5°", "45°", "67.5°", "90°"))
) %>%
ggplot(aes(x = contribution_angle, y = ratio, colour = condition)) +
geom_jitter(width = 0.15, alpha = 0.5, size = 2) +
stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
              position = position_dodge(0.5), linewidth = 0.8) +
stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
              position = position_dodge(0.5)) +
geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
facet_wrap(~ phase) +
scale_colour_manual(values = c("AI" = "#9B59B6", "Playback" = "#27AE60")) +
labs(
  title = "DIST ratio (Individual / Condition) by contribution angle",
  subtitle = "Dashed line = no difference from individual | >1 = condition shorter",
  x = "Contribution angle", y = "DIST ratio", colour = "Condition"
) +
theme_minimal(base_size = 13) +
theme(legend.position = "bottom", strip.text = element_text(face = "bold"))

```

DIST ratio (Individual / Condition) by contribution angle

Dashed line = no difference from individual | >1 = condition shorter



```
#Condition x phase ANOVA RT
```

```
library(afex)
```

```
## Warning: package 'afex' was built under R version 4.4.3
```

```

## Loading required package: lme4

## Warning: package 'lme4' was built under R version 4.4.3

## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## *****
## Welcome to afex. For support visit: http://afex.singmann.science/

## - Functions for ANOVAs: aov_car(), aov_ez(), and aov_4()
## - Methods for calculating p-values with mixed(): 'S', 'KR', 'LRT', and 'PB'
## - 'afex_aov' and 'mixed' objects can be passed to emmeans() for follow-up tests
## - Get and set global package options with: afex_options()
## - Set sum-to-zero contrasts globally: set_sum_contrasts()
## - For example analyses see: browseVignettes("afex")
## *****

##
## Attaching package: 'afex'

## The following object is masked from 'package:lme4':
##
##     lmer

#participant level dataset collapse over contribution angles
cond_phase_RT <- analysis_data %>%
  filter(condition %in% c("individual", "ai", "playback")) %>%
  group_by(participant_id, condition, phase) %>%
  summarise(mean_RT = mean(RT, na.rm = TRUE), .groups = "drop") %>%
  mutate(
    condition = factor(condition, levels = c("individual", "ai", "playback")),
    phase      = factor(phase)
  )

#mixed ANOVA
aov_cond_phase_RT <- aov_ez(
  id      = "participant_id",
  dv      = "mean_RT",
  data    = cond_phase_RT,
  within  = c("condition", "phase"),
  type    = 3
)
cat("  Condition × Phase ANOVA (RT)  \n")

##    Condition × Phase ANOVA (RT)

```

```
print(summary(aov_cond_phase_RT))
```

```
##
## Univariate Type III Repeated-Measures ANOVA Assuming Sphericity
##
##               Sum Sq num Df Error SS den Df  F value    Pr(>F)
## (Intercept)    521.36     1  25.2018     11 227.5600 1.074e-08 ***
## condition      23.32     2   8.9418     22  28.6851 7.420e-07 ***
## phase          83.68     1   8.6254     11 106.7199 5.337e-07 ***
## condition:phase  6.27     2   9.2414     22   7.4627 0.003358 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Mauchly Tests for Sphericity
##
##               Test statistic p-value
## condition           0.62810 0.097753
## condition:phase      0.29737 0.002325
##
##
## Greenhouse-Geisser and Huynh-Feldt Corrections
## for Departure from Sphericity
##
##               GG eps Pr(>F[GG])
## condition      0.72891 1.647e-05 ***
## condition:phase 0.58733 0.0143 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               HF eps  Pr(>F[HF])
## condition      0.8118661 6.358945e-06
## condition:phase 0.6155413 1.293746e-02
```

```
effectsize::eta_squared(aov_cond_phase_RT, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
##
## Parameter      | Eta2 (partial) |      95% CI
## -----
## condition      |          0.72 | [0.53, 1.00]
## phase          |          0.91 | [0.78, 1.00]
## condition:phase |          0.40 | [0.12, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
#post-hoc: pairwise condition comparisons within each phase
for (ph in levels(cond_phase_RT$phase)) {
  cat("\n Phase:", ph, "\n")
  d <- cond_phase_RT %>% filter(phase == ph)
  print(pairwise.t.test(
    x = d$mean_RT,
```

```

    g                = d$condition,
    paired           = TRUE,
    p.adjust.method = "bonferroni"
  ))
}

```

```

##
##   Phase: Approach
##
##   Pairwise comparisons using paired t tests
##
## data:  d$mean_RT and d$condition
##
##           individual ai
## ai      3.7e-05      -
## playback 0.0042    1.5e-06
##
## P value adjustment method: bonferroni
##
##   Phase: Homing-in
##
##   Pairwise comparisons using paired t tests
##
## data:  d$mean_RT and d$condition
##
##           individual ai
## ai      0.0017      -
## playback 0.7007    3.8e-05
##
## P value adjustment method: bonferroni

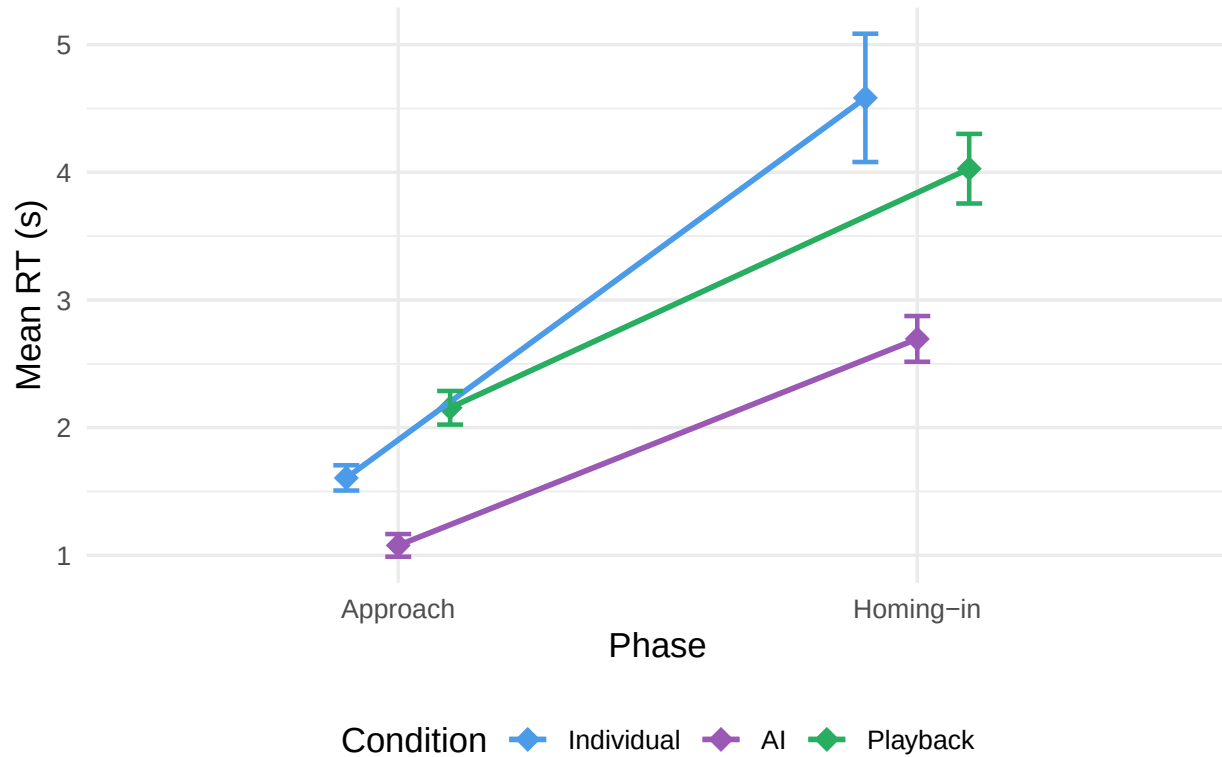
```

```

#plot
cond_phase_RT %>%
  ggplot(aes(x = phase, y = mean_RT, colour = condition, group = condition)) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
               position = position_dodge(0.3)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1,
               position = position_dodge(0.3)) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.15,
               position = position_dodge(0.3), linewidth = 0.8) +
  scale_colour_manual(
    values = c("individual" = "#4C9BE8", "ai" = "#9B59B6", "playback" = "#27AE60"),
    labels = c("Individual", "AI", "Playback")
  ) +
  labs(
    title = "Mean RT by Condition × Phase",
    x = "Phase", y = "Mean RT (s)", colour = "Condition"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")

```

Mean RT by Condition × Phase



```
#Condition x phase ANOVA DIST
cond_phase_DIST <- analysis_data %>%
  filter(condition %in% c("individual", "ai", "playback")) %>%
  group_by(participant_id, condition, phase) %>%
  summarise(mean_DIST = mean(DIST, na.rm = TRUE), .groups = "drop") %>%
  mutate(
    condition = factor(condition, levels = c("individual", "ai", "playback")),
    phase = factor(phase)
  )

aov_cond_phase_DIST <- aov_car(
  mean_DIST ~ condition * phase + Error(participant_id / (condition * phase)),
  data = cond_phase_DIST
)
print(summary(aov_cond_phase_DIST))
```

```
##
## Univariate Type III Repeated-Measures ANOVA Assuming Sphericity
##
##               Sum Sq num Df Error SS den Df  F value    Pr(>F)
## (Intercept)    7418249      1   399090    11 204.4668 1.884e-08 ***
## condition       88174      2   131887    22   7.3541 0.0035830 **
## phase           58801      1   341373    11   1.8947 0.1960403
## condition:phase  82029      2    87174    22  10.3508 0.0006788 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Mauchly Tests for Sphericity
##
##              Test statistic    p-value
## condition          0.22248 0.00054507
## condition:phase    0.23141 0.00066362
##
##
## Greenhouse-Geisser and Huynh-Feldt Corrections
## for Departure from Sphericity
##
##              GG eps Pr(>F[GG])
## condition          0.56258  0.016243 *
## condition:phase    0.56542  0.005893 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              HF eps  Pr(>F[HF])
## condition          0.5823872 0.015155892
## condition:phase    0.5861767 0.005308307
```

```
effectsize::eta_squared(aov_cond_phase_DIST, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
##
## Parameter      | Eta2 (partial) |      95% CI
## -----
## condition      |          0.40 | [0.11, 1.00]
## phase          |          0.15 | [0.00, 1.00]
## condition:phase |          0.48 | [0.20, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

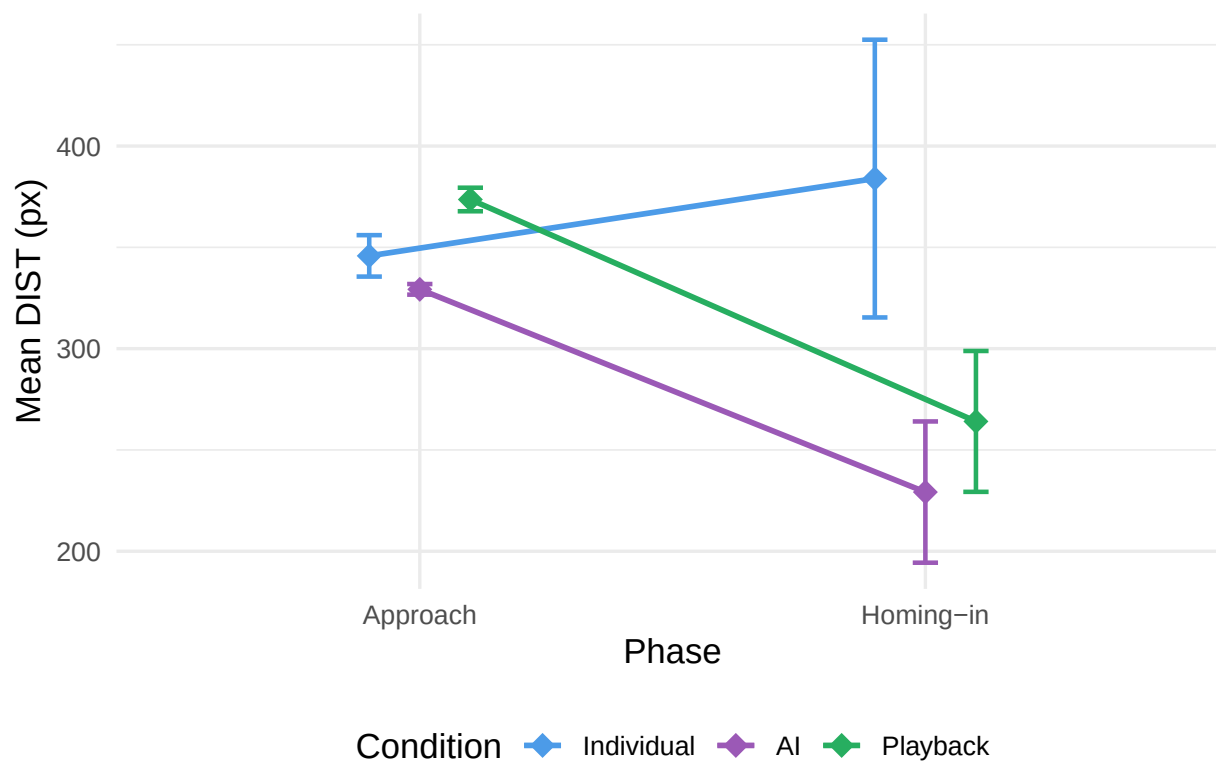
```
for (ph in levels(cond_phase_DIST$phase)) {
  cat("\n Phase:", ph, "\n")
  d <- cond_phase_DIST %>% filter(phase == ph)
  print(pairwise.t.test(
    x = d$mean_DIST,
    g = d$condition,
    paired = TRUE,
    p.adjust.method = "bonferroni"
  ))
}
```

```
##
## Phase: Approach
##
## Pairwise comparisons using paired t tests
##
## data: d$mean_DIST and d$condition
##
```

```
##           individual ai
## ai       0.56585      -
## playback 0.07297      0.00023
##
## P value adjustment method: bonferroni
##
## Phase: Homing-in
##
## Pairwise comparisons using paired t tests
##
## data: d$mean_DIST and d$condition
##
##           individual ai
## ai       0.011      -
## playback 0.122      0.143
##
## P value adjustment method: bonferroni
```

```
cond_phase_DIST %>%
  ggplot(aes(x = phase, y = mean_DIST, colour = condition, group = condition)) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
               position = position_dodge(0.3)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1,
               position = position_dodge(0.3)) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.15,
               position = position_dodge(0.3), linewidth = 0.8) +
  scale_colour_manual(
    values = c("individual" = "#4C9BE8", "ai" = "#9B59B6", "playback" = "#27AE60"),
    labels = c("Individual", "AI", "Playback")
  ) +
  labs(
    title = "Mean DIST by Condition × Phase",
    x = "Phase", y = "Mean DIST (px)", colour = "Condition"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```


Mean DIST by Condition × Phase



```
#Condition-contribution analysis (phase x contribution angle ANOVAs)
run_contribution_anova <- function(cond_label, dv_col, dv_name) {

  d <- analysis_data %>%
    filter(condition == cond_label,
           contribution_angle %in% c(0, 22.5, 45, 67.5, 90)) %>%
    group_by(participant_id, phase, contribution_angle) %>%
    summarise(dv = mean(.data[[dv_col]], na.rm = TRUE), .groups = "drop") %>%
    mutate(
      phase = factor(phase),
      contribution_angle = factor(contribution_angle)
    )

  cat("\n", cond_label, "-", dv_name, "Phase × Contribution Angle ANOVA", "\n")

  anova_out <- ezANOVA(
    data = d,
    dv = dv,
    wid = participant_id,
    within = .(phase, contribution_angle),
    type = 3,
    detailed = TRUE,
    return_aov = TRUE
  )
  print(anova_out)
  print(effectsize::eta_squared(anova_out$aov, partial = TRUE))
}
```

```

#pairwise comparisons of contribution angle (collapsed over phase)
cat("\n Pairwise angle comparisons (Bonferroni) \n")
pw_data <- d %>%
  group_by(participant_id, contribution_angle) %>%
  summarise(mean_dv = mean(dv, na.rm = TRUE), .groups = "drop")

pw <- pairwise.t.test(
  x = pw_data$mean_dv,
  g = pw_data$contribution_angle,
  paired = TRUE,
  p.adjust.method = "bonferroni"
)
print(pw)

#plot
p <- d %>%
  mutate(
    contribution_angle = factor(paste0(contribution_angle, "°"),
                                levels = c("0°", "22.5°", "45°", "67.5°", "90°"))
  ) %>%
  ggplot(aes(x = contribution_angle, y = dv, colour = phase, group = phase)) +
  geom_jitter(width = 0.1, alpha = 0.4, size = 2) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
               position = position_dodge(0.3)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1,
               position = position_dodge(0.3)) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
               position = position_dodge(0.3), linewidth = 0.8) +
  scale_colour_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  labs(
    title = paste(cond_label, "-", dv_name, "by Phase × Contribution Angle"),
    x = "Contribution angle", y = dv_name, colour = "Phase"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold"))

print(p)
invisible(list(anova = anova_out, pairwise = pw))
}

#RT
res_ai_RT <- run_contribution_anova("ai", "RT", "Mean RT (s)")

```

```

##
##      ai - Mean RT (s) Phase × Contribution Angle ANOVA

## Warning: Converting "participant_id" to factor for ANOVA.

```

```

## $ANOVA
##           Effect DFn DFd      SSn      SSd      F      p
## 1 (Intercept)    1   11 426.890726 19.744159 237.832267 8.508324e-09
## 2      phase     1   11  78.507911   6.630915 130.236486 1.946085e-07

```

```

## 3      contribution_angle  4 44  4.295211 23.161526  2.039905 1.052299e-01
## 4 phase:contribution_angle  4 44  4.002802 25.396727  1.733720 1.596527e-01
##    p<.05      ges
## 1      * 0.85067809
## 2      * 0.51164806
## 3      0.05421293
## 4      0.05070937
##
## $`Mauchly's Test for Sphericity`
##              Effect              W              p p<.05
## 3      contribution_angle 0.40178377 0.48366234
## 4 phase:contribution_angle 0.09465847 0.00926161      *
##
## $`Sphericity Corrections`
##              Effect      GGe      p[GG] p[GG]<.05      HFe      p[HF]
## 3      contribution_angle 0.6409671 0.1384067      0.8524767 0.1177325
## 4 phase:contribution_angle 0.6138337 0.1904800      0.8035427 0.1748325
##    p[HF]<.05
## 3
## 4
##
## $aov
##
## Call:
## aov(formula = formula(aov_formula), data = data)
##
## Grand Mean: 1.886113
##
## Stratum 1: participant_id
##
## Terms:
##              Residuals
## Sum of Squares  19.74416
## Deg. of Freedom    11
##
## Residual standard error: 1.339748
##
## Stratum 2: participant_id:phase
##
## Terms:
##              phase Residuals
## Sum of Squares 78.50791  6.63091
## Deg. of Freedom    1      11
##
## Residual standard error: 0.7764087
## 4 out of 5 effects not estimable
## Estimated effects are balanced
##
## Stratum 3: participant_id:contribution_angle
##
## Terms:
##              contribution_angle Residuals
## Sum of Squares      4.295211 23.161526
## Deg. of Freedom      4      44

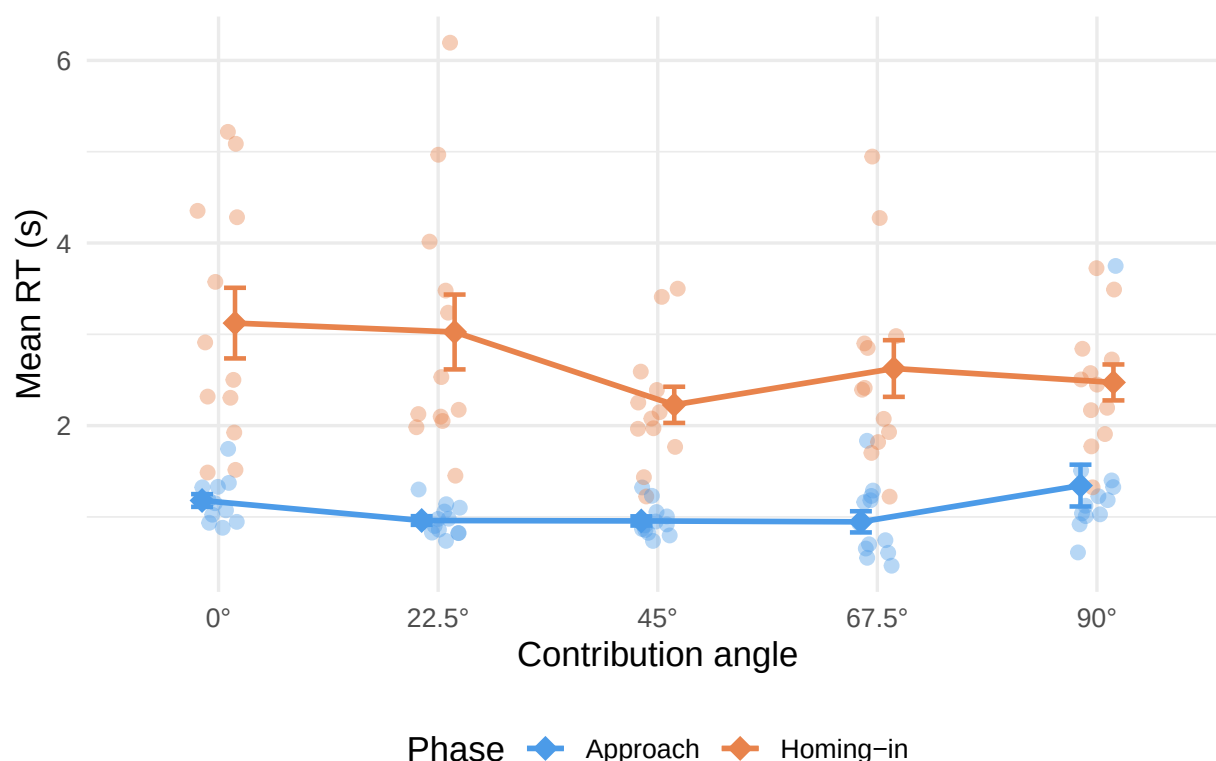
```

```

##
## Residual standard error: 0.7255331
## 4 out of 8 effects not estimable
## Estimated effects may be unbalanced
##
## Stratum 4: participant_id:phase:contribution_angle
##
## Terms:
##               phase:contribution_angle Residuals
## Sum of Squares           4.002802 25.396727
## Deg. of Freedom           4         44
##
## Residual standard error: 0.7597357
## Estimated effects may be unbalanced
##
## # Effect Size for ANOVA (Type I)
##
## Group | Parameter
## -----|-----
## participant_id:phase | phase
## participant_id:contribution_angle | contribution_angle
## participant_id:phase:contribution_angle | phase:contribution_angle
##
## Group | Eta2 (partial) | 95% CI
## -----|-----|-----
## participant_id:phase | 0.92 | [0.82, 1.00]
## participant_id:contribution_angle | 0.16 | [0.00, 1.00]
## participant_id:phase:contribution_angle | 0.14 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
## Pairwise angle comparisons (Bonferroni)
##
## Pairwise comparisons using paired t tests
##
## data: pw_data$mean_dv and pw_data$contribution_angle
##
##      0      22.5 45      67.5
## 22.5 1.00 - - -
## 45   0.35 0.61 - -
## 67.5 0.70 1.00 1.00 -
## 90   1.00 1.00 0.67 1.00
##
## P value adjustment method: bonferroni

```

ai – Mean RT (s) by Phase × Contribution Angle



```
res_pb_RT <- run_contribution_anova("playback", "RT", "Mean RT (s)")
```

```
##
## playback - Mean RT (s) Phase × Contribution Angle ANOVA
```

```
## Warning: Converting "participant_id" to factor for ANOVA.
```

```
## $ANOVA
##           Effect DFn DFd      SSn      SSd      F
## 1      (Intercept)   1  11 1147.214711 49.25251 256.2176232
## 2             phase   1  11  105.255530 11.26387  102.7898112
## 3  contribution_angle   4  44   2.950942 44.49268   0.7295664
## 4 phase:contribution_angle   4  44   1.566071 40.37201   0.4267011
##           p p<.05      ges
## 1 5.741062e-09      * 0.88752782
## 2 6.442281e-07      * 0.41995276
## 3 5.766160e-01      0.01989417
## 4 7.885283e-01      0.01065738
##
## $`Mauchly's Test for Sphericity`
##           Effect      W      p p<.05
## 3  contribution_angle 0.5097549 0.7100715
## 4 phase:contribution_angle 0.4508640 0.5916109
##
## $`Sphericity Corrections`
```

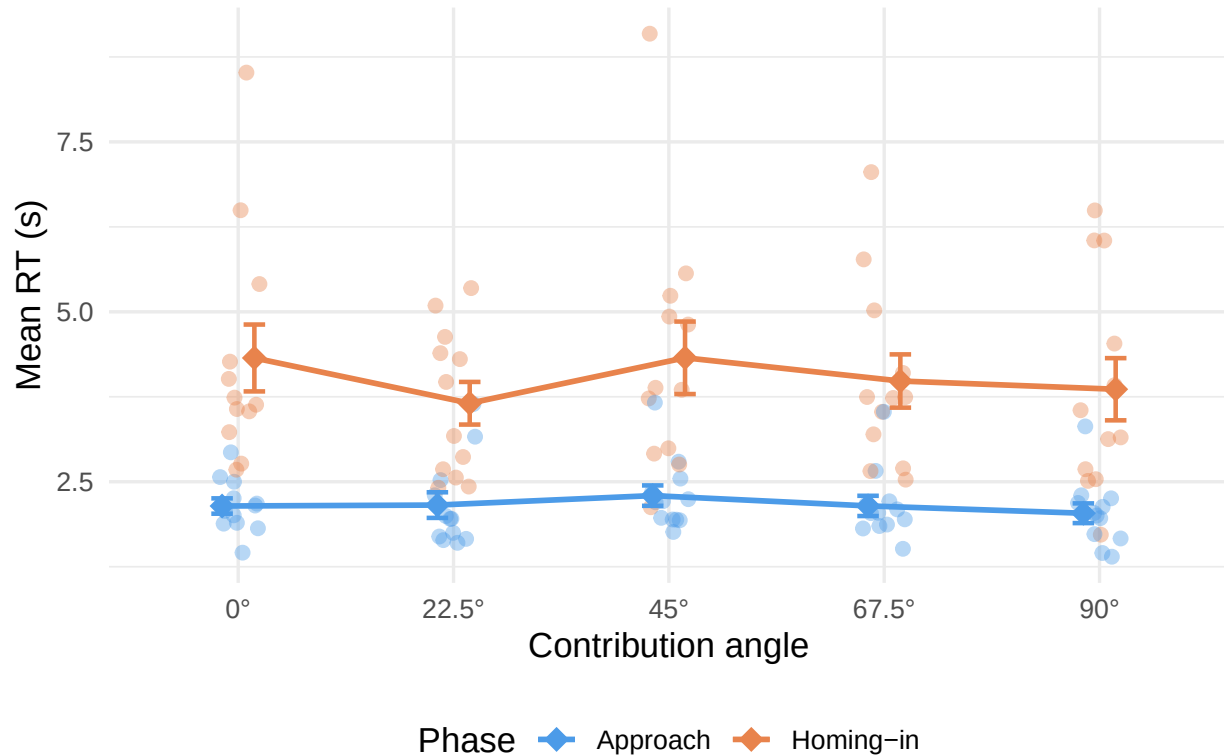
```

##               Effect      GGe      p[GG] p[GG]<.05      HFe      p[HF]
## 3      contribution_angle 0.7357693 0.5394219      1.033798 0.5766160
## 4 phase:contribution_angle 0.7200258 0.7274645      1.002514 0.7885283
##      p[HF]<.05
## 3
## 4
##
## $aov
##
## Call:
## aov(formula = formula(aov_formula), data = data)
##
## Grand Mean: 3.091945
##
## Stratum 1: participant_id
##
## Terms:
##              Residuals
## Sum of Squares   49.25251
## Deg. of Freedom      11
##
## Residual standard error: 2.116011
##
## Stratum 2: participant_id:phase
##
## Terms:
##              phase Residuals
## Sum of Squares 105.25553 11.26387
## Deg. of Freedom      1      11
##
## Residual standard error: 1.011923
## 4 out of 5 effects not estimable
## Estimated effects are balanced
##
## Stratum 3: participant_id:contribution_angle
##
## Terms:
##              contribution_angle Residuals
## Sum of Squares      2.95094 44.49268
## Deg. of Freedom      4      44
##
## Residual standard error: 1.005583
## 4 out of 8 effects not estimable
## Estimated effects may be unbalanced
##
## Stratum 4: participant_id:phase:contribution_angle
##
## Terms:
##              phase:contribution_angle Residuals
## Sum of Squares      1.56607 40.37201
## Deg. of Freedom      4      44
##
## Residual standard error: 0.957886
## Estimated effects may be unbalanced

```

```
##
## # Effect Size for ANOVA (Type I)
##
## Group | Parameter
## -----
## participant_id:phase | phase
## participant_id:contribution_angle | contribution_angle
## participant_id:phase:contribution_angle | phase:contribution_angle
##
## Group | Eta2 (partial) | 95% CI
## -----
## participant_id:phase | 0.90 | [0.77, 1.00]
## participant_id:contribution_angle | 0.06 | [0.00, 1.00]
## participant_id:phase:contribution_angle | 0.04 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
## Pairwise angle comparisons (Bonferroni)
##
## Pairwise comparisons using paired t tests
##
## data: pw_data$mean_dv and pw_data$contribution_angle
##
## 0 22.5 45 67.5
## 22.5 1 - - -
## 45 1 1 - -
## 67.5 1 1 1 -
## 90 1 1 1 1
##
## P value adjustment method: bonferroni
```

playback – Mean RT (s) by Phase × Contribution Angle



```
#DIST
res_ai_DIST <- run_contribution_anova("ai", "DIST", "Mean DIST (px)")

##
## ai - Mean DIST (px) Phase × Contribution Angle ANOVA

## Warning: Converting "participant_id" to factor for ANOVA.

## $ANOVA
##          Effect DFn DFd      SSn      SSd      F      p
## 1 (Intercept)    1  11 9356403.94 392568.1 262.172203 5.083683e-09
## 2      phase      1  11  300442.91 414261.5   7.977744 1.653433e-02
## 3 contribution_angle 4  44   36003.43 341014.3   1.161352 3.408202e-01
## 4 phase:contribution_angle 4  44   55375.61 326546.1   1.865377 1.335178e-01
## p<.05 ges
## 1 * 0.86387056
## 2 * 0.16927955
## 3 0.02383712
## 4 0.03619875
##
## $`Mauchly's Test for Sphericity`
##          Effect      W      p p<.05
## 3 contribution_angle 0.08446349 0.006354271 *
## 4 phase:contribution_angle 0.11896252 0.019344396 *
```



```

## `$`Sphericity Corrections`
##           Effect      GGe      p[GG] p[GG]<.05      HFe      p[HF]
## 3      contribution_angle 0.5057233 0.3318283      0.6203201 0.3363945
## 4 phase:contribution_angle 0.5325421 0.1753341      0.6641056 0.1626361
##      p[HF]<.05
## 3
## 4
##
## $aov
##
## Call:
## aov(formula = formula(aov_formula), data = data)
##
## Grand Mean: 279.2311
##
## Stratum 1: participant_id
##
## Terms:
##           Residuals
## Sum of Squares  392568.1
## Deg. of Freedom      11
##
## Residual standard error: 188.9127
##
## Stratum 2: participant_id:phase
##
## Terms:
##           phase Residuals
## Sum of Squares 300442.9 414261.5
## Deg. of Freedom      1      11
##
## Residual standard error: 194.0622
## 4 out of 5 effects not estimable
## Estimated effects are balanced
##
## Stratum 3: participant_id:contribution_angle
##
## Terms:
##           contribution_angle Residuals
## Sum of Squares           36003.4 341014.3
## Deg. of Freedom           4      44
##
## Residual standard error: 88.03593
## 4 out of 8 effects not estimable
## Estimated effects may be unbalanced
##
## Stratum 4: participant_id:phase:contribution_angle
##
## Terms:
##           phase:contribution_angle Residuals
## Sum of Squares           55375.6 326546.1
## Deg. of Freedom           4      44
##
## Residual standard error: 86.14814

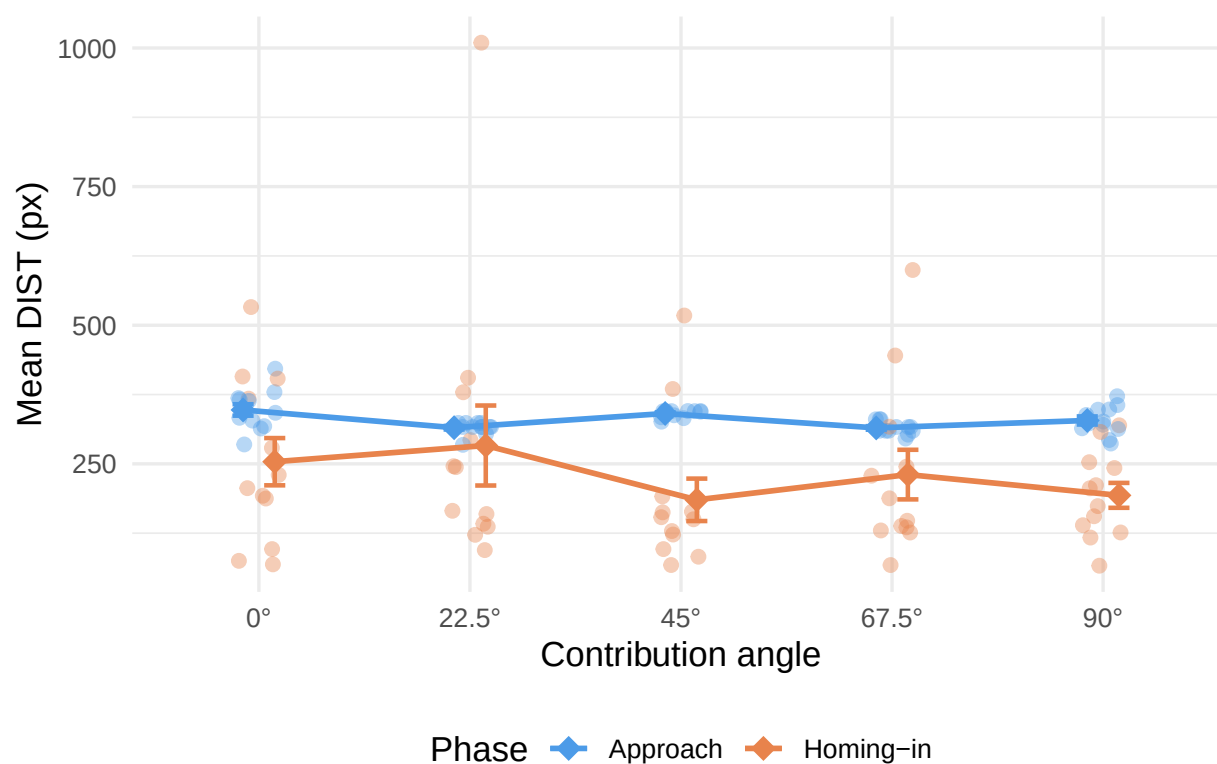
```

```

## Estimated effects may be unbalanced
##
## # Effect Size for ANOVA (Type I)
##
## Group | Parameter
## -----
## participant_id:phase | phase
## participant_id:contribution_angle | contribution_angle
## participant_id:phase:contribution_angle | phase:contribution_angle
##
## Group | Eta2 (partial) | 95% CI
## -----
## participant_id:phase | 0.42 | [0.06, 1.00]
## participant_id:contribution_angle | 0.10 | [0.00, 1.00]
## participant_id:phase:contribution_angle | 0.14 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
## Pairwise angle comparisons (Bonferroni)
##
## Pairwise comparisons using paired t tests
##
## data: pw_data$mean_dv and pw_data$contribution_angle
##
## 0 22.5 45 67.5
## 22.5 1 - - -
## 45 1 1 - -
## 67.5 1 1 1 -
## 90 1 1 1 1
##
## P value adjustment method: bonferroni

```

ai – Mean DIST (px) by Phase × Contribution Angle



```
res_pb_DIST <- run_contribution_anova("playback", "DIST", "Mean DIST (px)")
```

```
##
## playback - Mean DIST (px) Phase × Contribution Angle ANOVA
```

```
## Warning: Converting "participant_id" to factor for ANOVA.
```

```
## $ANOVA
##           Effect DFn DFd      SSn      SSd      F
## 1      (Intercept)   1  11 12200023.63 401539.5 334.2143642
## 2             phase   1  11  360027.31 418576.7   9.4613504
## 3  contribution_angle   4  44  32979.87 423504.0   0.8566120
## 4 phase:contribution_angle   4  44   32877.90 423077.6   0.8548241
##           p p<.05      ges
## 1 1.397964e-09      * 0.87980593
## 2 1.054764e-02      * 0.17763994
## 3 4.974081e-01      0.01940360
## 4 4.984678e-01      0.01934477
##
## $`Mauchly's Test for Sphericity`
##           Effect      W      p p<.05
## 3  contribution_angle 0.4455248 0.5801640
## 4 phase:contribution_angle 0.2717917 0.2055864
##
## $`Sphericity Corrections`
```

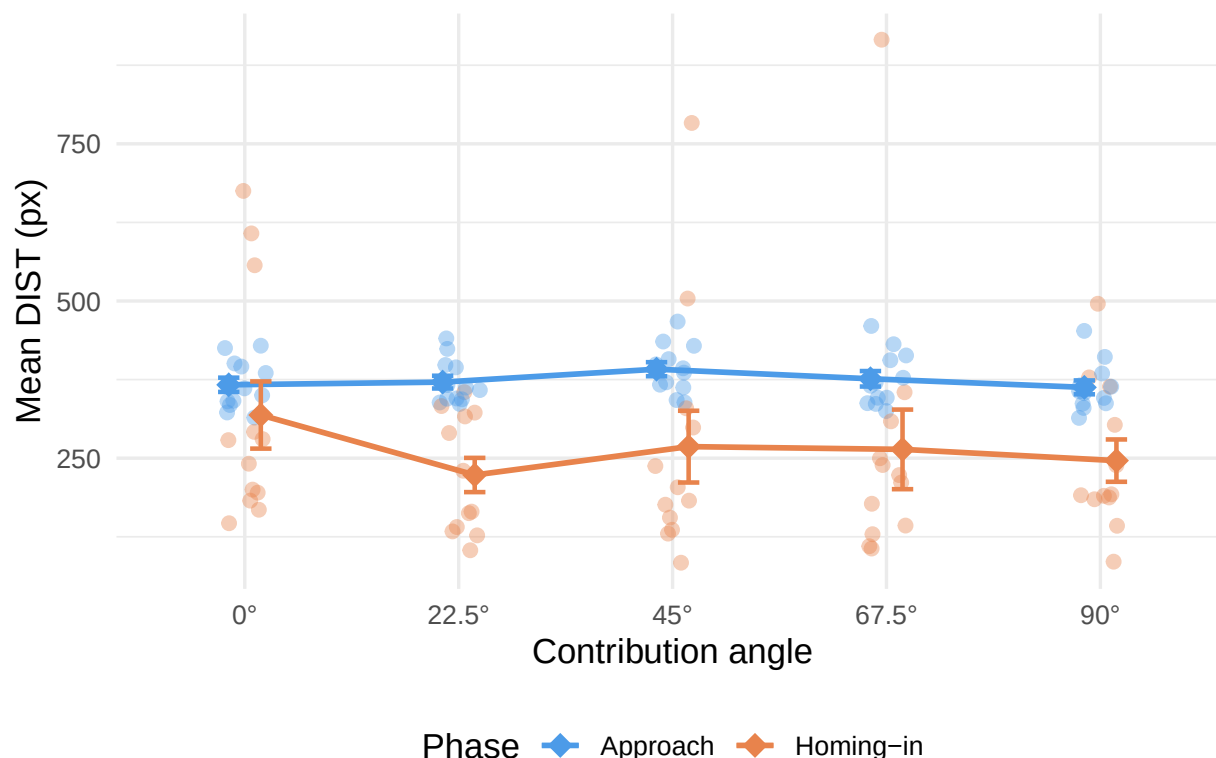
```

##               Effect      GGe      p[GG] p[GG]<.05      HFe      p[HF]
## 3      contribution_angle 0.7179717 0.4695193      0.9984680 0.4972798
## 4 phase:contribution_angle 0.5993962 0.4548796      0.7780087 0.4772494
##      p[HF]<.05
## 3
## 4
##
## $aov
##
## Call:
## aov(formula = formula(aov_formula), data = data)
##
## Grand Mean: 318.8524
##
## Stratum 1: participant_id
##
## Terms:
##              Residuals
## Sum of Squares   401539.5
## Deg. of Freedom      11
##
## Residual standard error: 191.0591
##
## Stratum 2: participant_id:phase
##
## Terms:
##              phase Residuals
## Sum of Squares 360027.3 418576.7
## Deg. of Freedom      1      11
##
## Residual standard error: 195.0703
## 4 out of 5 effects not estimable
## Estimated effects are balanced
##
## Stratum 3: participant_id:contribution_angle
##
## Terms:
##              contribution_angle Residuals
## Sum of Squares           32979.9 423504.0
## Deg. of Freedom           4      44
##
## Residual standard error: 98.10755
## 4 out of 8 effects not estimable
## Estimated effects may be unbalanced
##
## Stratum 4: participant_id:phase:contribution_angle
##
## Terms:
##              phase:contribution_angle Residuals
## Sum of Squares           32877.9 423077.6
## Deg. of Freedom           4      44
##
## Residual standard error: 98.05815
## Estimated effects may be unbalanced

```

```
##
## # Effect Size for ANOVA (Type I)
##
## Group | Parameter
## -----
## participant_id:phase | phase
## participant_id:contribution_angle | contribution_angle
## participant_id:phase:contribution_angle | phase:contribution_angle
##
## Group | Eta2 (partial) | 95% CI
## -----
## participant_id:phase | 0.46 | [0.09, 1.00]
## participant_id:contribution_angle | 0.07 | [0.00, 1.00]
## participant_id:phase:contribution_angle | 0.07 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
## Pairwise angle comparisons (Bonferroni)
##
## Pairwise comparisons using paired t tests
##
## data: pw_data$mean_dv and pw_data$contribution_angle
##
## 0 22.5 45 67.5
## 22.5 1 - - -
## 45 1 1 - -
## 67.5 1 1 1 -
## 90 1 1 1 1
##
## P value adjustment method: bonferroni
```

playback – Mean DIST (px) by Phase × Contribution Angle



#Adaptiveness analysis - adaptive ai vs non-adaptive playback (for RT)

AI vs Playback adaptiveness benefit: RT and DIST

```
ai_vs_pb <- analysis_data %>%
  filter(condition %in% c("ai", "playback")) %>%
  group_by(participant_id, condition, phase, contribution_angle) %>%
  summarise(
    mean_RT = mean(RT, na.rm = TRUE),
    mean_DIST = mean(DIST, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_wider(
    names_from = condition,
    values_from = c(mean_RT, mean_DIST)
  ) %>%
  mutate(
    adaptiveness_RT = mean_RT_playback / mean_RT_ai, # >1 means AI faster
    adaptiveness_DIST = mean_DIST_playback / mean_DIST_ai # >1 means AI shorter
  )
```

One-sample t-tests vs 1 per phase × angle

```
adapt_tests <- ai_vs_pb %>%
  group_by(phase, contribution_angle) %>%
  summarise(
    n = sum(!is.na(adaptiveness_RT)),
    M_RT = mean(adaptiveness_RT, na.rm = TRUE),
```

```

SE_RT      = sd(adaptiveness_RT,      na.rm = TRUE) / sqrt(n),
t_RT       = tryCatch(t.test(adaptiveness_RT, mu = 1)$statistic, error = function(e) NA),
p_RT       = tryCatch(t.test(adaptiveness_RT, mu = 1)$p.value,   error = function(e) NA),
d_RT       = (M_RT - 1) / sd(adaptiveness_RT, na.rm = TRUE),
M_DIST     = mean(adaptiveness_DIST, na.rm = TRUE),
SE_DIST    = sd(adaptiveness_DIST, na.rm = TRUE) / sqrt(n),
t_DIST     = tryCatch(t.test(adaptiveness_DIST, mu = 1)$statistic, error = function(e) NA),
p_DIST     = tryCatch(t.test(adaptiveness_DIST, mu = 1)$p.value,   error = function(e) NA),
d_DIST     = (M_DIST - 1) / sd(adaptiveness_DIST, na.rm = TRUE),
.groups = "drop"
) %>%
mutate(
  sig_RT  = case_when(p_RT < .001 ~ "***", p_RT < .01 ~ "**", p_RT < .05 ~ "*", p_RT < .10 ~ ".", p_RT > .10 ~ NA),
  sig_DIST = case_when(p_DIST < .001 ~ "***", p_DIST < .01 ~ "**", p_DIST < .05 ~ "*", p_DIST < .10 ~ ".", p_DIST > .10 ~ NA)
)

print(adapt_tests)

```

```

## # A tibble: 10 x 15
##   phase contribution_angle      n M_RT SE_RT t_RT p_RT d_RT M_DIST SE_DIST
##   <fct> <fct>              <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Appr~ 0                12  1.86 0.111  7.76 8.76e-6 2.24  1.07 0.0499
## 2 Appr~ 22.5             12  2.22 0.102 11.9  1.26e-7 3.44  1.18 0.0309
## 3 Appr~ 45               12  2.42 0.113 12.5  7.61e-8 3.61  1.15 0.0349
## 4 Appr~ 67.5             12  2.56 0.269  5.82 1.15e-4 1.68  1.20 0.0431
## 5 Appr~ 90               12  1.76 0.193  3.92 2.37e-3 1.13  1.11 0.0499
## 6 Homi~ 0                12  1.55 0.186  2.96 1.30e-2 0.855  1.53 0.271
## 7 Homi~ 22.5             12  1.38 0.162  2.32 4.09e-2 0.668  1.10 0.188
## 8 Homi~ 45               12  2.00 0.206  4.88 4.91e-4 1.41  1.66 0.253
## 9 Homi~ 67.5             12  1.67 0.207  3.21 8.25e-3 0.928  1.28 0.221
## 10 Homi~ 90              12  1.72 0.280  2.58 2.58e-2 0.743  1.53 0.291
## # i 5 more variables: t_DIST <dbl>, p_DIST <dbl>, d_DIST <dbl>, sig_RT <chr>,
## #   sig_DIST <chr>

```

```

# Plot
adapt_long <- ai_vs_pb %>%
  pivot_longer(
    cols      = c(adaptiveness_RT, adaptiveness_DIST),
    names_to  = "dv",
    values_to = "benefit"
  ) %>%
  mutate(
    dv = factor(dv,
      levels = c("adaptiveness_RT", "adaptiveness_DIST"),
      labels = c("RT ratio", "DIST ratio")),
    contribution_angle = factor(paste0(contribution_angle, "°"),
      levels = c("0°", "22.5°", "45°", "67.5°", "90°"))
  )

adapt_tests_long <- adapt_tests %>%
  mutate(contribution_angle = factor(paste0(contribution_angle, "°"),
    levels = c("0°", "22.5°", "45°", "67.5°", "90°"))) %>%
  pivot_longer(

```

```

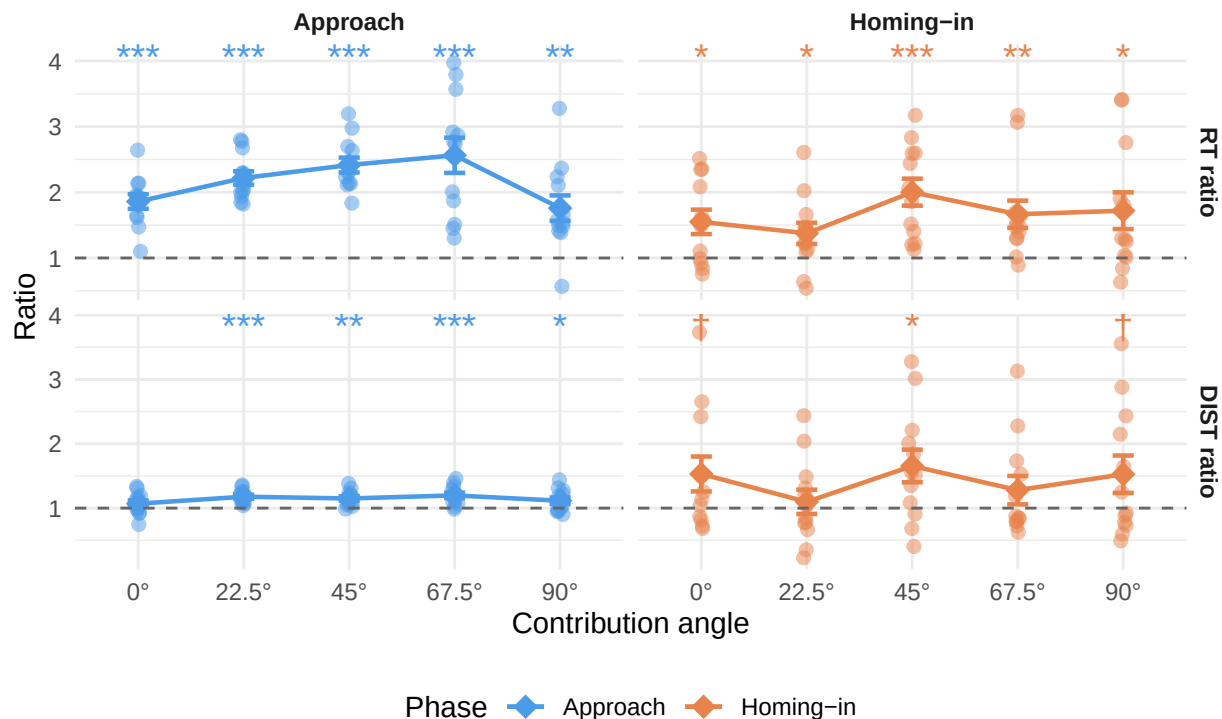
cols      = c(sig_RT, sig_DIST),
names_to  = "dv",
values_to = "sig"
) %>%
mutate(
  dv      = factor(dv,
                    levels = c("sig_RT", "sig_DIST"),
                    labels = c("RT ratio", "DIST ratio")),
  y_pos   = ifelse(dv == "RT ratio",
                    max(ai_vs_pb$adaptiveness_RT, na.rm = TRUE) + 0.1,
                    max(ai_vs_pb$adaptiveness_DIST, na.rm = TRUE) + 0.1)
)

ggplot(adapt_long, aes(x = contribution_angle, y = benefit, colour = phase)) +
  geom_jitter(alpha = 0.5, size = 2,
              position = position_jitterdodge(jitter.width = 0.1, dodge.width = 0.4)) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
              position = position_dodge(0.4)) +
  stat_summary(fun = mean, geom = "line", aes(group = phase),
              position = position_dodge(0.4), linewidth = 0.8) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
              position = position_dodge(0.4), linewidth = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
  geom_text(data = adapt_tests_long,
            aes(x = contribution_angle, y = y_pos, label = sig, colour = phase),
            position = position_dodge(0.4), size = 5, show.legend = FALSE) +
  facet_grid(dv ~ phase, scales = "free_y") +
  scale_colour_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  labs(
    title      = "Adaptiveness benefit: Playback / AI",
    subtitle   = "Dashed line = no benefit | >1 = AI outperforms playback",
    x          = "Contribution angle", y = "Ratio", colour = "Phase"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold"))

```


Adaptiveness benefit: Playback / AI

Dashed line = no benefit | >1 = AI outperforms playback



Interpretation: The results of the condition-contribution analysis and the adaptiveness analysis look very similar but there are key differences:

- Condition-contribution analysis: For each condition (AI, playback) separately, it just takes the raw mean RT/DIST at each contribution angle and phase, runs an ANOVA to see if angle and phase affect performance, and plots those means. It describes the shape of performance across angles within each condition.
- Adaptiveness benefit: Takes the ratio of playback/AI at each matched angle and phase within the same participant, so it's a direct within-person measure of what the AI's adaptiveness buys you over a non-adaptive partner doing the exact same movements. Values above 1 mean the AI condition was faster/shorter than playback.
- Human cooperative mode had to be taken out here because their data is grouped (it belongs to 2 people instead of just one) and therefore cannot be added to the analysis here

```
#for AI and playback: average the two dyad members per dyad as baseline
#for cooperative: best dyad member is used as baseline
cb_ai_dyad_RT <- analysis_data %>%
  filter(condition == "ai") %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  rename(dyad_id = dyad_id.y) %>%
  filter(!is.na(dyad_id)) %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(part_RT = mean(RT, na.rm = TRUE), .groups = "drop") %>%
  left_join(best_individual_RT, by = c("dyad_id", "phase", "contribution_angle")) %>%
  mutate(collective_benefit = best_individual_RT / part_RT,
```

```

    partner_type = "adaptive_ai")

cb_pb_dyad_RT <- analysis_data %>%
  filter(condition == "playback") %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  rename(dyad_id = dyad_id.y) %>%
  filter(!is.na(dyad_id)) %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(part_RT = mean(RT, na.rm = TRUE), .groups = "drop") %>%
  left_join(best_individual_RT, by = c("dyad_id", "phase", "contribution_angle")) %>%
  mutate(collective_benefit = best_individual_RT / part_RT,
         partner_type = "nonadaptive_playback")

cb_coop_dyad_RT <- collective_benefit_RT %>%
  mutate(partner_type = "adaptive_human")

#bind all three - now all at dyad level, n = 6 each
cb_all_dyad_RT <- bind_rows(
  cb_coop_dyad_RT %>% select(dyad_id, phase, contribution_angle, collective_benefit, partner_type),
  cb_ai_dyad_RT,
  cb_pb_dyad_RT
) %>%
  mutate(
    partner_type = factor(partner_type,
                          levels = c("adaptive_human", "adaptive_ai", "nonadaptive_playback")),
    contribution_angle = factor(paste0(contribution_angle, "°"),
                                levels = c("0°", "22.5°", "45°", "67.5°", "90°"))
  )

#pairwise comparisons per phase (paired by dyad)
for (ph in levels(cb_all_dyad_RT$phase)) {
  cat("\n Phase:", ph, "\n")
  d <- cb_all_dyad_RT %>% filter(phase == ph) %>%
    group_by(dyad_id, partner_type) %>%
    summarise(mean_cb = mean(collective_benefit, na.rm = TRUE), .groups = "drop")
  print(pairwise.t.test(
    x = d$mean_cb,
    g = d$partner_type,
    paired = TRUE,
    p.adjust.method = "bonferroni"
  ))
}

```

```

##
## Phase: Approach
##
## Pairwise comparisons using paired t tests
##
## data: d$mean_cb and d$partner_type
##
##               adaptive_human adaptive_ai
## adaptive_ai      0.00902      -
## nonadaptive_playback 0.00072      0.00011

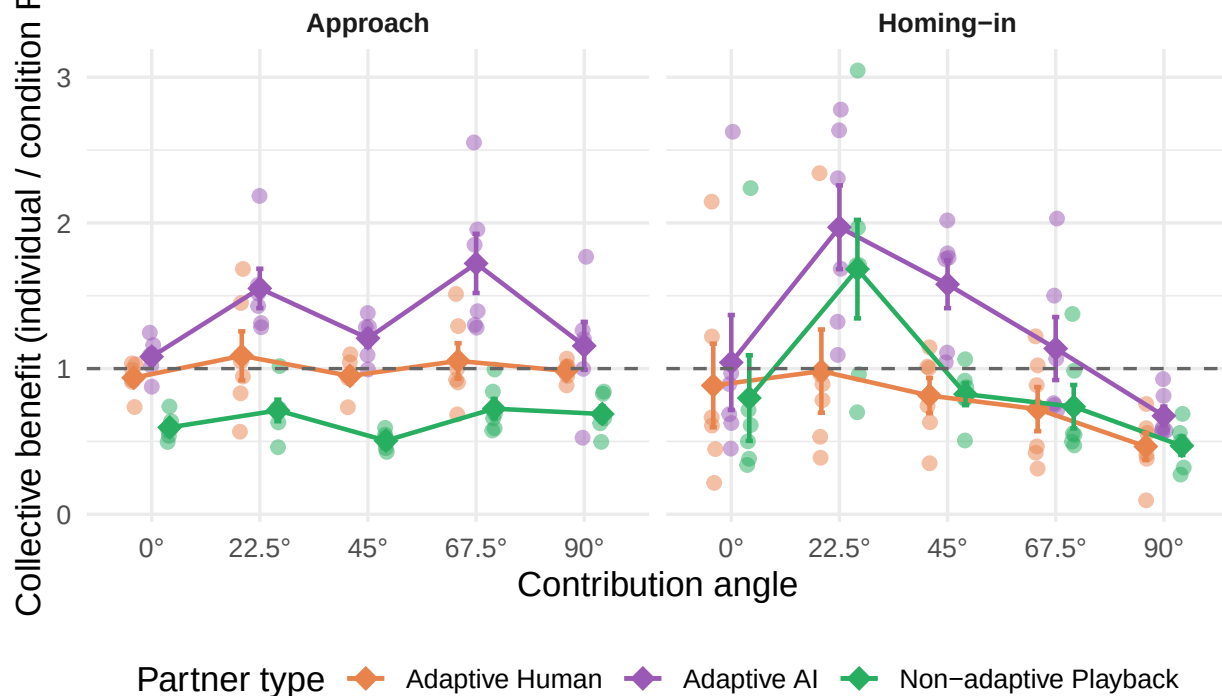
```

```
##
## P value adjustment method: bonferroni
##
## Phase: Homing-in
##
## Pairwise comparisons using paired t tests
##
## data: d$mean_cb and d$partner_type
##
##               adaptive_human adaptive_ai
## adaptive_ai      0.0045      -
## nonadaptive_playback 0.4874      0.0059
##
## P value adjustment method: bonferroni
```

```
#plot
ggplot(cb_all_dyad_RT,
  aes(x = contribution_angle, y = collective_benefit, colour = partner_type)) +
  geom_jitter(alpha = 0.5, size = 2,
    position = position_jitterdodge(jitter.width = 0.1, dodge.width = 0.5)) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
    position = position_dodge(0.5)) +
  stat_summary(fun = mean, geom = "line", aes(group = partner_type),
    position = position_dodge(0.5), linewidth = 0.8) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
    position = position_dodge(0.5), linewidth = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
  facet_wrap(~ phase) +
  scale_colour_manual(
    values = c("adaptive_human" = "#E8834C",
      "adaptive_ai" = "#9B59B6",
      "nonadaptive_playback" = "#27AE60"),
    labels = c("Adaptive Human", "Adaptive AI", "Non-adaptive Playback")
  ) +
  labs(
    title = "Collective benefit (RT) by partner type",
    subtitle = "Dyad-level comparison, n = 6 | Dashed line = no benefit over individual",
    x = "Contribution angle", y = "Collective benefit (individual / condition RT)",
    colour = "Partner type"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold"))
```

Collective benefit (RT) by partner type

Dyad-level comparison, n = 6 | Dashed line = no benefit over individual



Now all three conditions are at dyad level with the same $n = 6$, so the paired t-tests are valid — each dyad contributes one observation per condition. Flag in the write-up that the cooperative collective benefit uses the better of the two members as the individual baseline, while AI and playback average across both members, which is a slight asymmetry.

We use the best individual performance of the two dyad members as the baseline for all three conditions, we accept the small asymmetry in the write-up, and note that a human-contribution-only reconstruction would require frame-level simulation beyond the scope of the current analysis.

```
#doing the same for DIST
#for AI and playback: average the two dyad members per dyad as baseline
#for cooperative: best dyad member is used as baseline
cb_ai_dyad_DIST <- analysis_data %>%
  filter(condition == "ai") %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  rename(dyad_id = dyad_id.y) %>%
  filter(!is.na(dyad_id)) %>%
  group_by(dyad_id, phase, contribution_angle) %>%
  summarise(part_DIST = mean(DIST, na.rm = TRUE), .groups = "drop") %>%
  left_join(best_individual_DIST, by = c("dyad_id", "phase", "contribution_angle")) %>%
  mutate(collective_benefit = best_individual_DIST / part_DIST,
         partner_type = "adaptive_ai")

cb_pb_dyad_DIST <- analysis_data %>%
  filter(condition == "playback") %>%
  left_join(participant_dyad_map, by = "participant_id") %>%
  rename(dyad_id = dyad_id.y) %>%

```

```

filter(!is.na(dyad_id)) %>%
group_by(dyad_id, phase, contribution_angle) %>%
summarise(part_DIST = mean(DIST, na.rm = TRUE), .groups = "drop") %>%
left_join(best_individual_DIST, by = c("dyad_id", "phase", "contribution_angle")) %>%
mutate(collective_benefit = best_individual_DIST / part_DIST,
       partner_type = "nonadaptive_playback")

cb_coop_dyad_DIST <- collective_benefit_DIST %>%
mutate(partner_type = "adaptive_human")

#bind all three - now all at dyad level, n = 6 each
cb_all_dyad_DIST <- bind_rows(
  cb_coop_dyad_DIST %>% select(dyad_id, phase, contribution_angle, collective_benefit, partner_type),
  cb_ai_dyad_DIST,
  cb_pb_dyad_DIST
) %>%
mutate(
  partner_type = factor(partner_type,
                        levels = c("adaptive_human", "adaptive_ai", "nonadaptive_playback")),
  contribution_angle = factor(paste0(contribution_angle, "°"),
                              levels = c("0°", "22.5°", "45°", "67.5°", "90°"))
)

#pairwise comparisons per phase (paired by dyad)
for (ph in levels(cb_all_dyad_DIST$phase)) {
  cat("\n Phase:", ph, "\n")
  d <- cb_all_dyad_DIST %>% filter(phase == ph) %>%
    group_by(dyad_id, partner_type) %>%
    summarise(mean_cb = mean(collective_benefit, na.rm = TRUE), .groups = "drop")
  print(pairwise.t.test(
    x = d$mean_cb,
    g = d$partner_type,
    paired = TRUE,
    p.adjust.method = "bonferroni"
  ))
}

```

```

##
## Phase: Approach
##
## Pairwise comparisons using paired t tests
##
## data: d$mean_cb and d$partner_type
##
##               adaptive_human adaptive_ai
## adaptive_ai      0.1794      -
## nonadaptive_playback 0.0628      0.0066
##
## P value adjustment method: bonferroni
##
## Phase: Homing-in
##
## Pairwise comparisons using paired t tests

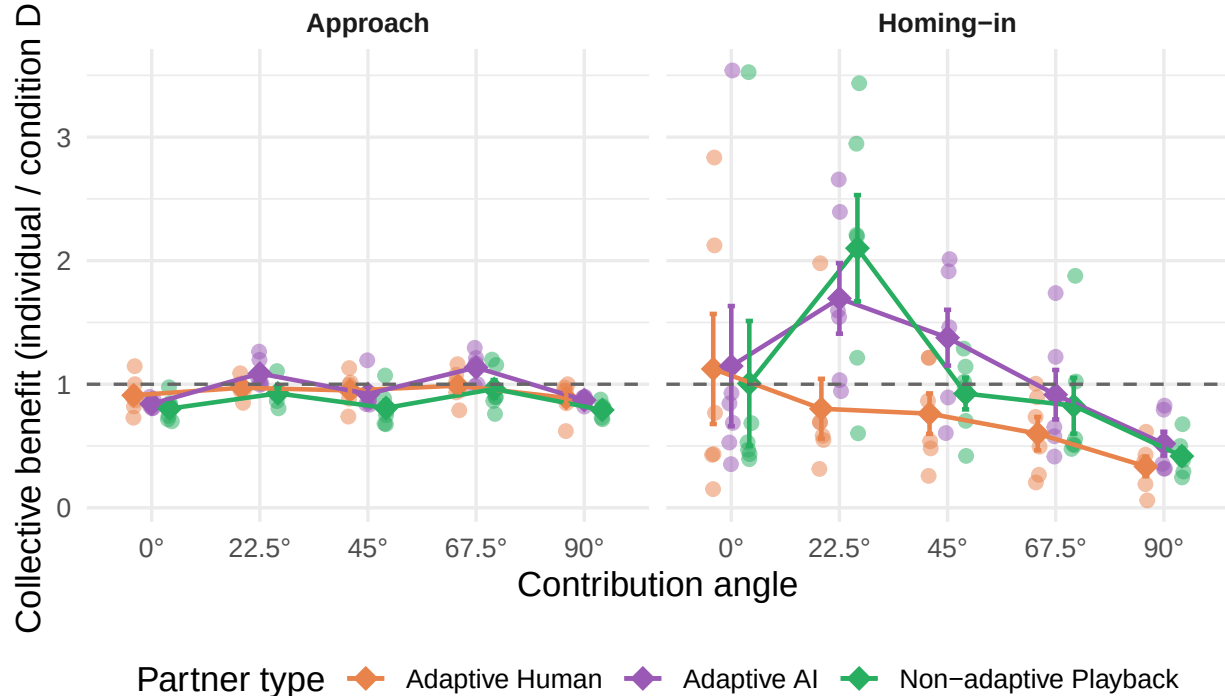
```

```
##
## data: d$mean_cb and d$partner_type
##
##               adaptive_human adaptive_ai
## adaptive_ai      0.20          -
## nonadaptive_playback 0.16          1.00
##
## P value adjustment method: bonferroni
```

```
#plot
ggplot(cb_all_dyad_DIST,
  aes(x = contribution_angle, y = collective_benefit, colour = partner_type)) +
  geom_jitter(alpha = 0.5, size = 2,
    position = position_jitterdodge(jitter.width = 0.1, dodge.width = 0.5)) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18,
    position = position_dodge(0.5)) +
  stat_summary(fun = mean, geom = "line", aes(group = partner_type),
    position = position_dodge(0.5), linewidth = 0.8) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2,
    position = position_dodge(0.5), linewidth = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey40") +
  facet_wrap(~ phase) +
  scale_colour_manual(
    values = c("adaptive_human" = "#E8834C",
      "adaptive_ai" = "#9B59B6",
      "nonadaptive_playback" = "#27AE60"),
    labels = c("Adaptive Human", "Adaptive AI", "Non-adaptive Playback")
  ) +
  labs(
    title = "Collective benefit (DIST) by partner type",
    subtitle = "Dyad-level comparison, n = 6 | Dashed line = no benefit over individual",
    x = "Contribution angle", y = "Collective benefit (individual / condition DIST)",
    colour = "Partner type"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold"))
```

Collective benefit (DIST) by partner type

Dyad-level comparison, n = 6 | Dashed line = no benefit over individual



Within the AI and playback conditions separately, and within each phase, does a participant's average reaction time depend on whether their (artificial) partner was assigned the horizontal or the vertical axis?

```
#Axis-control analysis RT
#join frame_clean axis_control variable to summary trial data

axis_trial <- frame_clean %>%
  mutate(mode = tolower(mode)) %>%
  filter(mode %in% c("ai", "playback")) %>%
  group_by(participant_id, mode, iteration, phase) %>%
  summarise(axis_control = first(na.omit(axis_control)), .groups = "drop") %>%
  mutate(phase = str_replace_all(tolower(phase), "[a-z0-9]+", "_"))

# analysis_data side: identical normalization before the join
analysis_axis <- analysis_data %>%
  filter(condition %in% c("ai", "playback")) %>%
  mutate(
    condition = as.character(condition),
    phase = str_replace_all(tolower(as.character(phase)), "[a-z0-9]+", "_")
  ) %>%
  left_join(axis_trial %>% rename(condition = mode),
    by = c("participant_id", "condition", "iteration", "phase")) %>%
  filter(!is.na(axis_control)) %>%
  mutate(
    partner_axis = case_when(
      str_detect(axis_control, regex("horiz", ignore_case = TRUE)) ~ "horizontal",
```

```

    str_detect(axis_control, regex("vert", ignore_case = TRUE)) ~ "vertical",
    TRUE ~ NA_character_
  ),
  partner_axis = factor(partner_axis, levels = c("horizontal", "vertical"))
) %>%
filter(!is.na(partner_axis))

#participant-level means by condition × phase × axis
axis_means_RT <- analysis_axis %>%
  group_by(participant_id, condition, phase, partner_axis) %>%
  summarise(mean_RT = mean(RT, na.rm = TRUE), .groups = "drop")

#paired t-test + Cohen's d: horizontal vs vertical within each condition × phase
axis_tests_RT <- axis_means_RT %>%
  pivot_wider(names_from = partner_axis, values_from = mean_RT) %>%
  group_by(condition, phase) %>%
  summarise(
    n          = sum(!is.na(horizontal) & !is.na(vertical)),
    M_horiz    = mean(horizontal, na.rm = TRUE),
    M_vert     = mean(vertical, na.rm = TRUE),
    mean_diff  = mean(vertical - horizontal, na.rm = TRUE),
    t_stat     = tryCatch(t.test(vertical[!is.na(vertical) & !is.na(horizontal)],
                                horizontal[!is.na(vertical) & !is.na(horizontal)],
                                paired = TRUE)$statistic, error = function(e) NA),
    p_value    = tryCatch(t.test(vertical[!is.na(vertical) & !is.na(horizontal)],
                                horizontal[!is.na(vertical) & !is.na(horizontal)],
                                paired = TRUE)$p.value, error = function(e) NA),

    cohens_d   = tryCatch(
      mean(vertical - horizontal, na.rm = TRUE) /
      sd(vertical - horizontal, na.rm = TRUE),
      error = function(e) NA
    ),
    .groups = "drop"
  ) %>%
mutate(
  sig = case_when(
    p_value < .001 ~ "***",
    p_value < .01  ~ "**",
    p_value < .05  ~ "*",
    p_value < .10  ~ "†",
    TRUE          ~ ""
  )
)

cat("  Axis-control tests (RT)  \n")

```

```
##  Axis-control tests (RT)
```

```
print(axis_tests_RT)
```

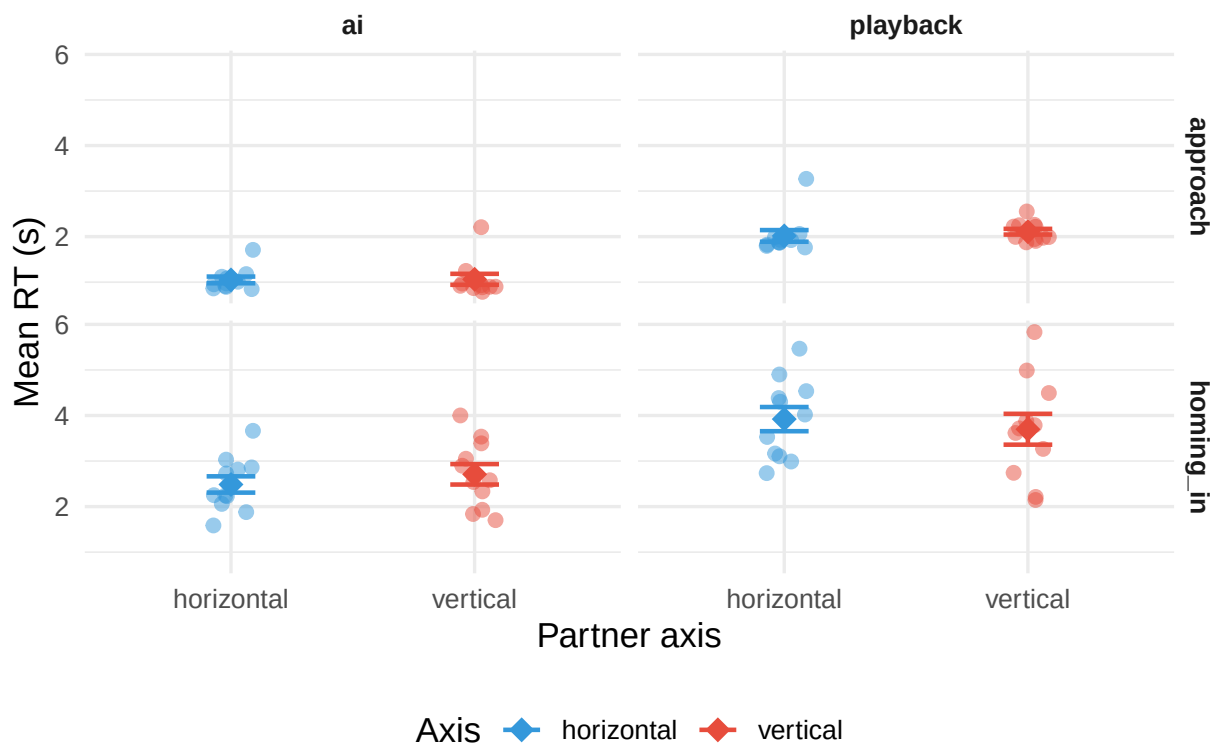
```
## # A tibble: 4 x 10
##   condition phase      n M_horiz M_vert mean_diff t_stat p_value cohens_d sig
##   <chr>      <chr> <int>  <dbl>  <dbl>      <dbl>  <dbl>  <dbl>  <dbl> <chr>
```



```
## 1 ai      approa~    11    1.05    1.06    0.0134    0.215    0.834    0.0648    ""
## 2 ai      homing~    11    2.49    2.71    0.223    0.939    0.370    0.283    ""
## 3 playback approa~    11    2.02    2.11    0.0920    0.740    0.476    0.223    ""
## 4 playback homing~    11    3.92    3.70   -0.223   -0.576    0.578   -0.174    ""
```

```
#plot
axis_means_RT %>%
  ggplot(aes(x = partner_axis, y = mean_RT, colour = partner_axis)) +
  geom_jitter(width = 0.1, alpha = 0.5, size = 2) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2, linewidth = 0.8) +
  facet_grid(phase ~ condition) +
  scale_colour_manual(values = c("horizontal" = "#3498DB", "vertical" = "#E74C3C")) +
  labs(
    title = "RT by Axis Control (Horizontal vs Vertical)",
    x = "Partner axis", y = "Mean RT (s)", colour = "Axis"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold"))
```

RT by Axis Control (Horizontal vs Vertical)



```
#Axis-control analysis DIST
#participant-level means by condition × phase × axis
axis_means_DIST <- analysis_axis %>%
  group_by(participant_id, condition, phase, partner_axis) %>%
  summarise(mean_DIST = mean(DIST, na.rm = TRUE), .groups = "drop")
```

```

#paired t-test + Cohen's d: horizontal vs vertical within each condition × phase
axis_tests_DIST <- axis_means_DIST %>%
  pivot_wider(names_from = partner_axis, values_from = mean_DIST) %>%
  group_by(condition, phase) %>%
  summarise(
    n          = sum(!is.na(horizontal) & !is.na(vertical)),
    M_horiz    = mean(horizontal, na.rm = TRUE),
    M_vert     = mean(vertical,   na.rm = TRUE),
    mean_diff  = mean(vertical - horizontal, na.rm = TRUE),
    t_stat     = tryCatch(t.test(vertical[!is.na(vertical) & !is.na(horizontal)],
                                horizontal[!is.na(vertical) & !is.na(horizontal)],
                                paired = TRUE)$statistic, error = function(e) NA),
    p_value    = tryCatch(t.test(vertical[!is.na(vertical) & !is.na(horizontal)],
                                horizontal[!is.na(vertical) & !is.na(horizontal)],
                                paired = TRUE)$p.value,   error = function(e) NA),

    cohens_d   = tryCatch(
      mean(vertical - horizontal, na.rm = TRUE) /
      sd(vertical - horizontal, na.rm = TRUE),
      error = function(e) NA
    ),
    .groups = "drop"
  ) %>%
  mutate(
    sig = case_when(
      p_value < .001 ~ "***",
      p_value < .01  ~ "**",
      p_value < .05  ~ "*",
      p_value < .10  ~ "†",
      TRUE          ~ ""
    )
  )

cat("  Axis-control tests (DIST)  \n")

```

```
##  Axis-control tests (DIST)
```

```
print(axis_tests_DIST)
```

```
## # A tibble: 4 x 10
##   condition phase      n M_horiz M_vert mean_diff t_stat p_value cohens_d sig
##   <chr>      <chr> <int>  <dbl>  <dbl>    <dbl>  <dbl>  <dbl>  <dbl> <chr>
## 1 ai        approa~   11   336.   323.   -12.3  -2.58   0.0273  -0.779 "*"
## 2 ai        homing~   11   194.   208.    14.3   0.632   0.542    0.191 ""
## 3 playback  approa~   11   377.   372.    -4.39 -0.489   0.636   -0.147 ""
## 4 playback  homing~   11   239.   231.    -7.72 -0.340   0.741   -0.102 ""
```

```

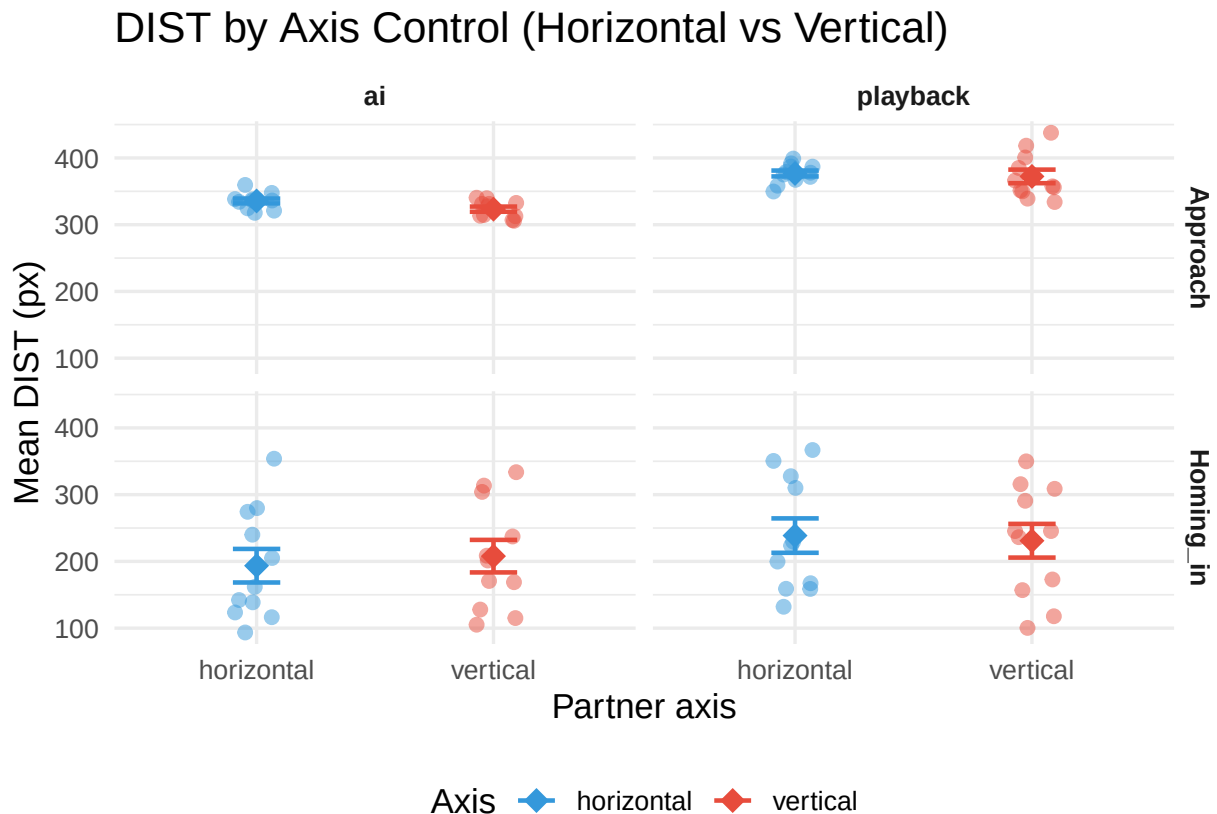
#plot
axis_means_DIST %>%
  ggplot(aes(x = partner_axis, y = mean_DIST, colour = partner_axis)) +
  geom_jitter(width = 0.1, alpha = 0.5, size = 2) +
  stat_summary(fun = mean, geom = "point", size = 4, shape = 18) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2, linewidth = 0.8) +

```

```

facet_grid(phase ~ condition, labeller = labeller(phase = str_to_title)) + # restore phase capitali
scale_colour_manual(values = c("horizontal" = "#3498DB", "vertical" = "#E74C3C")) +
labs(
  title = "DIST by Axis Control (Horizontal vs Vertical)",
  x = "Partner axis", y = "Mean DIST (px)", colour = "Axis"
) +
theme_minimal(base_size = 13) +
theme(legend.position = "bottom", strip.text = element_text(face = "bold"))

```



#Run learning analysis

The approach: for each cooperative trial, collective benefit is computed as the dyad's reaction time on that trial relative to the fixed best-individual baseline for the relevant contribution angle. Trial order within each dyad and phase is then entered as a continuous predictor in a linear mixed model with a random intercept and slope per dyad, so a positive trial slope indicates that collective benefit grows with practice (i.e. learning). Because there are only a handful of cooperative trials per contribution angle, trials are pooled across angle (the angle-specific baseline keeps the measure comparable) and binned only for the plot. linear mixed model with block as the learning predictor and a random effect per dyad.

The question behind it: do dyads get better at working together the more they practise? → “better together” here means that their joint performance beats what the stronger of the two could do alone (the collective benefit).

Interpretation for the plots: we divide the cooperative RT/DIST by the better member's RT/DIST → above 1: the pair did better than the better individual, below 1: the pair did worse than the better individual. Plots: dashed line at 1 means no benefit, it chops the trials into 4 chunks from early to late trials.

```

#Run learning analysis
#does collective benefit (RT) grow over cooperative trials?

library(lme4)
library(lmerTest)

## Warning: package 'lmerTest' was built under R version 4.4.3

##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##      lmer

## The following object is masked from 'package:stats':
##
##      step

#Per-trial collective benefit for the cooperative condition
#numerator   = fixed best-individual baseline
#denominator = the dyad's RT on that single cooperative trial
coop_trials_RT <- analysis_data %>%
  filter(condition == "cooperative") %>%
  left_join(best_individual_RT, by = c("dyad_id", "phase", "contribution_angle")) %>%
  filter(!is.na(best_individual_RT), !is.na(RT), RT > 0) %>%
  mutate(cb_trial = best_individual_RT / RT)

#re-index trials chronologically within each dyad x phase, then center & scale
coop_trials_RT <- coop_trials_RT %>%
  group_by(dyad_id, phase) %>%
  arrange(iteration, .by_group = TRUE) %>%
  mutate(trial_order = row_number()) %>%
  ungroup() %>%
  mutate(
    trial_z = as.numeric(scale(trial_order)), # stabilises the random slope
    phase   = factor(phase)
  )

#Learning model: trial order predicting collective benefit -----
#continuous trial predictor, random intercept + slope per dyad
m_learn <- lmer(cb_trial ~ trial_z * phase + (trial_z | dyad_id), data = coop_trials_RT)

## boundary (singular) fit: see help('isSingular')

summary(m_learn)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: cb_trial ~ trial_z * phase + (trial_z | dyad_id)
##      Data: coop_trials_RT

```

```
##
## REML criterion at convergence: 433.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8670 -0.5016 -0.0951  0.2839  6.2726
##
## Random effects:
##   Groups   Name      Variance Std.Dev. Corr
## dyad_id  (Intercept) 0.01151  0.1073
##          trial_z     0.00308  0.0555  1.00
## Residual                0.33104  0.5754
## Number of obs: 240, groups: dyad_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      1.032289   0.068393  10.391422  15.094 2.11e-08 ***
## trial_z           0.050976   0.057302  24.380700   0.890   0.382
## phaseHoming-in    -0.030970   0.074279  230.999977  -0.417   0.677
## trial_z:phaseHoming-in -0.008537   0.074434  230.999977  -0.115   0.909
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) trl_z phsHm-
## trial_z       0.253
## phaseHmng-n  -0.543  0.000
## trl_z:phsH-   0.000 -0.649  0.000
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

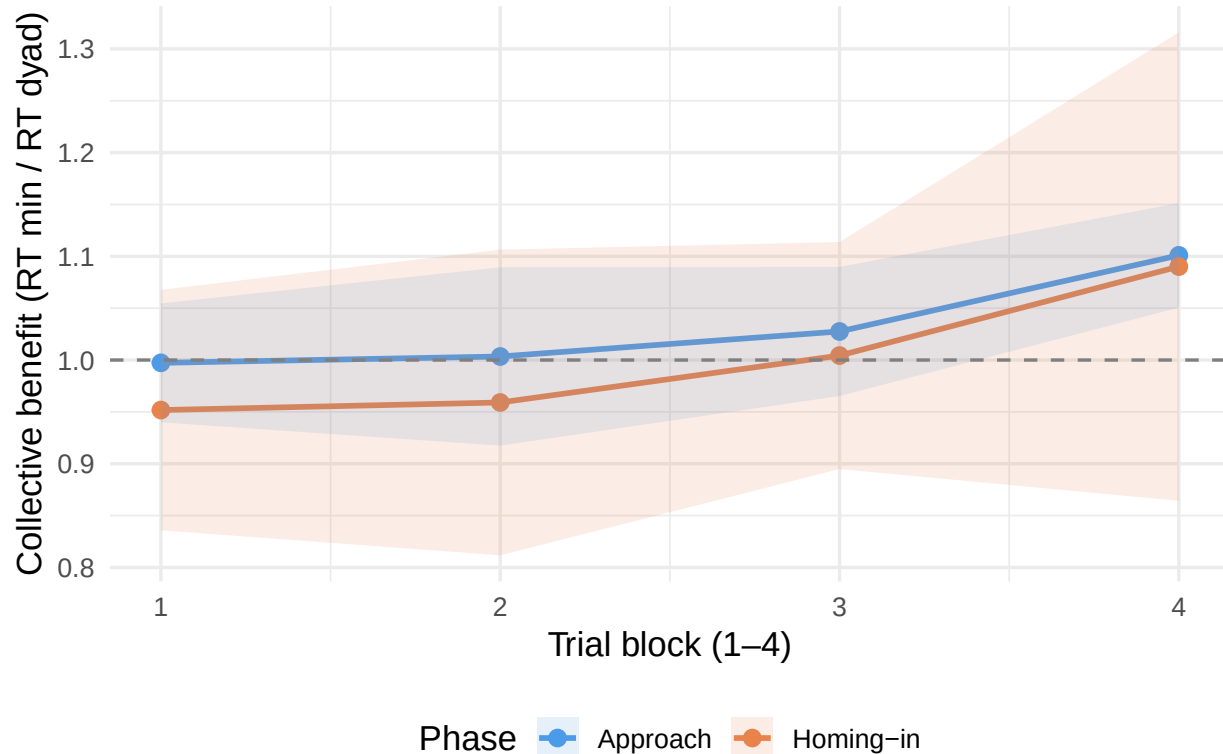
#plot: block means across the experiment, per phase
n_blocks <- 4 # blocks are for the PLOT only; the model above uses raw trial order

cb_blocks_RT <- coop_trials_RT %>%
  group_by(dyad_id, phase) %>%
  mutate(block = ntile(trial_order, n_blocks)) %>%
  group_by(dyad_id, phase, block) %>%
  summarise(collective_benefit = mean(cb_trial, na.rm = TRUE), .groups = "drop")

ggplot(cb_blocks_RT, aes(block, collective_benefit, colour = phase, fill = phase)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1) +
  stat_summary(fun = mean, geom = "point", size = 2.5) +
  stat_summary(fun.data = mean_se, geom = "ribbon", alpha = 0.15, colour = NA) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey50") +
  scale_colour_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  scale_fill_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  scale_x_continuous(breaks = 1:n_blocks) +
  labs(
    title = "Learning: collective benefit (RT) across cooperative trials",
    x = paste0("Trial block (1\u2013", n_blocks, ")"),
    y = "Collective benefit (RT min / RT dyad)",
    colour = "Phase", fill = "Phase"
  ) +
```

```
theme_minimal(base_size = 13) +
theme(legend.position = "bottom")
```

Learning: collective benefit (RT) across cooperative trials



```
# Run learning analysis: does collective benefit (DIST) grow over cooperative trials?

library(lme4)
library(lmerTest)

#per-trial collective benefit (DIST) for the cooperative condition
#numerator = fixed best-individual DIST baseline (angle-specific)
#denominator = the dyad's DIST on that single cooperative trial
coop_trials_DIST <- analysis_data %>%
  filter(condition == "cooperative") %>%
  left_join(best_individual_DIST, by = c("dyad_id", "phase", "contribution_angle")) %>%
  filter(!is.na(best_individual_DIST), !is.na(DIST), DIST > 0) %>%
  mutate(cb_trial = best_individual_DIST / DIST)

#re-index trials chronologically within each dyad x phase, then center & scale
coop_trials_DIST <- coop_trials_DIST %>%
  group_by(dyad_id, phase) %>%
  arrange(iteration, .by_group = TRUE) %>%
  mutate(trial_order = row_number()) %>%
  ungroup() %>%
  mutate(
    trial_z = as.numeric(scale(trial_order)), # stabilises the random slope
```

```

    phase = factor(phase)
  )

#Learning model: trial order predicting collective benefit (DIST)
m_learn_DIST <- lmer(cb_trial ~ trial_z * phase + (trial_z | dyad_id), data = coop_trials_DIST)

## boundary (singular) fit: see help('isSingular')

summary(m_learn_DIST)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: cb_trial ~ trial_z * phase + (trial_z | dyad_id)
## Data: coop_trials_DIST
##
## REML criterion at convergence: 599
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.7543 -0.4693 -0.0624  0.1714  7.6467
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## dyad_id (Intercept) 0.025681 0.16025
## trial_z 0.008089 0.08994 1.00
## Residual 0.667485 0.81700
## Number of obs: 240, groups: dyad_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.96090    0.09921    9.87002   9.686 2.35e-06 ***
## trial_z         0.01330    0.08327   19.16407   0.160  0.875
## phaseHoming-in  0.14538    0.10547  230.99997   1.378  0.169
## trial_z:phaseHoming-in 0.04799    0.10569  230.99997   0.454  0.650
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) tril_z phsHm-
## trial_z      0.291
## phaseHmng-n -0.532  0.000
## trl_z:phsH-  0.000 -0.635  0.000
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

#Plot: block means across the experiment, per phase

n_blocks <- 4 # blocks are for the PLOT only; the model above uses raw trial order

cb_blocks_DIST <- coop_trials_DIST %>%
  group_by(dyad_id, phase) %>%
  mutate(block = ntile(trial_order, n_blocks)) %>%

```

```

group_by(dyad_id, phase, block) %>%
  summarise(collective_benefit = mean(cb_trial, na.rm = TRUE), .groups = "drop")

ggplot(cb_blocks_DIST, aes(block, collective_benefit, colour = phase, fill = phase)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1) +
  stat_summary(fun = mean, geom = "point", size = 2.5) +
  stat_summary(fun.data = mean_se, geom = "ribbon", alpha = 0.15, colour = NA) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey50") +
  scale_colour_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  scale_fill_manual(values = c("Approach" = "#4C9BE8", "Homing-in" = "#E8834C")) +
  scale_x_continuous(breaks = 1:n_blocks) +
  labs(
    title = "Learning: collective benefit (DIST) across cooperative trials",
    x = paste0("Trial block (1\u2013", n_blocks, ")"),
    y = "Collective benefit (DIST min / DIST dyad)",
    colour = "Phase", fill = "Phase"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")

```

